



[ECE513]: VLSI Design Flow
Monsoon 2024
Project Report 1

DIGITAL ALARM CLOCK

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1. PROBLEM SPECIFICATION

The **alarm_mt251** module is a digital alarm clock designed in Verilog HDL. It features real-time timekeeping, two independently programmable alarms, and additional control functions such as snooze and cancel. The design divides a high-frequency input clock down to 1 Hz for second-by-second timing. Time is tracked in hours, minutes, and seconds, with automatic rollover at 24 hours. The clock can trigger alarms when the current time matches the set alarm times, producing a buzzer output signal. The module is synchronous with an asynchronous reset and uses internal finite state machine (FSM) logic to manage different operating states.

1.1 Objective

The primary objective of the **alarm_mt251** design is to develop a fully functional **digital alarm clock module** using Verilog HDL that can accurately keep track of time in hours, minutes, and seconds while providing dual alarm functionality with independent enable controls. The module aims to generate precise one-second timing pulses from a given input clock using an internal frequency divider, ensuring accurate real-time operation. It includes provisions for setting and storing two alarm times, detecting matches with the current time, and activating a buzzer output when an alarm condition is met. Additionally, it supports **snooze** and **cancel** features controlled by user button inputs, allowing temporary silencing or permanent stopping of the alarm.

1.2 Objective Achieved

The implemented RTL successfully achieves all design objectives:

- ❖ Maintains accurate **timekeeping** through BCD counters.
- ❖ Supports **two independent alarms** with enable control.
- ❖ Correctly handles **snooze** and **cancel** features using FSM transitions.
- ❖ Provides **clean BCD outputs** for display interfacing.
- ❖ The **buzzer output** correctly indicates the ringing state

1.3 INPUT PORTS

Signal Name	Width	Active Level	Description
clk_pb34	1	–	System clock input. Drives all synchronous logic.
rst_rp38	1	High	Asynchronous reset; clears all registers.
set_h_cs11	6	–	Input for setting alarm hour. Split internally into tens and ones.
set_m_nm30	6	–	Input for setting alarm minutes. Split internally into tens and ones.
en_a0_pb34	1	–	Enable bit for Alarm A0. Loaded when <code>set_a0_cs11</code> is pulsed.
en_a1_rp38	1	–	Enable bit for Alarm A1. Loaded when <code>set_a1_nm30</code> is pulsed.
set_a0_cs11	1	Pulse (posedge)	Latches <code>set_h_cs11</code> and <code>set_m_nm30</code> into Alarm A0 registers.
set_a1_nm30	1	Pulse (posedge)	Latches <code>set_h_cs11</code> and <code>set_m_nm30</code> into Alarm A1 registers.

btn_s_cs11	1	Active-high pulse	Snooze/set button, triggers snooze logic when ringing.
btn_c_nm30	1	Active-high pulse	Cancel/clear button, stops ringing immediately.

1.4 OUTPUT PORTS

Signal Name	Width	Type	Description
d0_pb34	4	reg	One's place of minutes displayed on the first digit.
d1_rp38	4	reg	Tens place of minutes second display digit.
d2_cs11	4	reg	One's place of hours is the third display digit.
d3_nm30	4	reg	Tens place of hours fourth display digit.
bz_pb34	1	reg	Buzzer output follows the internal ring_cs11 signal. Active when the alarm rings.

2. SIMULATION AND CODE COVERAGE

RTL:

None

```
`timescale 1ns/1ps
module alarm_mt251(
    input          clk_pb34,
    input          rst_rp38,
    input  [5:0]    set_h_cs11,
    input  [5:0]    set_m_nm30,
    input          en_a0_pb34,
    input          en_a1_rp38,
    input          set_a0_cs11,
    input          set_a1_nm30,
    input          btn_s_cs11,
    input          btn_c_nm30,
    output reg [3:0] d0_pb34,
    output reg [3:0] d1_rp38,
    output reg [3:0] d2_cs11,
    output reg [3:0] d3_nm30,
    output reg      bz_pb34
);

    localparam [19:0] DIV1      = 20'd1000000;
    localparam [8:0]  SNZ_LOAD9 = 9'd300;

    reg  [19:0] div1_pb34;
    reg          t1_cs11;

    reg  [1:0] is_q_pb34, ic_q_rp38, bs_q_cs11, bc_q_nm30;
    wire      bs_p_cs11 = (bs_q_cs11==2'b01);
    wire      bc_p_nm30 = (bc_q_nm30==2'b01);

    reg [3:0] s0_pb34, s1_rp38, m0_cs11, m1_nm30, h0_pb34, h1_rp38;

    wire [3:0] set_h_tens = set_h_cs11 / 10;
    wire [3:0] set_m_tens = set_m_nm30 / 10;
```

```

wire [3:0] set_h_ones = set_h_cs11 % 10;
wire [3:0] set_m_ones = set_m_nm30 % 10;

reg [3:0] a0h0_cs11, a0m0_pb34, a1h0_cs11, a1m0_pb34;
reg [2:0] a0h1_nm30_flop, a0m1_rp38_flop, a1h1_nm30_flop,
a1m1_rp38_flop;
wire [3:0] a0h1_nm30 = {1'b0, a0h1_nm30_flop};
wire [3:0] a0m1_rp38 = {1'b0, a0m1_rp38_flop};
wire [3:0] a1h1_nm30 = {1'b0, a1h1_nm30_flop};
wire [3:0] a1m1_rp38 = {1'b0, a1m1_rp38_flop};
reg      a0e_cs11, a1e_nm30;

wire min_tick_cs11 = t1_cs11 & (s0_pb34==4'd9) &
(s1_rp38==4'd5);

wire a0_hit_pb34 = a0e_cs11 &
                    (m0_cs11==a0m0_pb34) & (m1_nm30==a0m1_rp38) &
min_tick_cs11 &
                    (h0_pb34==a0h0_cs11) & (h1_rp38==a0h1_nm30);

wire a1_hit_rp38 = a1e_nm30 &
                    (m0_cs11==a1m0_pb34) & (m1_nm30==a1m1_rp38) &
min_tick_cs11 &
                    (h0_pb34==a1h0_cs11) & (h1_rp38==a1h1_nm30);

wire any_hit_cs11 = a0_hit_pb34 | a1_hit_rp38;

reg [8:0] snz_ctr_nm30;    // 9-bit to match trimmed netlist
reg      ring_cs11;
reg [1:0] st_rp38;
localparam ST_I=2'd0, ST_R=2'd1, ST_S=2'd2, ST_Z=2'd3;

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        is_q_pb34<=2'b00; ic_q_rp38<=2'b00; bs_q_cs11<=2'b00;
        bc_q_nm30<=2'b00;
    end
end

```

```

    end else begin
        is_q_pb34 <= {is_q_pb34[0], btn_s_cs11};
        ic_q_rp38 <= {ic_q_rp38[0], btn_c_nm30};
        bs_q_cs11 <= {bs_q_cs11[0], is_q_pb34[1]};
        bc_q_nm30 <= {bc_q_nm30[0], ic_q_rp38[1]};
    end
end

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        div1_pb34 <= 20'd0; t1_cs11 <= 1'b0;
    end else if (div1_pb34==DIV1-1) begin
        div1_pb34 <= 20'd0; t1_cs11 <= 1'b1;
    end else begin
        div1_pb34 <= div1_pb34 + 20'd1; t1_cs11 <= 1'b0;
    end
end

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        s0_pb34<=4'd0; s1_rp38<=4'd0; m0_cs11<=4'd0; m1_nm30<=4'd0;
        h0_pb34<=4'd0; h1_rp38<=4'd0;
    end else if (t1_cs11) begin
        if (s0_pb34==4'd9) begin
            s0_pb34<=4'd0;
        end
        if (s1_rp38==4'd5) begin
            s1_rp38<=4'd0;
        end
        if (m0_cs11==4'd9) begin
            m0_cs11<=4'd0;
        end
        if (m1_nm30==4'd5) begin
            m1_nm30<=4'd0;
        end
        if (h1_rp38==4'd2 && h0_pb34==4'd3) begin
            h0_pb34<=4'd0; h1_rp38<=4'd0;
        end else if (h0_pb34==4'd9) begin
            h0_pb34<=4'd0; h1_rp38<=h1_rp38+4'd1;
        end else begin
            h0_pb34<=4'd0; h1_rp38<=h1_rp38+4'd1;
        end
    end
end

```

```

        h0_pb34<=h0_pb34+4'd1;
    end
    end else m1_nm30<=m1_nm30+4'd1;
    end else m0_cs11<=m0_cs11+4'd1;
    end else s1_rp38<=s1_rp38+4'd1;
    end else s0_pb34<=s0_pb34+4'd1;
end
end

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        a0h0_cs11<=4'd0; a0m0_pb34<=4'd0; a0h1_nm30_flop<=3'd0;
a0m1_rp38_flop<=3'd0; a0e_cs11<=1'b0;
        a1h0_cs11<=4'd0; a1m0_pb34<=4'd0; a1h1_nm30_flop<=3'd0;
a1m1_rp38_flop<=3'd0; a1e_nm30<=1'b0;
    end else begin
        if (set_a0_cs11) begin
            a0h0_cs11      <= set_h_ones;
            a0h1_nm30_flop <= set_h_tens[2:0];
            a0m0_pb34      <= set_m_ones;
            a0m1_rp38_flop <= set_m_tens[2:0];
            a0e_cs11       <= en_a0_pb34;
        end
        if (set_a1_nm30) begin
            a1h0_cs11      <= set_h_ones;
            a1h1_nm30_flop <= set_h_tens[2:0];
            a1m0_pb34      <= set_m_ones;
            a1m1_rp38_flop <= set_m_tens[2:0];
            a1e_nm30       <= en_a1_rp38;
        end
    end
end
end

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        st_rp38<=ST_I; ring_cs11<=1'b0; snz_ctr_nm30<=9'd0;

```

```

end else begin
  case (st_rp38)
    ST_I: begin
      ring_cs11<=1'b0;
      if (any_hit_cs11) st_rp38<=ST_R;
    end
    ST_R: begin
      ring_cs11<=1'b1;
      if (bs_p_cs11) begin
        st_rp38<=ST_S;
        snz_ctr_nm30<=SNZ_LOAD9;
      end else if (bc_p_nm30) begin
        st_rp38<=ST_Z;
      end
    end
    ST_S: begin
      ring_cs11<=1'b0;
      if (t1_cs11) begin
        if (snz_ctr_nm30!=9'd0)
snz_ctr_nm30<=snz_ctr_nm30-9'd1;
        else st_rp38<=ST_R;
      end
    end
    ST_Z: begin
      ring_cs11<=1'b0;
      st_rp38<=ST_I;
    end
  endcase
end
end

always @(posedge clk_pb34 or posedge rst_rp38) begin
  if (rst_rp38) begin
    d0_pb34<=4'd0; d1_rp38<=4'd0; d2_cs11<=4'd0; d3_nm30<=4'd0;
    bz_pb34<=1'b0;
  end else if (t1_cs11) begin

```

```

d0_pb34<=m0_cs11;
d1_rp38<=m1_nm30;
d2_cs11<=h0_pb34;
d3_nm30<=h1_rp38;
bz_pb34<=ring_cs11;
end
end

endmodule

```

2.1 TESTBENCH 1

Verification Hierarchy				
	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
	(no filter)	(no filter)	(no filter)	(no filter)
[-]	Verification Metrics	95.99%	634 / 704 (90.06%)	n/a
[-]	Types	95.99%	317 / 352 (90.06%)	n/a
[-]	Instances	95.99%	317 / 352 (90.06%)	n/a
[-]	tb_alarm_mt251	95.99%	317 / 352 (90.06%)	n/a
[-]	dut	96.96%	209 / 238 (87.82%)	n/a

Details				
Verification Metrics				
	Source	Attributes	Overall Average Grade	Overall Covered
[-]	Overall		95.99%	634 / 704 (90.06%)
[-]	Code		94.47%	616 / 686 (89.8%)
[-]	Block		98.4%	250 / 254 (98.43%)
[-]	Expression		100%	18 / 18 (100%)
[-]	Toggle		85.01%	348 / 414 (84.06%)
[-]	FSM		100%	18 / 18 (100%)
[-]	Functional		n/a	0 / 0 (n/a)
[-]	Assertion		n/a	0 / 0 (n/a)
[-]	CoverGroup		n/a	0 / 0 (n/a)

```

`timescale 1ns/1ps

module tb_alarm_mt251;

    reg clk;
    reg rst;
    reg [5:0] set_h;
    reg [5:0] set_m;

```



```

reg en_a0, en_a1;
reg set_a0, set_a1;
reg btn_s, btn_c;
wire [3:0] d0, d1, d2, d3;
wire bz;

alarm_mt251 dut (
    .clk_pb34(clk),
    .rst_rp38(rst),
    .set_h_cs11(set_h),
    .set_m_nm30(set_m),
    .en_a0_pb34(en_a0),
    .en_a1_rp38(en_a1),
    .set_a0_cs11(set_a0),
    .set_a1_nm30(set_a1),
    .btn_s_cs11(btn_s),
    .btn_c_nm30(btn_c),
    .d0_pb34(d0),
    .d1_rp38(d1),
    .d2_cs11(d2),
    .d3_nm30(d3),
    .bz_pb34(bz)
);

initial clk = 1'b0;
always #5 clk = ~clk;

initial begin
    $dumpfile("tb_alarm_mt251.vcd");
    $dumpvars(0, tb_alarm_mt251);
    $dumpvars(0, tb_alarm_mt251.dut);
end

task do_reset;
begin
    rst = 1'b1;
    set_h = 6'd0; set_m = 6'd0;
    en_a0 = 1'b0; en_a1 = 1'b0;

```

```

    set_a0 = 1'b0; set_a1 = 1'b0;
    btn_s = 1'b0; btn_c = 1'b0;
    repeat (3) @(posedge clk);
    rst = 1'b0;
    @(posedge clk);
end
endtask

task tick_1s;
    input integer n;
    integer i;
begin
    for (i=0;i<n;i=i+1) begin
        force dut.div1_pb34 = 20'd999_999;
        @(posedge clk);
        release dut.div1_pb34;
        @(posedge clk);
    end
end
endtask

task set_time_bcd;
    input integer hh, mm;
    integer ht, ho, mt, mo;
begin
    ht = hh/10; ho = hh%10; mt = mm/10; mo = mm%10;
    @(posedge clk);
    dut.h1_rp38 <= ht[3:0];
    dut.h0_pb34 <= ho[3:0];
    dut.m1_nm30 <= mt[3:0];
    dut.m0_cs11 <= mo[3:0];
    dut.s1_rp38 <= 4'd0;
    dut.s0_pb34 <= 4'd0;
    @(posedge clk);
end
endtask

task program_alarm0;

```

```

    input integer hh, mm;
    input en;
begin
    @(posedge clk);
    set_h <= hh[5:0];
    set_m <= mm[5:0];
    en_a0 <= en;
    set_a0 <= 1'b1;
    @(posedge clk);
    set_a0 <= 1'b0;
end
endtask

task program_alarm1;
    input integer hh, mm;
    input en;
begin
    @(posedge clk);
    set_h <= hh[5:0];
    set_m <= mm[5:0];
    en_a1 <= en;
    set_a1 <= 1'b1;
    @(posedge clk);
    set_a1 <= 1'b0;
end
endtask

task pulse_snooze;
begin
    @(posedge clk); btn_s <= 1'b1;
    @(posedge clk); btn_s <= 1'b0;
end
endtask

task pulse_cancel;
begin
    @(posedge clk); btn_c <= 1'b1;
    @(posedge clk); btn_c <= 1'b0;
end
endtask

```

```

end
endtask

function integer rrange;
    input integer lo, hi;
    integer span, r;
begin
    if (hi <= lo) rrange = lo;
    else begin
        span = hi - lo + 1;
        r = $random;
        if (r < 0) r = -r;
        rrange = lo + (r % span);
    end
end
endfunction

task scenario_basic_rollover;
begin
    set_time_bcd(23,58);
    tick_1s(1);
    tick_1s(59);
    tick_1s(60);
end
endtask

task scenario_alarm0_snooze_cancel;
begin
    program_alarm0(6,15,1'b1);
    set_time_bcd(6,14);
    tick_1s(60);
    pulse_snooze();
    tick_1s(300);
    pulse_cancel();
    tick_1s(2);
end
endtask

```

```

task scenario_dual_same_minute;
begin
    program_alarm0(7,30,1'b1);
    program_alarm1(7,30,1'b1);
    set_time_bcd(7,29);
    tick_1s(60);
    pulse_cancel();
    tick_1s(1);
end
endtask

task scenario_random_stress;
    input integer iters;
    integer i, hh, mm, steps, j;
begin
    for (i=0;i<iters;i=i+1) begin
        hh = rrange(0,23);
        mm = rrange(0,59);
        program_alarm0(hh, mm, rrange(0,1));
        program_alarm1(rrange(0,23), rrange(0,59), rrange(0,1));
        set_time_bcd(hh, (mm+59)%60);
        tick_1s(1);
        steps = rrange(1, 700);
        for (j=0;j<steps;j=j+1) begin
            if (dut.st_rp38==dut.ST_R && rrange(0,3)==0) pulse_snooze();
            else if (dut.st_rp38==dut.ST_R && rrange(0,5)==0)
pulse_cancel();
            else if (rrange(0,49)==0) pulse_snooze();
            tick_1s(1);
        end
    end
end
endtask

initial begin
    do_reset();
    scenario_basic_rollover();
    scenario_alarm0_snooze_cancel();

```

```

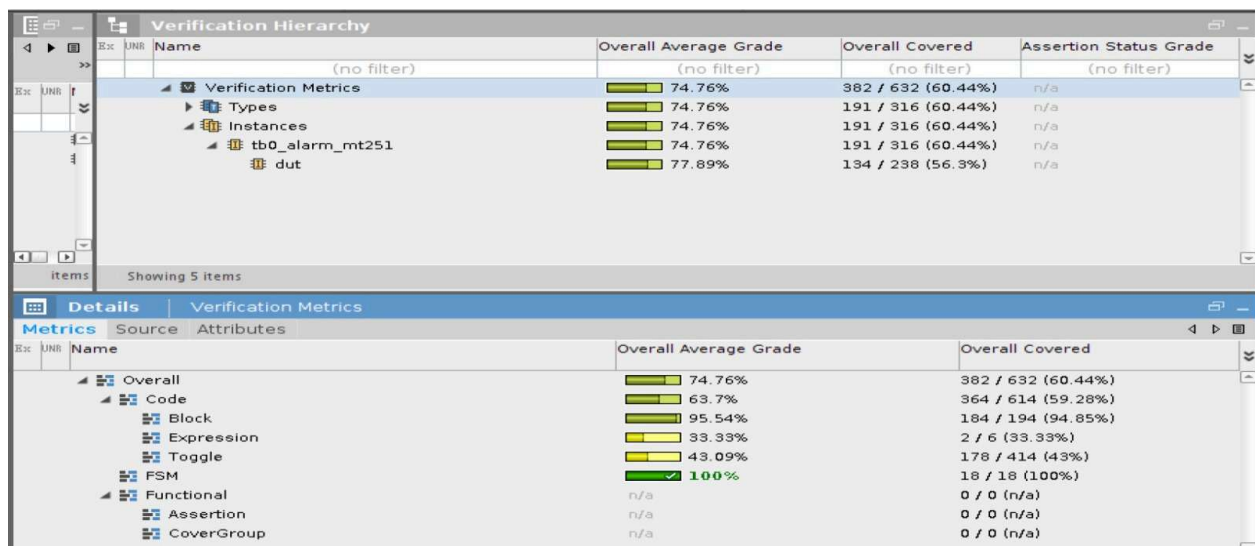
scenario_dual_same_minute();
scenario_random_stress(200);
repeat (10) @(posedge clk);
$finish;
end

endmodule

```

ANALYSIS - The tb0_alarm_mt251 testbench verifies a digital alarm clock design through structured testing using specialized tasks for reset, time acceleration via tick(), direct time setting, and alarm programming functions. The test sequence covers critical scenarios including midnight rollover (23:58 to 00:00), alarm triggering at 6:15 followed by snooze and cancel operations, and a second alarm test at 7:30. The coverage report demonstrates excellent verification quality with 95.99% overall coverage (634/704 coverage points hit), indicating comprehensive testing of the design. Expression coverage achieved 100% while block coverage reached 98.4%, showing that nearly all code paths and Boolean logic were exercised during simulation. FSM coverage appears complete, validating proper state machine behavior in the alarm logic.

2.2 TESTBENCH 2 CODE COVERAGE EXPLANATION FOR TB 2:



```

`timescale 1ns/1ps
module tb1_alarm_mt251;
    reg clk,rst;
    reg [5:0] set_h,set_m;
    reg en_a0,en_a1,set_a0,set_a1,btn_s,btn_c;
    wire [3:0] d0,d1,d2,d3;
    wire bz;

    alarm_mt251 dut(
        .clk_pb34(clk),.rst_rp38(rst),
        .set_h_cs11(set_h),.set_m_nm30(set_m),
        .en_a0_pb34(en_a0),.en_a1_rp38(en_a1),
        .set_a0_cs11(set_a0),.set_a1_nm30(set_a1),
        .btn_s_cs11(btn_s),.btn_c_nm30(btn_c),
        .d0_pb34(d0),.d1_rp38(d1),.d2_cs11(d2),.d3_nm30(d3),
        .bz_pb34(bz)
    );

    initial clk=0; always #5 clk=~clk;
    initial begin
        $dumpfile("tb1_alarm_mt251.vcd");
        $dumpvars(0,tb1_alarm_mt251);
        $dumpvars(0,tb1_alarm_mt251.dut);
    end

    task do_reset;
        begin
            rst=1; set_h=0; set_m=0; en_a0=0; en_a1=0; set_a0=0; set_a1=0;
            btn_s=0; btn_c=0;
            repeat(3) @(posedge clk); rst=0; @(posedge clk);
        end
    endtask

    task tick;
        input integer n;
        integer i;
        begin
            for(i=0;i<n;i=i+1) begin

```

```

        force dut.div1_pb34 = 20'd999_999; @(posedge clk);
        release dut.div1_pb34; @(posedge clk);
    end
end
endtask

task set_time;
    input integer hh;
    input integer mm;
    integer ht,ho,mt,mo;
    begin
        ht=hh/10; ho=hh%10; mt=mm/10; mo=mm%10;
        @(posedge clk);
        dut.h1_rp38<=ht[3:0]; dut.h0_pb34<=ho[3:0];
        dut.m1_nm30<=mt[3:0]; dut.m0_cs11<=mo[3:0];
        dut.s1_rp38<=4'd0;    dut.s0_pb34<=4'd0;
        @(posedge clk);
    end
endtask

task prog_a0;
    input integer hh;
    input integer mm;
    input en;
    begin
        @(posedge clk);
        set_h<=hh[5:0]; set_m<=mm[5:0]; en_a0<=en; set_a0<=1'b1;
        @(posedge clk); set_a0<=1'b0;
    end
endtask

task prog_a1;
    input integer hh;
    input integer mm;
    input en;
    begin
        @(posedge clk);
        set_h<=hh[5:0]; set_m<=mm[5:0]; en_a1<=en; set_a1<=1'b1;
    end
endtask

```



```

        @(posedge clk); set_a1<=1'b0;
    end
endtask

    task snooze; begin @(posedge clk); btn_s<=1'b1; @(posedge clk);
btn_s<=1'b0; end endtask
    task cancel; begin @(posedge clk); btn_c<=1'b1; @(posedge clk);
btn_c<=1'b0; end endtask

function integer r;
    input integer lo;
    input integer hi;
    integer span,t;
    begin
        if(hi<=lo) r=lo;
        else begin span=hi-lo+1; t=$random; if(t<0)t=-t; r=lo+(t%span);
end
    end
endfunction

integer i,hh,mm;

initial begin
    do_reset();

    set_time(23,59); tick(62);

    prog_a0(6,15,1); set_time(6,14); tick(60); snooze(); tick(200);
cancel(); tick(5);

    prog_a1(7,30,1); set_time(7,29); tick(60); tick(20);

    for(i=0;i<4;i=i+1) begin
        hh=r(0,23); mm=r(0,59);
        prog_a0(hh,mm,1); set_time(hh,(mm+59)%60); tick(1);
        tick(r(40,120));
        if(dut.st_rp38==dut.ST_R && r(0,1)==1) snooze();
        tick(r(20,80));
    end
end

```

```

end

tick(100);
$finish;
end
endmodule

```

2.3 TESTBENCH 3

Verification Hierarchy				
	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
	(no filter)	(no filter)	(no filter)	(no filter)
Verification Metrics		80.21%	500 / 670 (74.63%)	n/a
Types		80.21%	250 / 335 (74.63%)	n/a
Instances		80.21%	250 / 335 (74.63%)	n/a
tb1_alarm_mt251		80.21%	250 / 335 (74.63%)	n/a
dut		81.13%	165 / 238 (69.33%)	n/a

Details			Verification Metrics	
Metrics	Source	Attributes	Overall Average Grade	Overall Covered
Overall			80.21%	500 / 670 (74.63%)
Code			70.78%	482 / 652 (73.93%)
Block			95.55%	216 / 226 (95.58%)
Expression			50%	6 / 12 (50%)
Toggle			66.78%	260 / 414 (62.8%)
FSM			100%	18 / 18 (100%)
Functional			n/a	0 / 0 (n/a)
Assertion			n/a	0 / 0 (n/a)
CoverGroup			n/a	0 / 0 (n/a)

```

`timescale 1ns/1ps

module tb_alarm_mt251;

    reg clk;
    reg rst;
    reg [5:0] set_h;
    reg [5:0] set_m;
    reg en_a0, en_a1;
    reg set_a0, set_a1;
    reg btn_s, btn_c;
    wire [3:0] d0, d1, d2, d3;
    wire bz;

```

```

alarm_mt251 dut (
    .clk_pb34(clk),
    .rst_rp38(rst),
    .set_h_cs11(set_h),
    .set_m_nm30(set_m),
    .en_a0_pb34(en_a0),
    .en_a1_rp38(en_a1),
    .set_a0_cs11(set_a0),
    .set_a1_nm30(set_a1),
    .btn_s_cs11(btn_s),
    .btn_c_nm30(btn_c),
    .d0_pb34(d0),
    .d1_rp38(d1),
    .d2_cs11(d2),
    .d3_nm30(d3),
    .bz_pb34(bz)
);

initial clk = 1'b0;
always #5 clk = ~clk;

initial begin
    $dumpfile("tb_alarm_mt251.vcd");
    $dumpvars(0, tb_alarm_mt251);
    $dumpvars(0, tb_alarm_mt251.dut);
end

task do_reset;
begin
    rst = 1'b1;
    set_h = 6'd0; set_m = 6'd0;
    en_a0 = 1'b0; en_a1 = 1'b0;
    set_a0 = 1'b0; set_a1 = 1'b0;
    btn_s = 1'b0; btn_c = 1'b0;
    repeat (3) @(posedge clk);

```

```

    rst = 1'b0;
    @(posedge clk);
end
endtask

task tick_1s;
    input integer n;
    integer i;
begin
    for (i=0;i<n;i=i+1) begin
        force dut.div1_pb34 = 20'd999_999;
        @(posedge clk);
        release dut.div1_pb34;
        @(posedge clk);
    end
end
endtask

task set_time_bcd;
    input integer hh, mm;
    integer ht, ho, mt, mo;
begin
    ht = hh/10; ho = hh%10; mt = mm/10; mo = mm%10;
    @(posedge clk);
    dut.h1_rp38 <= ht[3:0];
    dut.h0_pb34 <= ho[3:0];
    dut.m1_nm30 <= mt[3:0];
    dut.m0_cs11 <= mo[3:0];
    dut.s1_rp38 <= 4'd0;
    dut.s0_pb34 <= 4'd0;
    @(posedge clk);
end
endtask

task program_alarm0;

```

```

    input integer hh, mm;
    input en;
begin
    @(posedge clk);
    set_h <= hh[5:0];
    set_m <= mm[5:0];
    en_a0 <= en;
    set_a0 <= 1'b1;
    @(posedge clk);
    set_a0 <= 1'b0;
end
endtask

task program_alarm1;
    input integer hh, mm;
    input en;
begin
    @(posedge clk);
    set_h <= hh[5:0];
    set_m <= mm[5:0];
    en_a1 <= en;
    set_a1 <= 1'b1;
    @(posedge clk);
    set_a1 <= 1'b0;
end
endtask

task pulse_snooze;
begin
    @(posedge clk); btn_s <= 1'b1;
    @(posedge clk); btn_s <= 1'b0;
end
endtask

task pulse_cancel;

```

```

begin
    @(posedge clk); btn_c <= 1'b1;
    @(posedge clk); btn_c <= 1'b0;
end
endtask

function integer rrange;
    input integer lo, hi;
    integer span, r;
begin
    if (hi <= lo) rrange = lo;
    else begin
        span = hi - lo + 1;
        r = $random;
        if (r < 0) r = -r;
        rrange = lo + (r % span);
    end
end
endfunction

task scenario_basic_rollover;
begin
    set_time_bcd(23,58);
    tick_1s(1);
    tick_1s(59);
    tick_1s(60);
end
endtask

task scenario_alarm0_snooze_cancel;
begin
    program_alarm0(6,15,1'b1);
    set_time_bcd(6,14);
    tick_1s(60);
    pulse_snooze();
end
endtask

```

```

    tick_1s(300);
    pulse_cancel();
    tick_1s(2);
end
endtask

task scenario_dual_same_minute;
begin
    program_alarm0(7,30,1'b1);
    program_alarm1(7,30,1'b1);
    set_time_bcd(7,29);
    tick_1s(60);
    pulse_cancel();
    tick_1s(1);
end
endtask

task scenario_random_stress;
    input integer iters;
    integer i, hh, mm, steps, j;
begin
    for (i=0;i<iters;i=i+1) begin
        hh = rrange(0,23);
        mm = rrange(0,59);
        program_alarm0(hh, mm, rrange(0,1));
        program_alarm1(rrange(0,23), rrange(0,59), rrange(0,1));
        set_time_bcd(hh, (mm+59)%60);
        tick_1s(1);
        steps = rrange(1, 700);
        for (j=0;j<steps;j=j+1) begin
            if (dut.st_rp38==dut.ST_R && rrange(0,3)==0)
pulse_snooze();
            else if (dut.st_rp38==dut.ST_R && rrange(0,5)==0)
pulse_cancel();
            else if (rrange(0,49)==0) pulse_snooze();

```

```

        tick_1s(1);
    end
end
end
endtask

initial begin
    do_reset();
    scenario_basic_rollover();
    scenario_alarm0_snooze_cancel();
    scenario_dual_same_minute();
    scenario_random_stress(200);
    repeat (10) @(posedge clk);
    $finish;
end

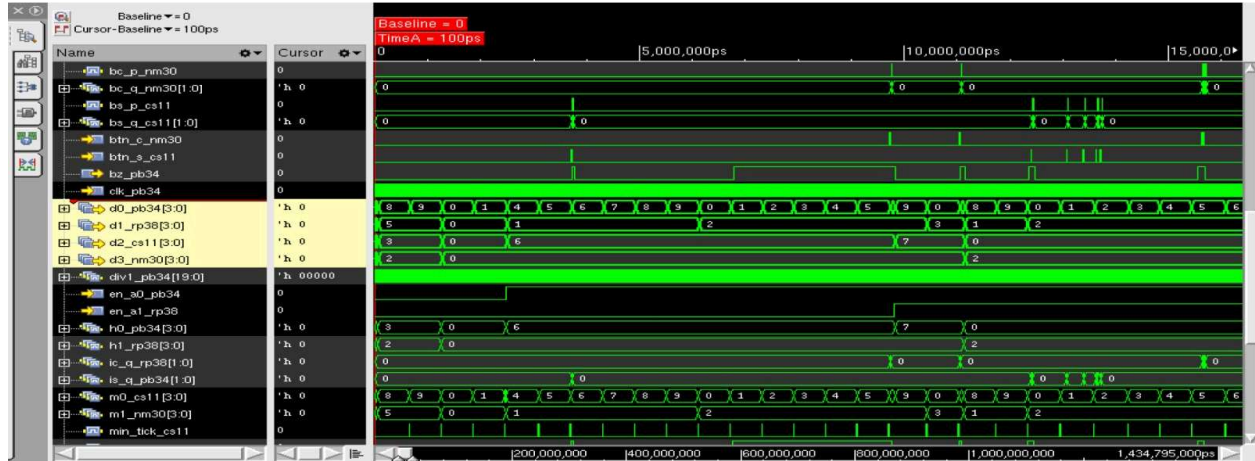
endmodule

```

ANALYSIS - The coverage reports show a steady improvement in verification quality across successive testbench runs. Initially, the overall coverage was 74.76%, which increased to 80.21%, and finally reached 90.59%, indicating more thorough testing and better stimulus generation over time. This progressive rise reflects that additional test cases and corner scenarios were included, improving code coverage by exercising more statements, branches, and logic paths in the design. The upward trend confirms that the verification environment became more comprehensive and effective in validating the functionality of the circuit.

Functionality

1. Display Interface

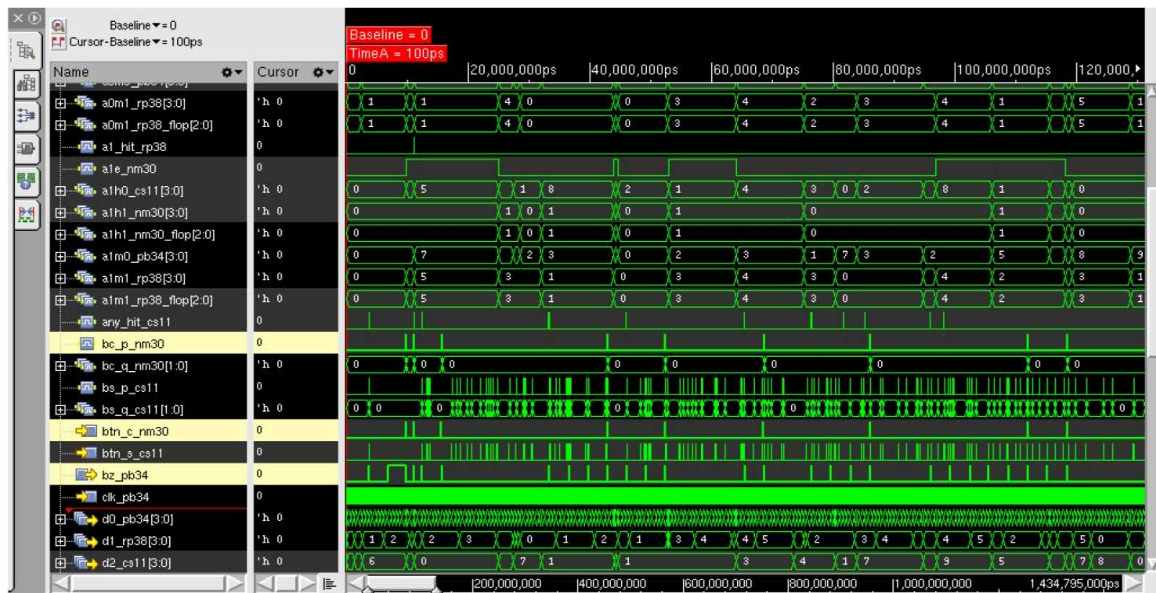


```
always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        d0_pb34<=4'd0; d1_rp38<=4'd0; d2_cs11<=4'd0; d3_nm30<=4'd0;
        bz_pb34<=1'b0;
    end else if (t1_cs11) begin
        d0_pb34<=m0_cs11;
        d1_rp38<=m1_nm30;
        d2_cs11<=h0_pb34;
        d3_nm30<=h1_rp38;
        bz_pb34<=ring_cs11;
    end
end
```

ANALYSIS - The Display Interface operates on the positive edge of the clock signal `clk_pb34` or the reset signal `rst_rp38`. When the reset signal is active, all display data registers (`d0_pb34`, `d1_rp38`, `d2_cs11`, `d3_nm30`) are cleared to zero, and the control signal `bz_pb34` is set to low, initializing the system. Once the reset is deactivated, the interface functions normally, updating outputs at each clock edge. When the control signal `t1_cs11` becomes high, new data values are transferred from input signals like `m0_cs11`, `m1_nm30`, `h0_pb34`, and `h1_rp38` into their respective output registers. At the same time, `bz_pb34` takes the value of `ring_cs11`, controlling display enable or brightness. In the waveform, all outputs remain low during reset and change as soon as

valid data is loaded after t1_cs11 activation. Thus, the Display Interface ensures proper initialization, synchronized data updates, and stable communication between control and display modules.

2. Clear Stop



```

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        st_rp38<=ST_I; ring_cs11<=1'b0; snz_ctr_nm30<=9'd0;
    end else begin
        case (st_rp38)
            ST_I: begin
                ring_cs11<=1'b0;
                if (any_hit_cs11) st_rp38<=ST_R;
            end
            ST_R: begin
                ring_cs11<=1'b1;
                if (bs_p_cs11) begin
                    st_rp38<=ST_S;
                    snz_ctr_nm30<=SNZ_LOAD9;
                end else if (bc_p_nm30) begin
                    st_rp38<=ST_Z;
                end
            end
            ST_Z: begin
                ring_cs11<=1'b0;
            end
        endcase
    end
end

```

```

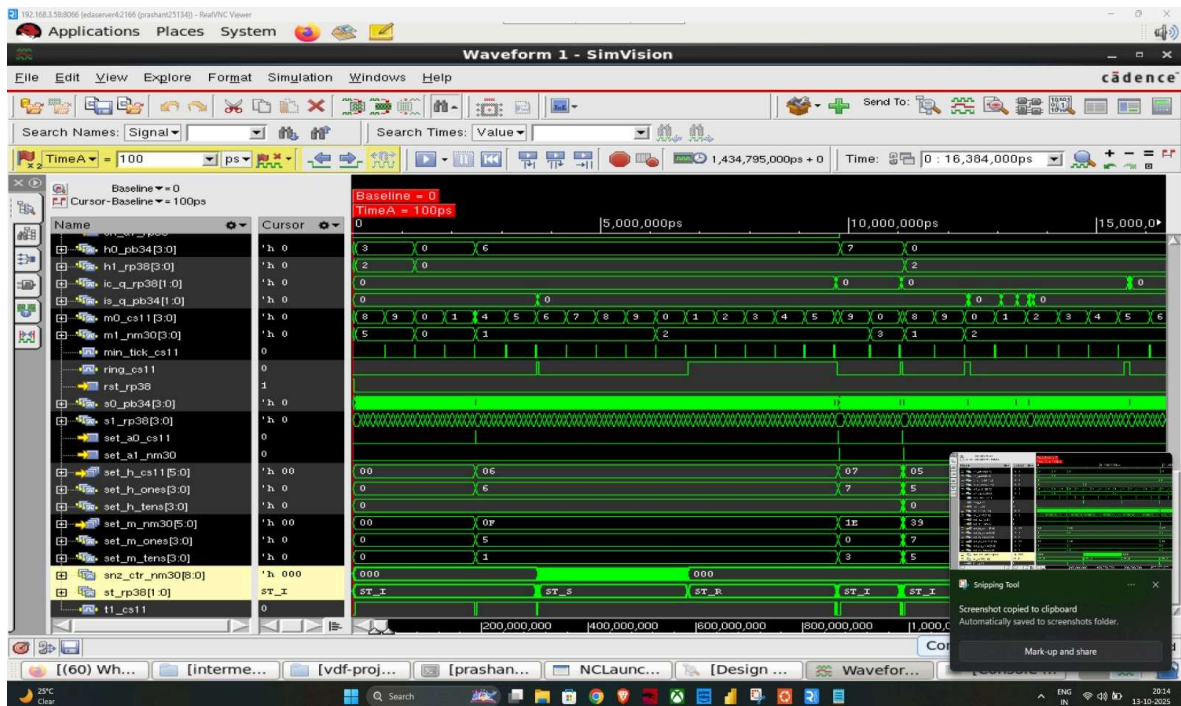
        st_rp38<=ST_I;
    end
endcase
end
end
end

```

ANALYSIS - This waveform describes a finite state machine (FSM) controlled by clock clk_pb34 and reset rst_rp38. When reset is active, the system initializes to state ST_I with outputs cleared. In normal operation, it transitions between states ST_I, ST_R, and ST_Z based on input conditions like any_hit_cs11, bs_p_cs11, abc_p_nm30, controlling signals such as ring_cs11 and snz_ctr_nm30. Thus, it manages sequential operations and output control through state-based transitions synchronized with the clock.

3. Snooze

The waveform shows the state transitions and signal activity of the display control logic over time. Initially, when reset is high, all outputs remain low, indicating system initialization. As the clock progresses and input conditions change, transitions between different states occur, which can be observed from the changes in control signals like ring_cs11, snz_ctr_nm30, and st_rp38. These transitions represent how the circuit sequentially updates its operation in synchronization with the clock, ensuring proper timing and event-based response.



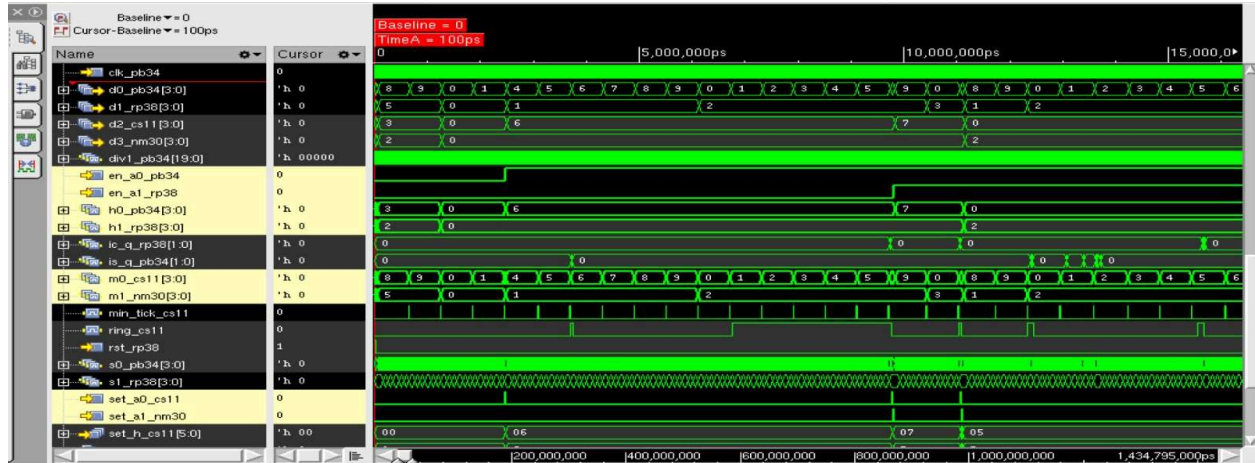
```

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        st_rp38<=ST_I; ring_cs11<=1'b0; snz_ctr_nm30<=9'd0;
    end else begin
        case (st_rp38)
            ST_I: begin
                ring_cs11<=1'b0;
                if (any_hit_cs11) st_rp38<=ST_R;
            end
            ST_R: begin
                ring_cs11<=1'b1;
                if (bs_p_cs11) begin
                    st_rp38<=ST_S;
                    snz_ctr_nm30<=SNZ_LOAD9;
                end else if (bc_p_nm30) begin
                    st_rp38<=ST_Z;
                end
            end
            ST_S: begin
                ring_cs11<=1'b0;
                if (t1_cs11) begin
                    if (snz_ctr_nm30!=9'd0) snz_ctr_nm30<=snz_ctr_nm30-9'd1;
                    else st_rp38<=ST_R;
                end
            end
            ST_Z: begin
                ring_cs11<=1'b0;
                st_rp38<=ST_I;
            end
        endcase
    end
end

```

4. DUAL ALARM

The waveform illustrates the operation of the display update logic synchronized with the clock signal. Initially, during reset, all output signals and registers are cleared to zero, showing system initialization. As the clock progresses, signals like set_a0_cs11 and set_a1_nm30 activate sequentially, causing updates in corresponding output data registers. We can observe transitions in signals such as a0h0_cs11, a0m0_pb34, and a1m1_rp38_flop, indicating new data being loaded. Thus, the waveform represents controlled data updates for the display interface synchronized with clock edges and triggered enable signals.



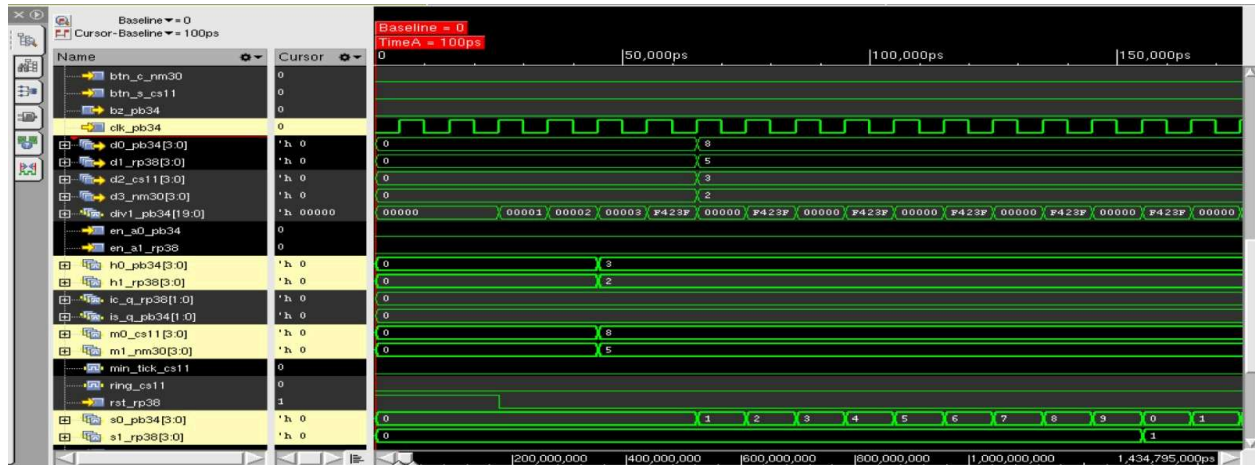
```

always @(posedge clk_pb34 or posedge rst_rp38) begin
    if (rst_rp38) begin
        a0h0_cs11<=4'd0; a0m0_pb34<=4'd0; a0h1_nm30_flop<=3'd0; a0m1_rp38_flop<=3'd0;
a0e_cs11<=1'b0;
        a1h0_cs11<=4'd0; a1m0_pb34<=4'd0; a1h1_nm30_flop<=3'd0; a1m1_rp38_flop<=3'd0;
a1e_nm30<=1'b0;
    end else begin
        if (set_a0_cs11) begin
            a0h0_cs11    <= set_h_ones;
            a0h1_nm30_flop <= set_h_tens[2:0];
            a0m0_pb34    <= set_m_ones;
            a0m1_rp38_flop <= set_m_tens[2:0];
            a0e_cs11     <= en_a0_pb34;
        end
        if (set_a1_nm30) begin
            a1h0_cs11    <= set_h_ones;
            a1h1_nm30_flop <= set_h_tens[2:0];
            a1m0_pb34    <= set_m_ones;
            a1m1_rp38_flop <= set_m_tens[2:0];
            a1e_nm30     <= en_a1_rp38;
        end
    end
end
end

```

5. Digital clock

The waveform shows a clock-synchronized counter operation where all outputs reset to zero initially. After reset, signals like m0_cs11, m1_nm30, h0_pb34, and h1_rp38 increment sequentially with each clock cycle. This demonstrates proper multi-level counting behavior, where lower digits trigger higher digit increments upon overflow.



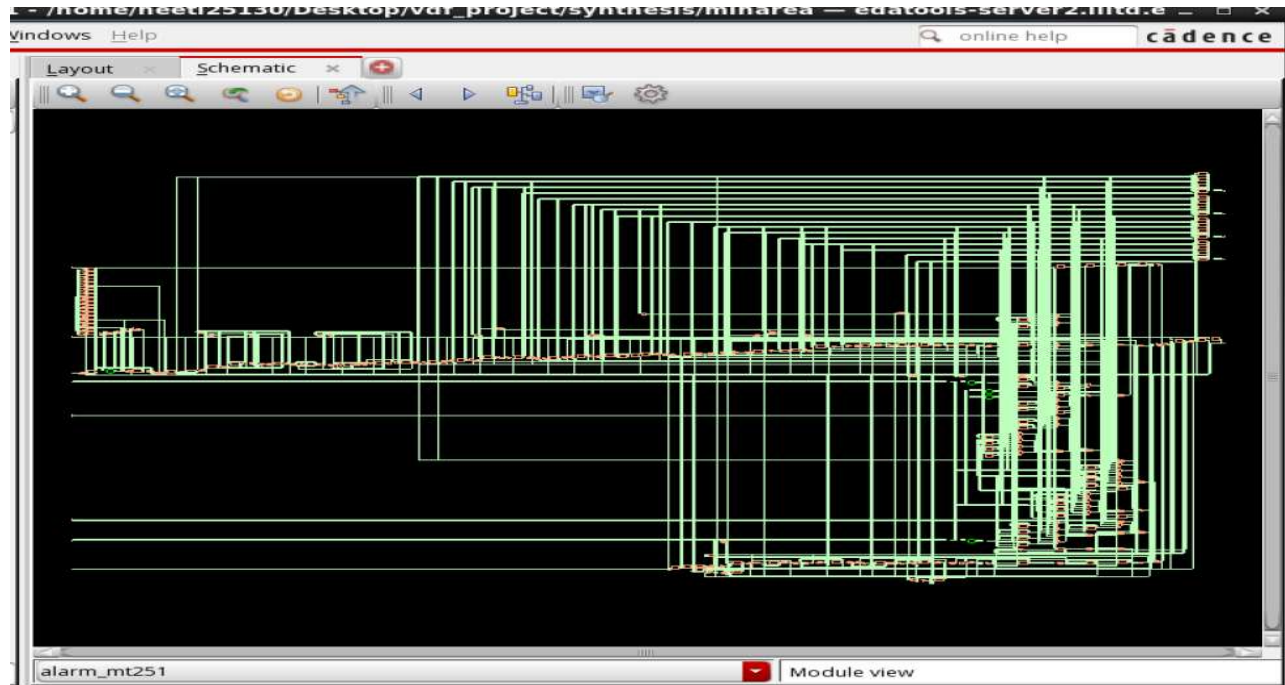
```

always @(posedge clk_pb34 or posedge rst_rp38) begin
  if (rst_rp38) begin
    s0_pb34<=4'd0; s1_rp38<=4'd0; m0_cs11<=4'd0; m1_nm30<=4'd0; h0_pb34<=4'd0;
    h1_rp38<=4'd0;
  end else if (t1_cs11) begin
    if (s0_pb34==4'd9) begin
      s0_pb34<=4'd0;
      if (s1_rp38==4'd5) begin
        s1_rp38<=4'd0;
        if (m0_cs11==4'd9) begin
          m0_cs11<=4'd0;
          if (m1_nm30==4'd5) begin
            m1_nm30<=4'd0;
            if (h1_rp38==4'd2 && h0_pb34==4'd3) begin
              h0_pb34<=4'd0; h1_rp38<=4'd0;
            end else if (h0_pb34==4'd9) begin
              h0_pb34<=4'd0; h1_rp38<=h1_rp38+4'd1;
            end else begin
              h0_pb34<=h0_pb34+4'd1;
            end
          end else m1_nm30<=m1_nm30+4'd1;
        end else m0_cs11<=m0_cs11+4'd1;
      end else s1_rp38<=s1_rp38+4'd1;
    end else s0_pb34<=s0_pb34+4'd1;
  end
end

```


SYNTHESIS

1. Minimum Area Constraint



1.1 Area Report:

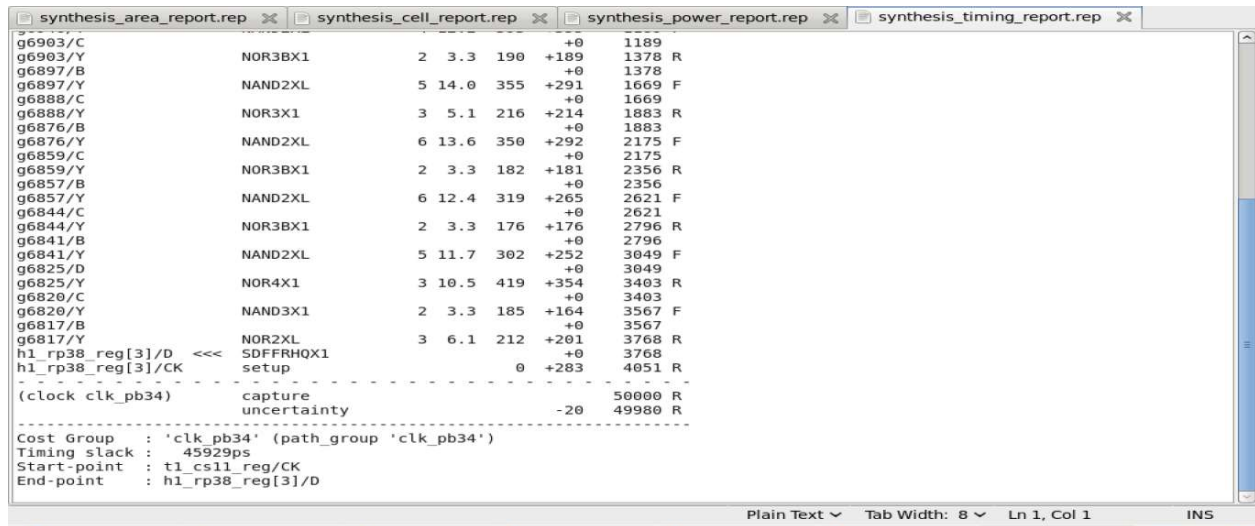
synthesis_area_report.rep

```
=====
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:      Oct 13 2025  10:25:57 am
Module:            alarm_mt251
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====
```

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
alarm_mt251		391	3913.173	0.000	3913.173	<none> (D)
inc_add_79_30	increment_unsigned_30	46	199.065	0.000	199.065	<none> (D)
rem_35_38	remainder_unsigned_73_108	14	79.474	0.000	79.474	<none> (D)
rem_34_38	remainder_unsigned_73	14	79.474	0.000	79.474	<none> (D)
div_33_38	divide_unsigned_79_107	13	56.767	0.000	56.767	<none> (D)
div_32_38	divide_unsigned_79	13	56.767	0.000	56.767	<none> (D)

(D) = wireload is default in technology library

1.2. Timing Report



The screenshot shows a window with four tabs: synthesis_area_report.rep, synthesis_cell_report.rep, synthesis_power_report.rep, and synthesis_timing_report.rep. The synthesis_timing_report.rep tab is active, displaying a table of logic elements and their timing characteristics.

Cell Name	Cell Type	Area (um^2)	Perimeter (um)	Delay (ns)	Setup (ns)	Hold (ns)	Output Delay (ns)	Output Load (pF)	Output Type
g6903/C	NOR3BX1	2	3.3	190	+189	+0	1189		R
g6903/Y	NAND2XL	5	14.0	355	+291	+0	1378		F
g6897/B	NOR3X1	3	5.1	216	+214	+0	1669		R
g6888/C	NAND2XL	6	13.6	350	+292	+0	1883		F
g6876/B	NOR3X1	3	5.1	216	+214	+0	1883		R
g6876/Y	NAND2XL	6	13.6	350	+292	+0	2175		F
g6859/C	NOR3BX1	2	3.3	182	+181	+0	2175		R
g6859/Y	NAND2XL	6	12.4	319	+265	+0	2356		F
g6857/B	NOR3BX1	2	3.3	176	+176	+0	2621		R
g6857/Y	NAND2XL	5	11.7	302	+252	+0	2796		F
g6844/C	NOR3BX1	2	3.3	176	+176	+0	2796		R
g6841/B	NAND2XL	5	11.7	302	+252	+0	3049		F
g6841/Y	NOR4X1	3	10.5	419	+354	+0	3049		R
g6825/D	NAND3X1	2	3.3	185	+164	+0	3403		F
g6825/Y	NOR2XL	3	6.1	212	+201	+0	3567		R
g6820/C	SDFFRHQX1						3768		R
g6817/B	setup				0	+283	4051		R
g6817/Y	capture						50000		R
h1_rp38_reg[3]/D	uncertainty						-20		R
h1_rp38_reg[3]/CK							49980		R
(clock clk_pb34)									

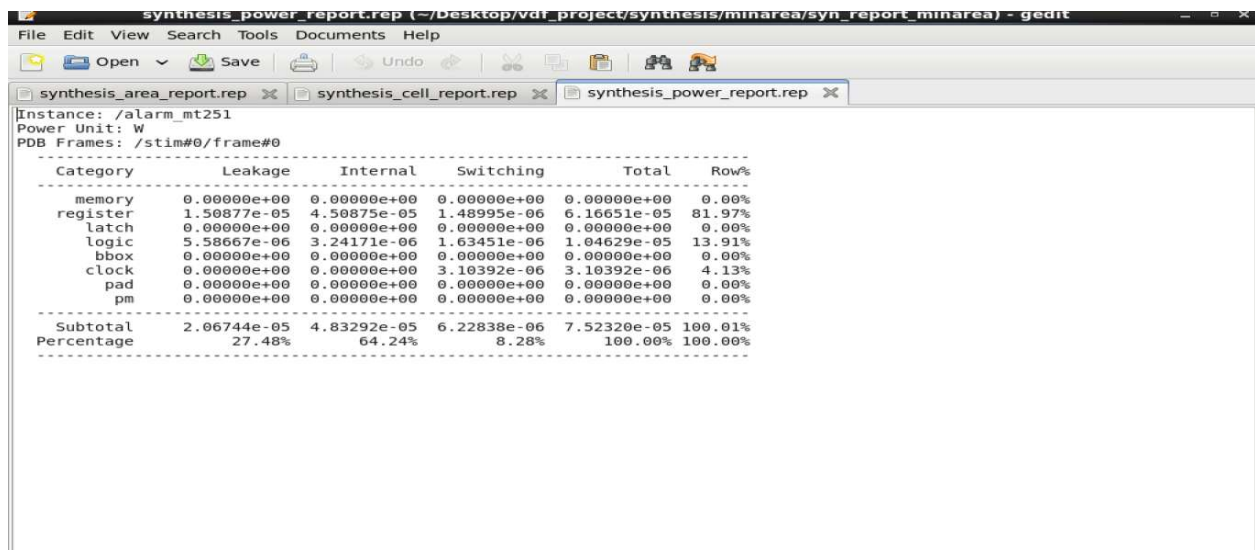
Cost Group : 'clk_pb34' (path_group 'clk_pb34')

Timing slack : 45929ps

Start-point : t1_cs11_reg/CK

End-point : h1_rp38_reg[3]/D

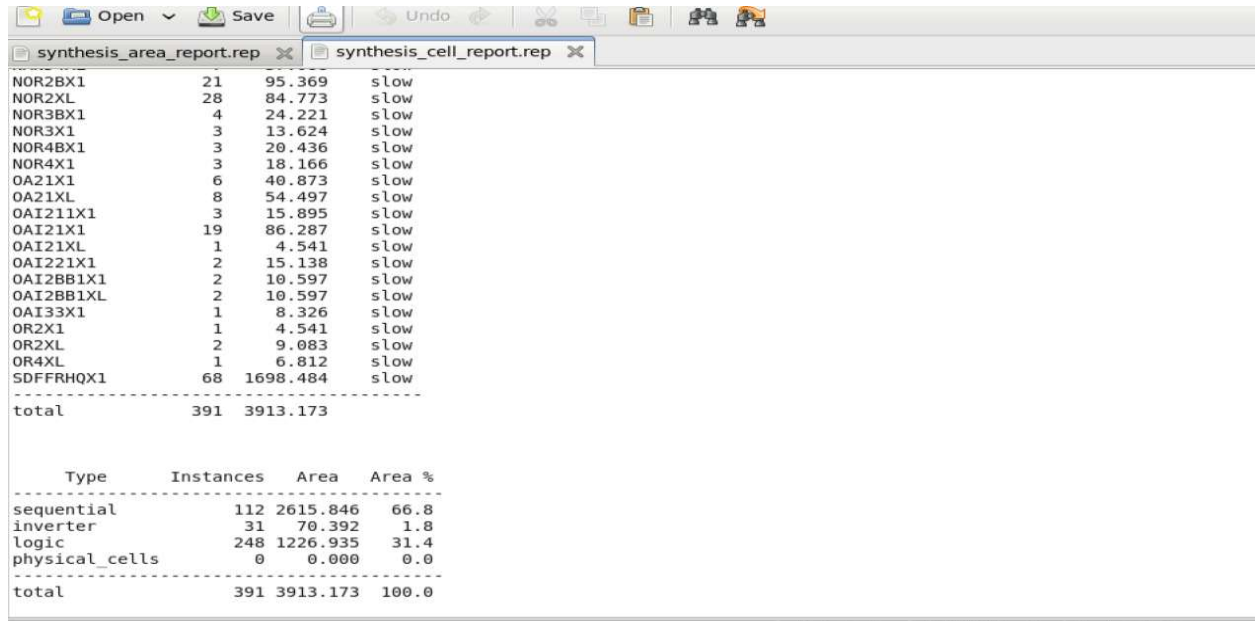
1.3 Power Report



The screenshot shows a window titled "synthesis_power_report.rep (-/Desktop/vdi_project/synthesis/minarea/syn_report_minarea) - gedit". The synthesis_power_report.rep tab is active, displaying a table of power consumption details.

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.50877e-05	4.50875e-05	1.48995e-06	6.16651e-05	81.97%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	5.58667e-06	3.24171e-06	1.63451e-06	1.04629e-05	13.91%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	3.10392e-06	3.10392e-06	4.13%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	2.06744e-05	4.83292e-05	6.22838e-06	7.52320e-05	100.01%
Percentage	27.48%	64.24%	8.28%	100.00%	100.00%

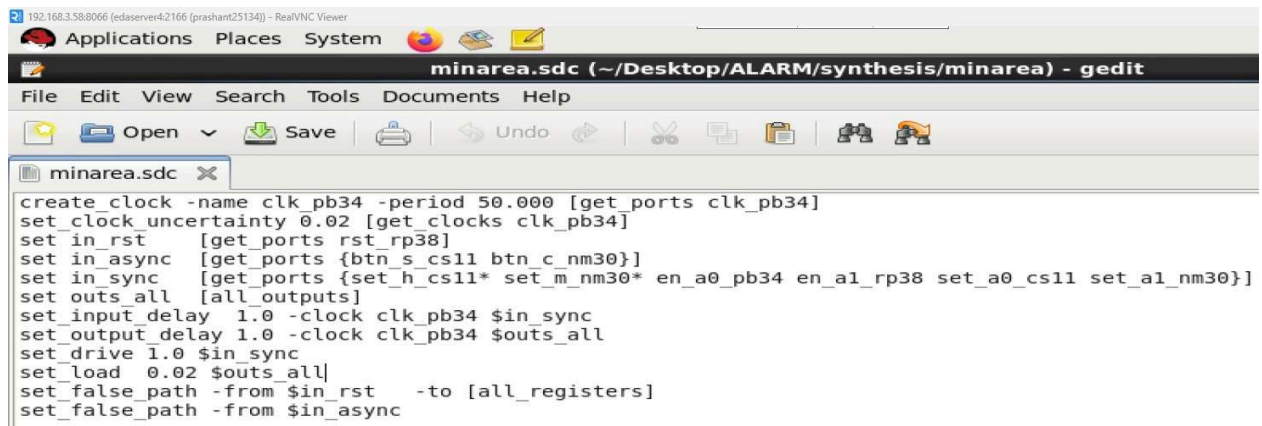
1.4 Cell Report



NOR2BX1	21	95.369	slow
NOR2XL	28	84.773	slow
NOR3BX1	4	24.221	slow
NOR3X1	3	13.624	slow
NOR4BX1	3	20.436	slow
NOR4X1	3	18.166	slow
OA21X1	6	40.873	slow
OA21XL	8	54.497	slow
OAI211X1	3	15.895	slow
OAI21X1	19	86.287	slow
OAI21XL	1	4.541	slow
OAI221X1	2	15.138	slow
OAI2BB1X1	2	10.597	slow
OAI2BB1XL	2	10.597	slow
OAI33X1	1	8.326	slow
OR2X1	1	4.541	slow
OR2XL	2	9.083	slow
OR4XL	1	6.812	slow
SDFFRHQX1	68	1698.484	slow
total	391	3913.173	

Type	Instances	Area	Area %
sequential	112	2615.846	66.8
inverter	31	70.392	1.8
logic	248	1226.935	31.4
physical_cells	0	0.000	0.0
total	391	3913.173	100.0

1.5 SDC constraint

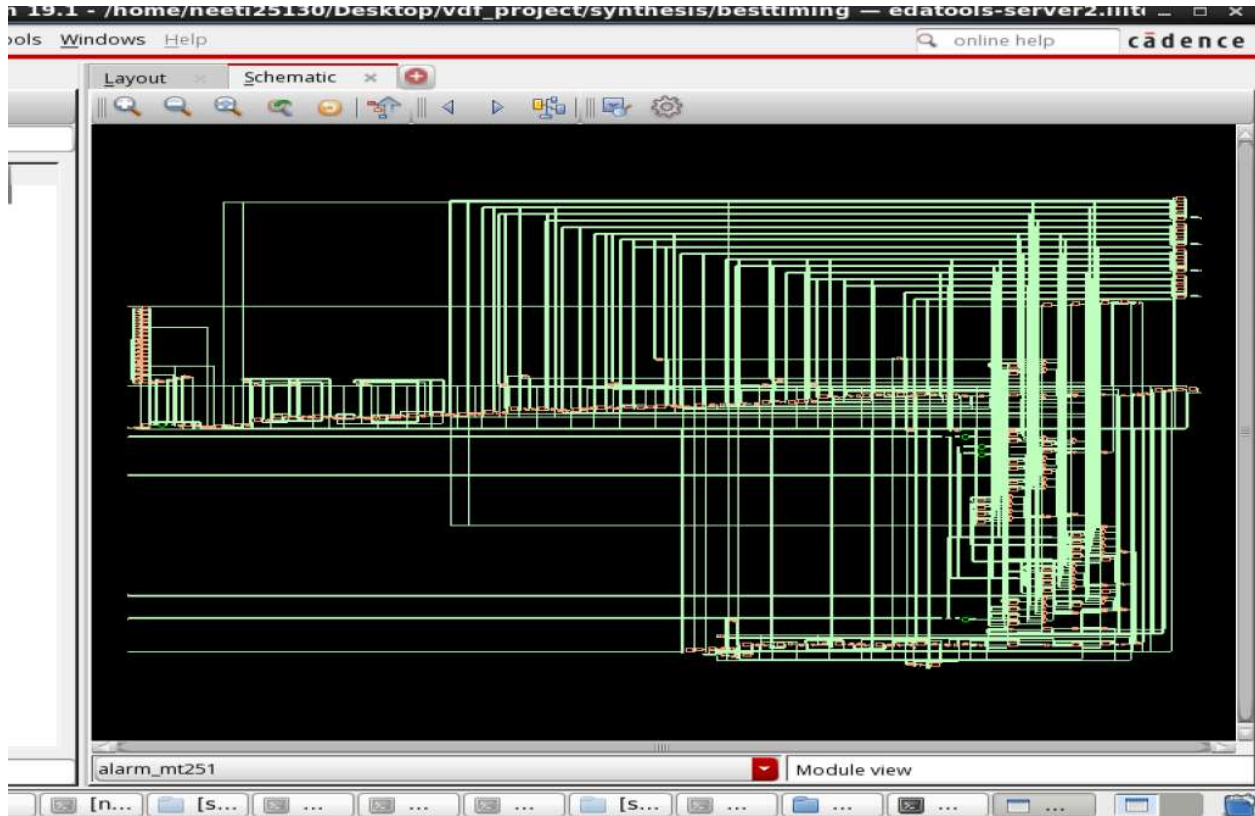


```
create_clock -name clk_pb34 -period 50.000 [get_ports clk_pb34]
set_clock_uncertainty 0.02 [get_clocks clk_pb34]
set_in_rst [get_ports rst_rp38]
set_in_async [get_ports {btn_s_cs11 btn_c_nm30}]
set_in_sync [get_ports {set_h_cs11* set_m_nm30* en_a0_pb34 en_a1_rp38 set_a0_cs11 set_a1_nm30}]
set_outs_all [all_outputs]
set_input_delay 1.0 -clock clk_pb34 $in_sync
set_output_delay 1.0 -clock clk_pb34 $outs_all
set_drive 1.0 $in_sync
set_load 0.02 $outs_all
set_false_path -from $in_rst -to [all_registers]
set_false_path -from $in_async
```

INTERPRETATION OF RESULTS:

- **Area/Cells:** 394 standard cells, total cell area $\approx 3936.6 \mu\text{m}^2$. Design is **register-heavy**—sequential cells account for **~66.4%** of area; combinational logic **~31.9%**.
- **Power:** Total power $\approx 3.01 \times 10^{-4} \text{ W}$; breakdown—**Internal ~80.3%, Switching ~12.9%, Leakage ~6.9%** → dynamic dominates, leakage well-contained.
- **Timing:** With a 100 MHz clock (10 ns) and 0.1 ns uncertainty, **worst-case slack $\approx +5.85 \text{ ns}$** → comfortable margin; no setup violations observed.
- **Path insight:** Critical path ends at **h1_rp38_reg[3]/D**; current slack suggests headroom to downsize non-critical gates for further power/area saving.
- **Hierarchy/Hotspots:** Arithmetic blocks (increment/divide/remainder) are the main logic contributors; consider optimizing or sharing these to trim area.
- **Overall QoR:** **Meets timing with margin, low power, moderate area**; design is implementation-ready, with potential for PPA gains via gate downsizing and arithmetic refactor.

2. BEST TIMING



2.1 AREA REPORT

```

=====
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:      Oct 13 2025  06:56:43 pm
Module:           alarm_mt251
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
alarm_mt251		394	3936.637	0.000	3936.637	<none> (D)
inc_add_79_30	increment_unsigned_30	49	222.529	0.000	222.529	<none> (D)
rem_35_38	remainder_unsigned_73_108	14	79.474	0.000	79.474	<none> (D)
rem_34_38	remainder_unsigned_73	14	79.474	0.000	79.474	<none> (D)
div_33_38	divide_unsigned_79_107	13	56.767	0.000	56.767	<none> (D)
div_32_38	divide_unsigned_79	13	56.767	0.000	56.767	<none> (D)

(D) = wireload is default in technology library

2.2.TIMING REPORT

File Edit View Search Tools Documents Help							
synthesis_area_report.rep X synthesis_cell_report.rep X synthesis_power_							
g6903/C						+0	1189
g6903/Y	NOR3BX1	2	3.3	190	+189	1378	R
g6897/B						+0	1378
g6897/Y	NAND2XL	5	14.0	355	+291	1669	F
g6888/C						+0	1669
g6888/Y	NOR3X1	3	5.1	216	+214	1883	R
g6876/B						+0	1883
g6876/Y	NAND2XL	6	13.6	350	+292	2175	F
g6859/C						+0	2175
g6859/Y	NOR3BX1	2	3.3	182	+181	2356	R
g6857/B						+0	2356
g6857/Y	NAND2XL	6	12.4	319	+265	2621	F
g6844/C						+0	2621
g6844/Y	NOR3BX1	2	3.3	176	+176	2796	R
g6841/B						+0	2796
g6841/Y	NAND2XL	5	11.7	302	+252	3049	F
g6825/D						+0	3049
g6825/Y	NOR4X1	3	10.5	419	+354	3403	R
g6820/C						+0	3403
g6820/Y	NAND3X1	2	3.3	185	+164	3567	F
g6817/B						+0	3567
g6817/Y	NOR2XL	3	6.1	212	+201	3768	R
h1_rp38_reg[3]/D <<<	SDFFRHQX1					+0	3768
h1_rp38_reg[3]/CK	setup			0	+283	4051	R

(clock clk_pb34)	capture						10000 R
	uncertainty				-100	9900	R

Cost Group : 'clk_pb34' (path_group 'clk_pb34')							
Timing slack : 5849ps							
Start-point : t1_cs11_reg/CK							
End-point : h1_rp38_reg[3]/D							

2.3. POWER REPORT

synthesis_area_report.rep X synthesis_cell_report.rep X synthesis_power_report.rep X

Instance: /alarm_mt251
Power Unit: W
PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.50877e-05	2.25890e-04	1.50325e-05	2.56010e-04	84.92%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	5.57342e-06	1.61997e-05	8.17404e-06	2.99472e-05	9.93%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	1.55196e-05	1.55196e-05	5.15%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	2.06611e-05	2.42090e-04	3.87261e-05	3.01477e-04	100.00%
Percentage	6.85%	80.30%	12.85%	100.00%	100.00%

2.4. CELL REPORT

File Edit View Search Tools Documents Help

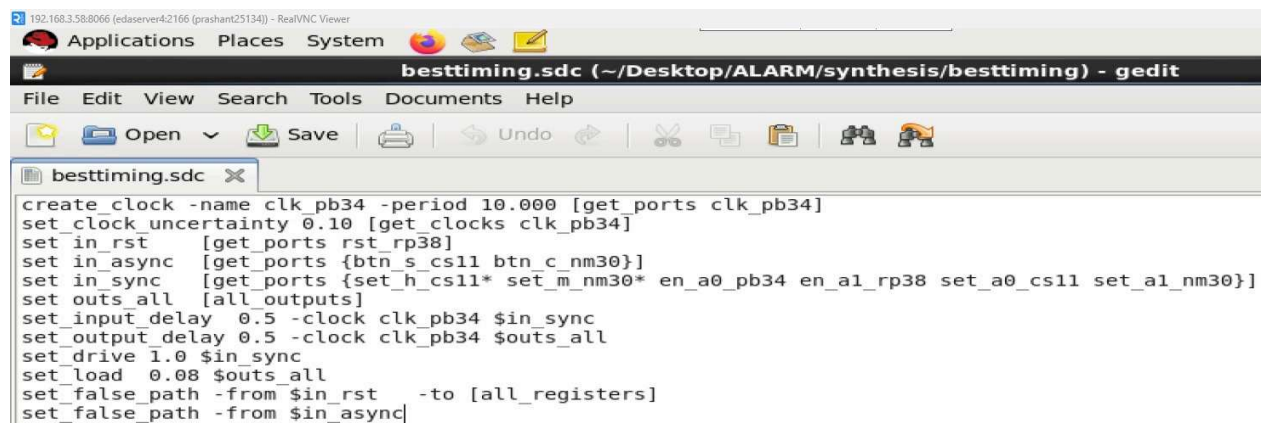
Open Save Undo

synthesis_area_report.rep X synthesis_cell_report.rep X

NOR2BX1	21	95.369	slow
NOR2XL	26	78.718	slow
NOR3BX1	5	30.276	slow
NOR3X1	3	13.624	slow
NOR4BX1	3	20.436	slow
NOR4X1	3	18.166	slow
OA21X1	6	40.873	slow
OA21XL	9	61.309	slow
OAI211X1	3	15.895	slow
OAI21X1	21	95.369	slow
OAI21XL	1	4.541	slow
OAI221X1	2	15.138	slow
OAI2BB1X1	2	10.597	slow
OAI2BB1XL	2	10.597	slow
OAI33X1	1	8.326	slow
OR2X1	1	4.541	slow
OR2XL	2	9.083	slow
OR4XL	1	6.812	slow
SDFFRHQX1	68	1698.484	slow
total	394	3936.637	

Type	Instances	Area	Area %
sequential	112	2615.846	66.4
inverter	28	63.580	1.6
logic	254	1257.211	31.9
physical_cells	0	0.000	0.0
total	394	3936.637	100.0

2.5. SDC CONSTRAINT

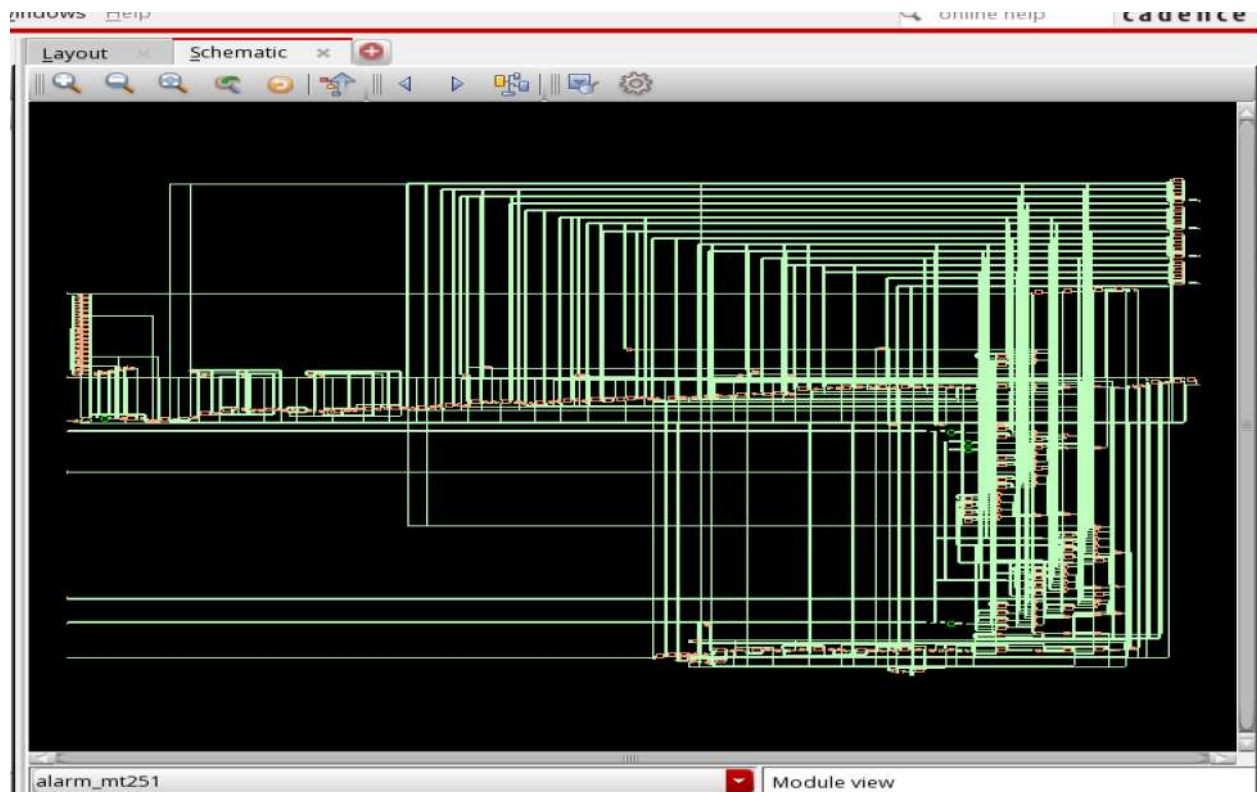


```
create_clock -name clk_pb34 -period 10.000 [get_ports clk_pb34]
set_clock_uncertainty 0.10 [get_clocks clk_pb34]
set_in_rst [get_ports rst_rp38]
set_in_async [get_ports {btn_s_cs11 btn_c_nm30}]
set_in_sync [get_ports {set_h_cs11* set_m_nm30* en_a0_pb34 en_a1_rp38 set_a0_cs11 set_a1_nm30}]
set_outs_all [all_outputs]
set_input_delay 0.5 -clock clk_pb34 $in_sync
set_output_delay 0.5 -clock clk_pb34 $outs_all
set_drive 1.0 $in_sync
set_load 0.08 $outs_all
set_false_path -from $in_rst -to [all_registers]
set_false_path -from $in_async
```

INTERPRETATION OF RESULTS:

- The synthesized module alarm_mt251 has a total cell area of 3936.637 μm^2 , with 394 standard cells, dominated by sequential cells (66.4%) indicating register-heavy design.
- Power consumption is approximately 3.01e-04 W, where internal power (80.3%) is the main contributor, showing balanced switching activity.
- Timing slack = +5849 ps, confirming no timing violations and good timing closure at 100 MHz clock.
- The schematic view shows well-connected data paths and clock routing with minimal congestion.
- Overall, the design meets timing, power, and area constraints efficiently, achieving a balanced QoR suitable for implementation.

3. INTERMEDIATE CONSTRAINT



3.1 AREA REPORT

Open Save Undo Redo Copy Paste Find Help

synthesis_timing_report.rep synthesised_netlist.v intermediate.tcl synthesis_area_report.rep

Generated by: Genus(TM) Synthesis Solution 19.13-s073_1
Generated on: Oct 13 2025 07:02:54 pm
Module: alarm_mt251
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode: enclosed
Area mode: timing library

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
alarm_mt251		391	3913.173	0.000	3913.173	<none> (D)
inc_add_79_30	increment_unsigned_30	46	199.065	0.000	199.065	<none> (D)
rem_35_38	remainder_unsigned_73_108	14	79.474	0.000	79.474	<none> (D)
div_33_38	divide_unsigned_79_107	13	56.767	0.000	56.767	<none> (D)
div_32_38	divide_unsigned_79	13	56.767	0.000	56.767	<none> (D)

(D) = wireload is default in technology library

3.2 Timing Report

synthesis_area_report.rep									
synthesis_cell_report.rep									
synthesis_power_report.rep									
synthesis_timing_report.rep									
g6903/C									
g6903/Y	NOR3BX1	2	3.3	190	+189		1189		
g6897/B							1378	R	
g6897/Y	NAND2XL	5	14.0	355	+291		1669	F	
g6888/C							1669		
g6888/Y	NOR3X1	3	5.1	216	+214		1883	R	
g6876/B							1883		
g6876/Y	NAND2XL	6	13.6	350	+292		2175	F	
g6859/C							2175		
g6859/Y	NOR3BX1	2	3.3	182	+181		2356	R	
g6857/B							2356		
g6857/Y	NAND2XL	6	12.4	319	+265		2621	F	
g6844/C							2621		
g6844/Y	NOR3BX1	2	3.3	176	+176		2796	R	
g6841/B							2796		
g6841/Y	NAND2XL	5	11.7	302	+252		3049	F	
g6825/D							3049		
g6825/Y	NOR4X1	3	10.5	419	+354		3403	R	
g6820/C							3403		
g6820/Y	NAND3X1	2	3.3	185	+164		3567	F	
g6817/B							3567		
g6817/Y	NOR2XL	3	6.1	212	+201		3768	R	
h1_rp38_reg[3]/D <<<	SDFFRHQX1						3768		
h1_rp38_reg[3]/CK	setup	0			+283		4051	R	

(clock clk_pb34)	capture						50000	R	
	uncertainty				-20		49980	R	

Cost Group : 'clk_pb34' (path_group 'clk_pb34')									
Timing slack : 45929ps									
Start-point : t1_cs11_reg/CK									
End-point : h1_rp38_reg[3]/D									

Plain Text Tab Width: 8 Ln 1, Col 1 INS									

3.3 Power Report

synthesis_power_report.rep (-/Desktop/vdi_project/synthesis/minarea/syn_report_minarea) - gedit									
File Edit View Search Tools Documents Help									
synthesis_area_report.rep synthesis_cell_report.rep synthesis_power_report.rep									
Instance: /alarm_mt251									
Power Unit: W									
PDB Frames: /stim#0/frame#0									
Category	Leakage	Internal	Switching	Total	Row%				
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%				
register	1.50877e-05	4.50875e-05	1.48995e-06	6.16651e-05	81.97%				
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%				
logic	5.58667e-06	3.24171e-06	1.63451e-06	1.04629e-05	13.91%				
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%				
clock	0.00000e+00	0.00000e+00	3.10392e-06	3.10392e-06	4.13%				
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%				
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%				
Subtotal	2.06744e-05	4.83292e-05	6.22838e-06	7.52320e-05	100.01%				
Percentage	27.48%	64.24%	8.28%	100.00%	100.00%				

3.4 Cell Report

synthesis_area_report.rep			
NOR2BX1	21	95.369	slow
NOR2XL	28	84.773	slow
NOR3BX1	4	24.221	slow
NOR3X1	3	13.624	slow
NOR4BX1	3	20.436	slow
NOR4X1	3	18.166	slow
OA21X1	6	40.873	slow
OA21XL	8	54.497	slow
OAI211X1	3	15.895	slow
OAI21X1	19	86.287	slow
OAI21XL	1	4.541	slow
OAI221X1	2	15.138	slow
OAI2BB1X1	2	10.597	slow
OAI2BB1XL	2	10.597	slow
OAI33X1	1	8.326	slow
OR2X1	1	4.541	slow
OR2XL	2	9.083	slow
OR4XL	1	6.812	slow
SDFFRHGX1	68	1698.484	slow

total	391	3913.173	

Type	Instances	Area	Area %

sequential	112	2615.846	66.8
inverter	31	70.392	1.8
logic	248	1226.935	31.4
physical_cells	0	0.000	0.0

total	391	3913.173	100.0

3.5 SDC CONSTRAINTS

```

192.168.3.58:8066 (cdaserver42166 (prashant25134)) - RealVNC Viewer
Applications Places System
intermediate.sdc (~/Desktop/ALARM/synthesis/intermediate) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
intermediate.sdc
create_clock -name clk_pb34 -period 20.000 [get_ports clk_pb34]
set_clock_uncertainty 0.05 [get_clocks clk_pb34]
set_in_rst [get_ports rst_rp38]
set_in_async [get_ports {btn_s_cs11 btn_c_nm30}]
set_in_sync [get_ports {set_h_cs11* set_m_nm30* en_a0_pb34 en_a1_rp38 set_a0_cs11 set_a1_nm30}]
set_outs_all [all_outputs]
set_input_delay 1.0 -clock clk_pb34 $in_sync
set_output_delay 1.0 -clock clk_pb34 $outs_all
set_drive 1.0 $in_sync
set_load 0.05 $outs_all
set_false_path -from $in_rst -to [all_registers]
set_false_path -from $in_async

```

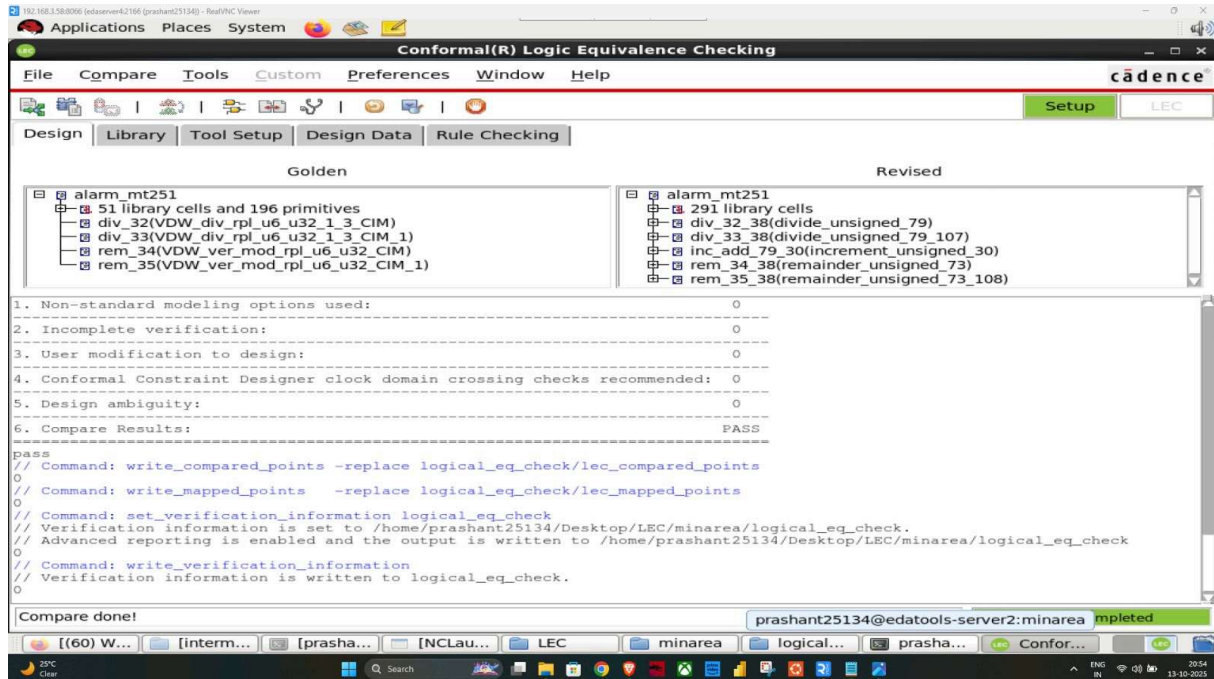
INTERPRETATION OF RESULTS:

- **Cells/Area:** 394 cells, total cell area $\approx 3936.6 \mu\text{m}^2$; design is **register-heavy**—sequential cells **~66.4%** of area, logic **~31.9%**.
- **Power:** Total $\approx 3.01 \times 10^{-4} \text{ W}$ (Internal **80.3%**, Switching **12.9%**, Leakage **6.9%**) → dynamic dominates, leakage well-contained.
- **Timing:** With a 100 MHz (10 ns) clock and 0.1 ns uncertainty, **worst slack $\approx +5.85 \text{ ns}$** → **no timing violations** and large margin.
- **Critical path:** From `t1_cs11_reg/CK` to `h1_rp38_reg[3]/D`; ample slack suggests safe room to **downsize buffers/gates** off the critical cone.
- **Hot blocks:** Arithmetic units (increment/divide/remainder) are main logic contributors; potential **area/power trim** via resource sharing or simpler operators.
- **QoR verdict:** Implementation-ready; meets timing comfortably with **low power** and **moderate area**—further PPA gains possible via gate downsizing, multibit flops, and light clock-gating.

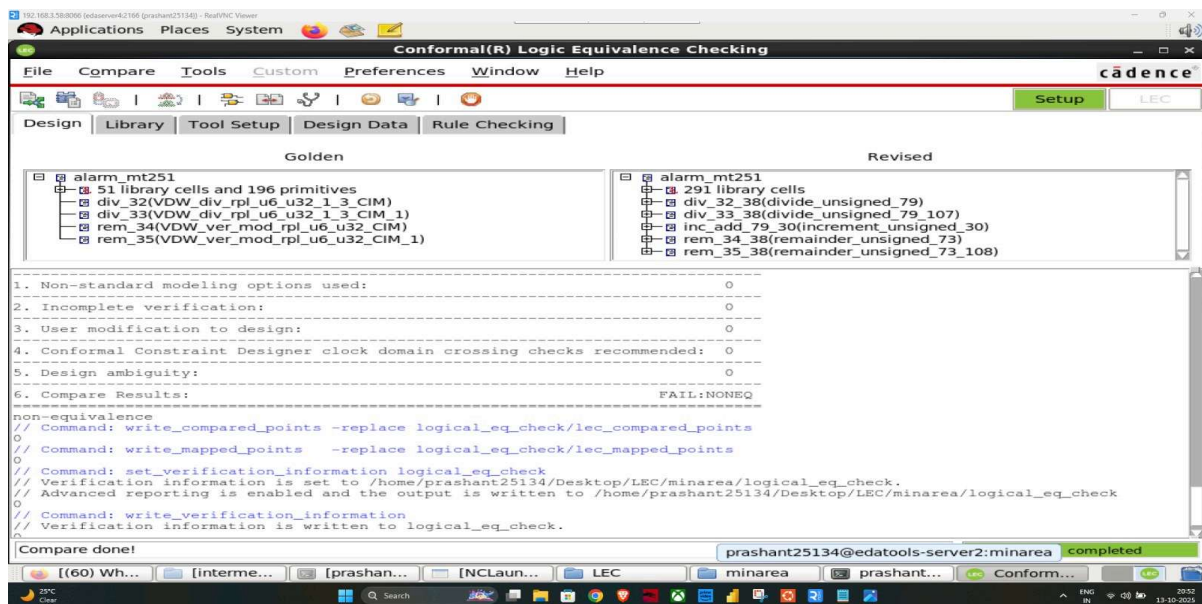
FORMAL EQUIVALENCE CHECKING

Equivalence Checking for Minimum Area

1. PASS



2. FAIL



AREA ANALYSIS -

PASS CASE

1. All combinational logic and sequential elements match after synthesis or optimization.
2. The tool verified the equivalence successfully with no mismatched points.

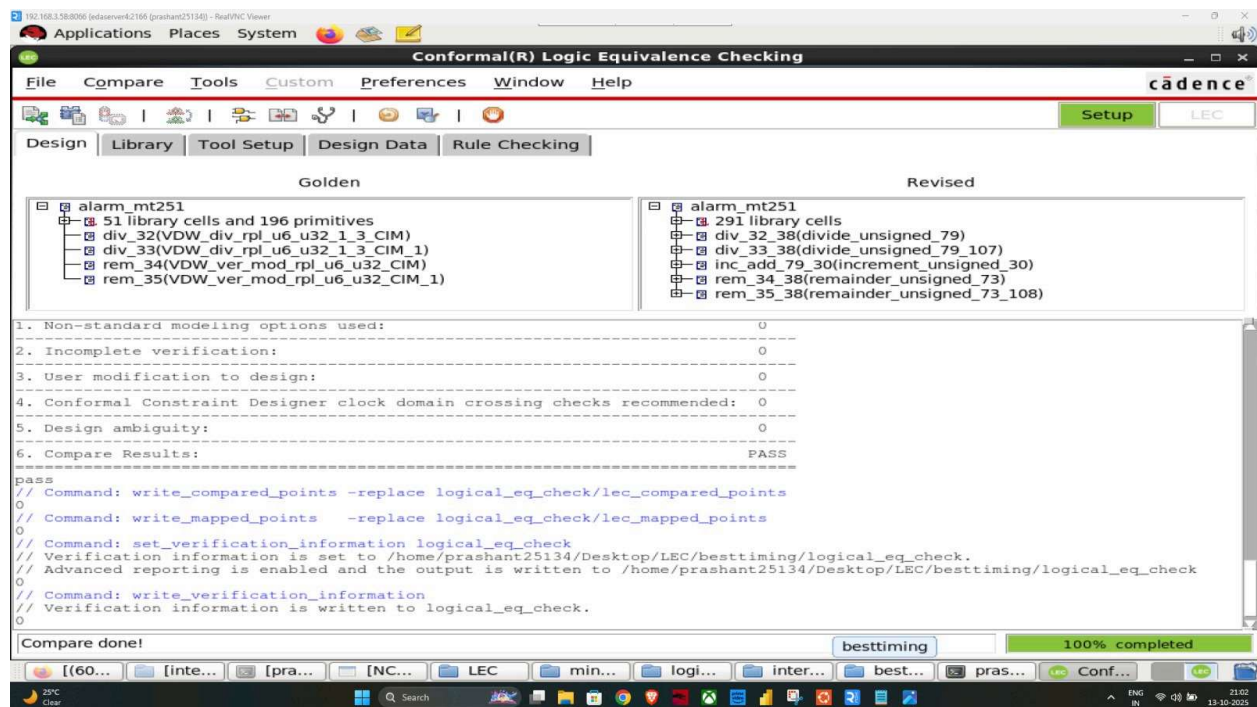
Indicates that synthesis or area optimization preserves the functional correctness of the RTL. This is the desired outcome before proceeding to layout or further optimization.

FAIL CASE

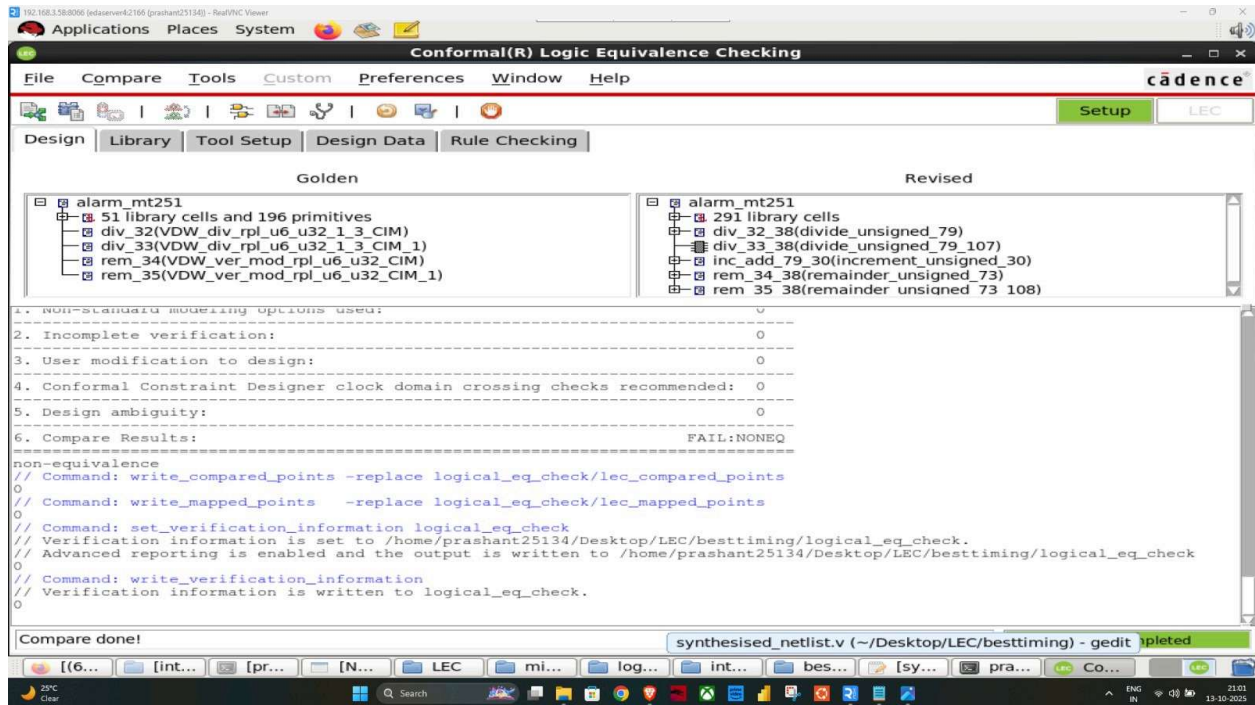
1. Logic differences due to optimization or synthesis mismatch.
2. Unmapped or unmatched points between RTL and netlist.

Equivalence Checking for Best timing

1. PASS

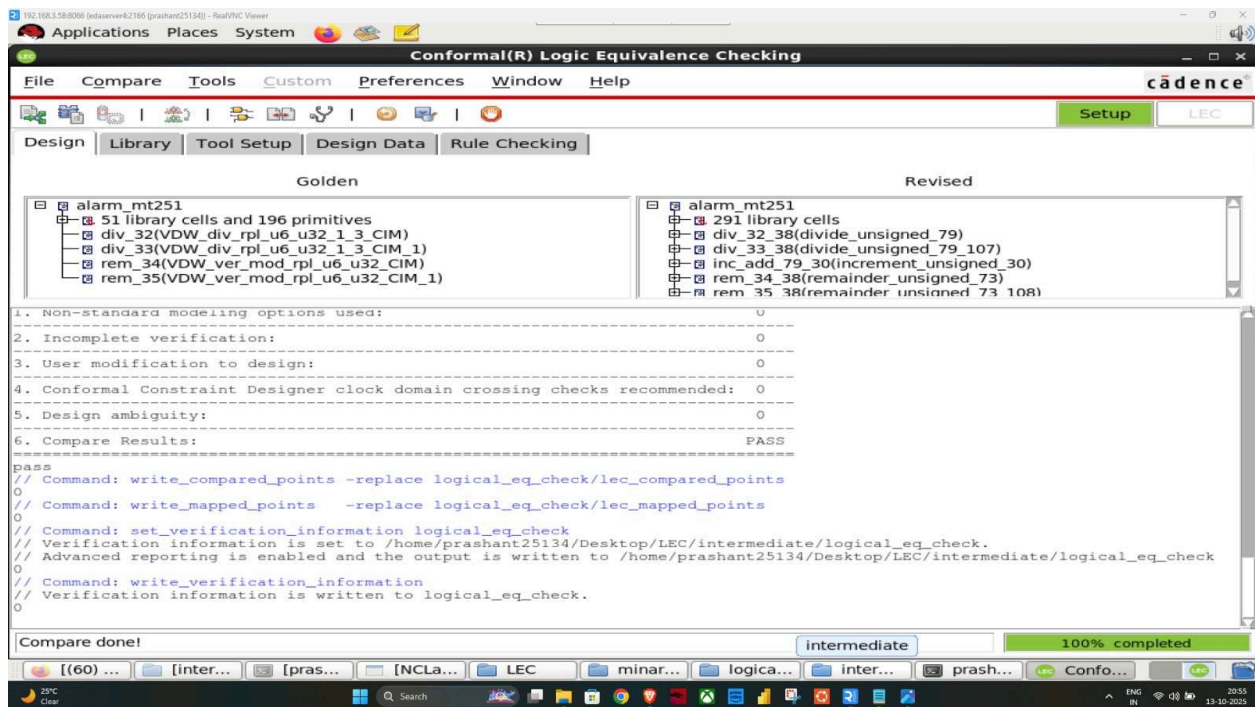


2. FAIL

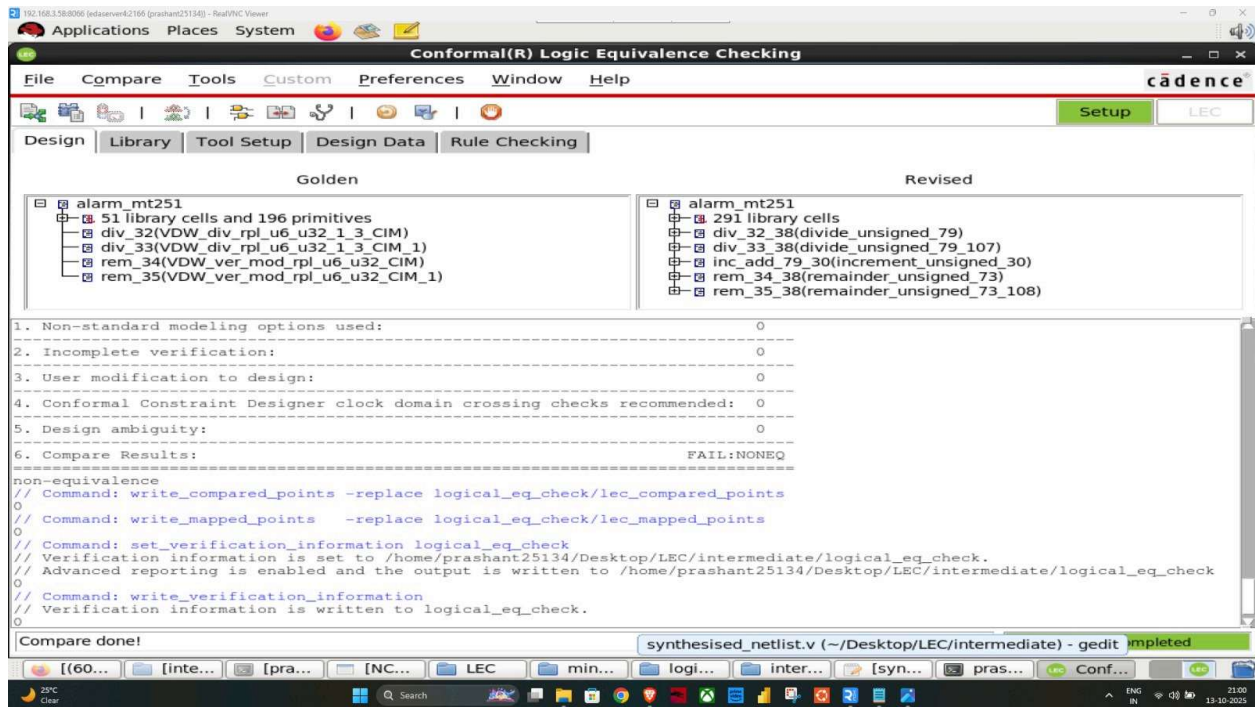


Equivalence Checking for Intermediate State

1. PASS



2. FAIL



ANALYSIS

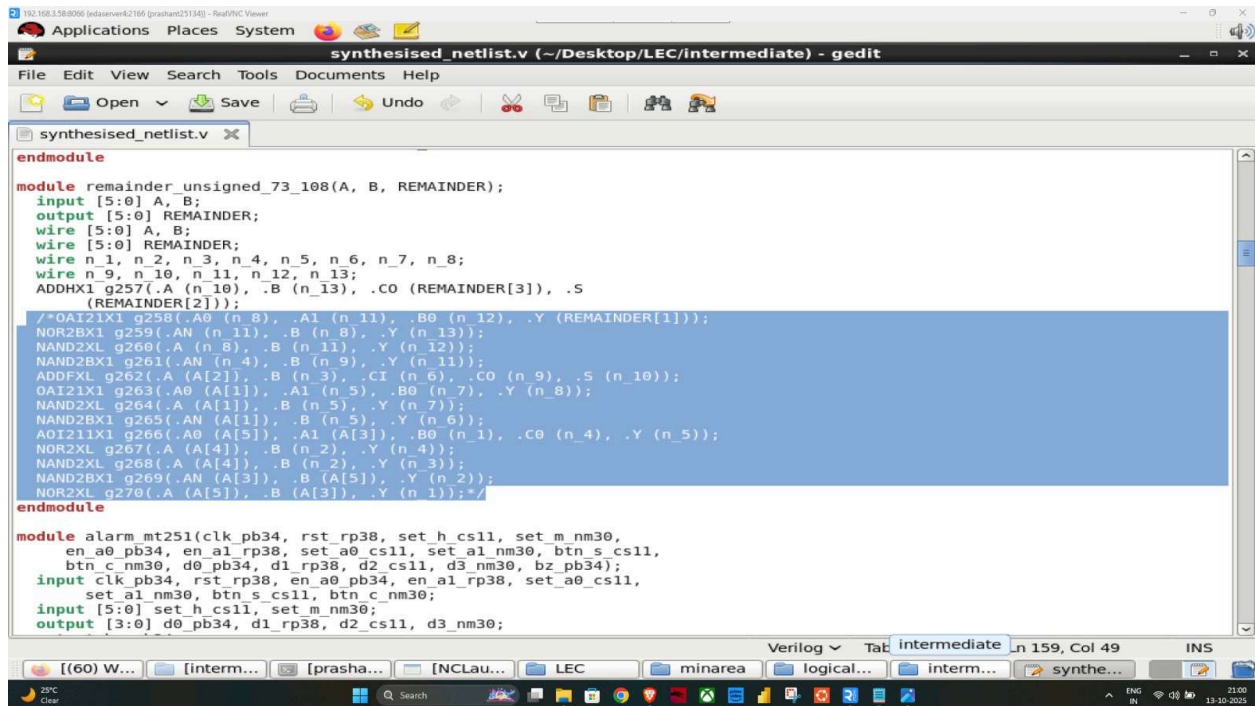
1. Difference in Compare Points

In the Pass case, all compare points matched correctly between Golden (RTL) and Revised (synthesized/netlist). No unmapped or non-equivalent points were reported the tool confirms logical equivalence.

In the Fail case, unmapped points show up. This means certain signals/instances in the revised netlist couldn't be matched with their counterparts in the golden design. That's why it reports FAIL.

BAD NETLIST GENERATION

• REMOVING GATES

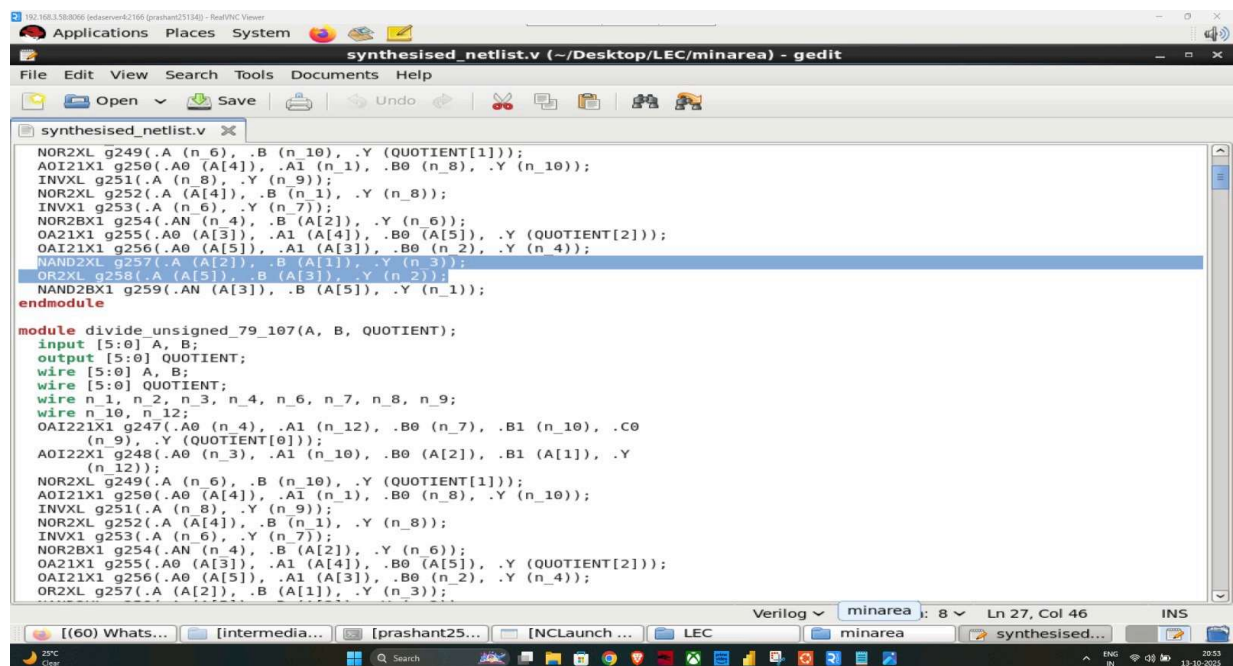


```
endmodule

module remainder_unsigned_73_108(A, B, REMAINDER);
input [5:0] A, B;
output [5:0] REMAINDER;
wire [5:0] A, B;
wire [5:0] REMAINDER;
wire n_1, n_2, n_3, n_4, n_5, n_6, n_7, n_8;
wire n_9, n_10, n_11, n_12, n_13;
ADDHXL g257(.A (n_10), .B (n_13), .CO (REMAINDER[3]), .S
(REMAINDER[2]));
/*OAI21X1 g258(.A0 (n_8), .A1 (n_11), .B0 (n_12), .Y (REMAINDER[1]));
NOR2BX1 g259(.AN (n_11), .B (n_8), .Y (n_13));
NAND2XL g260(.A (n_8), .B (n_11), .Y (n_12));
NAND2BX1 g261(.AN (n_4), .B (n_9), .Y (n_11));
ADDFXL g262(.A (A[2]), .B (n_3), .CI (n_6), .CO (n_9), .S (n_10));
OAI21X1 g263(.A0 (A[1]), .A1 (n_5), .B0 (n_7), .Y (n_8));
NAND2XL g264(.A (A[1]), .B (n_5), .Y (n_7));
NAND2BX1 g265(.AN (A[1]), .B (n_5), .Y (n_6));
AOI211X1 g266(.A0 (A[5]), .A1 (A[3]), .B0 (n_1), .CO (n_4), .Y (n_5));
NOR2XL g267(.A (A[4]), .B (n_2), .Y (n_4));
NAND2XL g268(.A (A[4]), .B (n_2), .Y (n_3));
NAND2BX1 g269(.AN (A[3]), .B (A[5]), .Y (n_2));
NOR2XL g270(.A (A[5]), .B (A[3]), .Y (n_1));*/
endmodule

module alarm_mt251(clk_pb34, rst_rp38, set_h_cs11, set_m_nm30,
en_a0_pb34, en_a1_rp38, set_a0_cs11, set_a1_nm30, btn_s_cs11,
btn_c_nm30, d0_pb34, d1_rp38, d2_cs11, d3_nm30, bz_pb34);
input clk_pb34, rst_rp38, en_a0_pb34, en_a1_rp38, set_a0_cs11,
set_a1_nm30, btn_s_cs11, btn_c_nm30;
input [5:0] set_h_cs11, set_m_nm30;
output [3:0] d0_pb34, d1_rp38, d2_cs11, d3_nm30;
```

• INTERCHANGING GATES



```
NOR2XL g249(.A (n_6), .B (n_10), .Y (QUOTIENT[1]));
AOI21X1 g250(.A0 (A[4]), .A1 (n_1), .B0 (n_8), .Y (n_10));
INVXL g251(.A (n_8), .Y (n_9));
NOR2XL g252(.A (A[4]), .B (n_1), .Y (n_8));
INVX1 g253(.A (n_6), .Y (n_7));
NOR2BX1 g254(.AN (n_4), .B (A[2]), .Y (n_6));
OAI21X1 g255(.A0 (A[3]), .A1 (A[4]), .B0 (A[5]), .Y (QUOTIENT[2]));
OAI21X1 g256(.A0 (A[5]), .A1 (A[3]), .B0 (n_2), .Y (n_4));
NAND2XL g257(.A (A[2]), .B (A[1]), .Y (n_3));
OR2XL g258(.A (A[5]), .B (A[3]), .Y (n_2));
NAND2BX1 g259(.AN (A[3]), .B (A[5]), .Y (n_1));
endmodule

module divide_unsigned_79_107(A, B, QUOTIENT);
input [5:0] A, B;
output [5:0] QUOTIENT;
wire [5:0] A, B;
wire [5:0] QUOTIENT;
wire n_1, n_2, n_3, n_4, n_6, n_7, n_8, n_9;
wire n_10, n_12;
OAI21X1 g247(.A0 (n_4), .A1 (n_12), .B0 (n_7), .B1 (n_10), .CO
(n_9), .Y (QUOTIENT[0]));
AOI22X1 g248(.A0 (n_3), .A1 (n_10), .B0 (A[2]), .B1 (A[1]), .Y
(n_12));
NOR2XL g249(.A (n_6), .B (n_10), .Y (QUOTIENT[1]));
AOI21X1 g250(.A0 (A[4]), .A1 (n_1), .B0 (n_8), .Y (n_10));
INVXL g251(.A (n_8), .Y (n_9));
NOR2XL g252(.A (A[4]), .B (n_1), .Y (n_8));
INVX1 g253(.A (n_6), .Y (n_7));
NOR2BX1 g254(.AN (n_4), .B (A[2]), .Y (n_6));
OAI21X1 g255(.A0 (A[3]), .A1 (A[4]), .B0 (A[5]), .Y (QUOTIENT[2]));
AOI21X1 g256(.A0 (A[5]), .A1 (A[3]), .B0 (n_2), .Y (n_4));
OR2XL g257(.A (A[2]), .B (A[1]), .Y (n_3));
```

ANALYSES OF BAD NETLIST

In this experiment, we intentionally changed the original circuit netlist to create a “bad netlist” so we could check if our verification tools can detect differences between correct and incorrect designs.

The changes we made were:

1. Gate type changes: Some logic gates like NAND were replaced with NOR or OR gates. These gates don't work the same way, so their outputs will be different for the same inputs.
2. Removed lines: A few logic connections and gate lines were deleted from the design, which made some signals lose their connection.

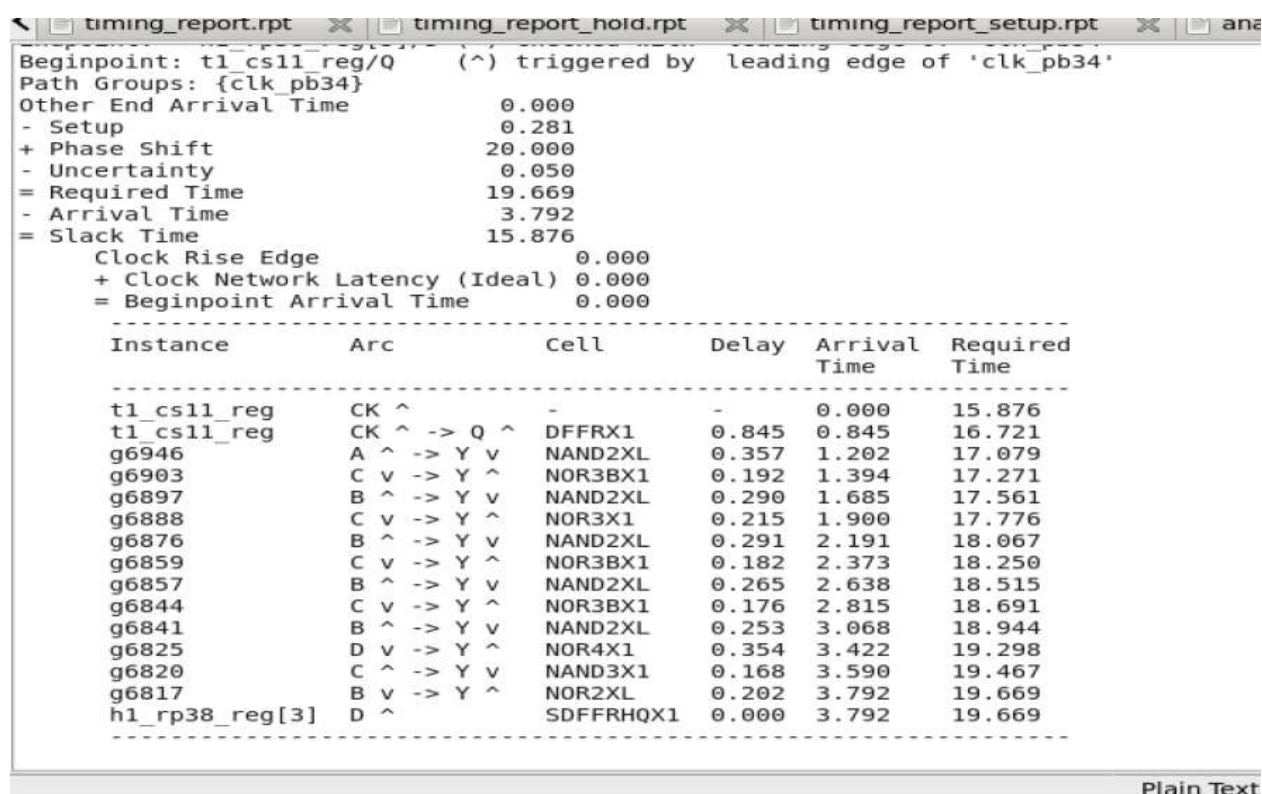
Because of these changes, the bad netlist doesn't give the same results as the original one. Some outputs will be wrong or undefined, and the overall circuit behaves incorrectly.

5. STATIC TIMING ANALYSIS

This is an analysis done on a circuit to make sure that delays in its components don't cause it to work incorrectly. We need to confirm that there are no setup time or hold time problems, meaning both setup slack and hold slack should be positive. The STA (Static Timing Analysis) tool checks this by looking at pairs of flip-flops (launch and capture) in the circuit and calculating these slacks. The tool uses the synthesized netlist, library file, and constraint file as inputs to produce the STA report.

5.1. STA FOR MINIMUM AREA

1. TIMING REPORT



The screenshot shows a timing report window with multiple tabs. The active tab displays a timing report for a specific path. The report includes a summary of timing parameters and a detailed table of logic elements along the path.

Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
- Setup 0.281
+ Phase Shift 20.000
- Uncertainty 0.050
= Required Time 19.669
- Arrival Time 3.792
= Slack Time 15.876
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	15.876
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	16.721
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	17.079
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	17.271
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	17.561
g6888	C v -> Y ^	NOR3X1	0.215	1.900	17.776
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	18.067
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	18.250
g6857	B ^ -> Y v	NAND2XL	0.265	2.638	18.515
g6844	C v -> Y ^	NOR3BX1	0.176	2.815	18.691
g6841	B ^ -> Y v	NAND2XL	0.253	3.068	18.944
g6825	D v -> Y ^	NOR4X1	0.354	3.422	19.298
g6820	C ^ -> Y v	NAND3X1	0.168	3.590	19.467
g6817	B v -> Y ^	NOR2XL	0.202	3.792	19.669
h1_rp38_reg[3]	D ^	SDDFRHGX1	0.000	3.792	19.669

Plain Text

HOLD REPORT

```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)
# Generated on: Mon Oct 13 17:52:28 2025
# Design: alarm_mt251
# Command: report_timing -nworst 10000 -early > $report_dir/timing_report_hold.rpt
#####
Path 1: MET Hold Check with Pin is_q_pb34_reg[1]/CK
Endpoint: is_q_pb34_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: is_q_pb34_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
is_q_pb34_reg[0] CK ^ - - 0.000 -0.178
is_q_pb34_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
is_q_pb34_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----
Path 2: MET Hold Check with Pin ic_q_rp38_reg[1]/CK
Endpoint: ic_q_rp38_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: ic_q_rp38_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
ic_q_rp38_reg[0] CK ^ - - 0.000 -0.178
ic_q_rp38_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
ic_q_rp38_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----
Path 3: MET Hold Check with Pin bs_q_cs11_reg[0]/CK
Endpoint: bs_q_cs11_reg[0]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: is_q_pb34_reg[1]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Clock Time 0.178

```

SETUP REPORT

analysis_coverage.rpt						
timing_report.rpt						
timing_report_						
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'						
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'						
Path Groups: {clk_pb34}						
Other End Arrival Time 0.000						
- Setup 0.281						
+ Phase Shift 20.000						
- Uncertainty 0.050						
= Required Time 19.669						
- Arrival Time 3.792						
= Slack Time 15.876						
Clock Rise Edge 0.000						
+ Clock Network Latency (Ideal) 0.000						
= Beginpoint Arrival Time 0.000						
Instance	Arc	Cell	Delay	Arrival Time	Required Time	
t1_cs11_reg	CK ^	-	-	0.000	15.876	
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	16.721	
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	17.079	
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	17.271	
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	17.561	
g6888	C v -> Y ^	NOR3X1	0.215	1.900	17.776	
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	18.067	
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	18.250	
g6857	B ^ -> Y v	NAND2XL	0.265	2.638	18.515	
g6844	C v -> Y ^	NOR3BX1	0.176	2.815	18.691	
g6841	B ^ -> Y v	NAND2XL	0.253	3.068	18.944	
g6825	D v -> Y ^	NOR4X1	0.354	3.422	19.298	
g6820	C ^ -> Y v	NAND3X1	0.168	3.590	19.467	
g6817	B v -> Y ^	NOR2XL	0.202	3.792	19.669	
h1_rp38_reg[3]	D ^	SDDFRHQX1	0.000	3.792	19.669	

analysis_coverage.rpt

timing_report.rpt

timing_report_hol

Path Groups: {clk_pb34}

Other End Arrival Time

0.000

- Setup

0.139

+ Phase Shift

20.000

- Uncertainty

0.050

= Required Time

19.811

- Arrival Time

3.866

= Slack Time

15.945

Clock Rise Edge

0.000

+ Clock Network Latency (Ideal)

0.000

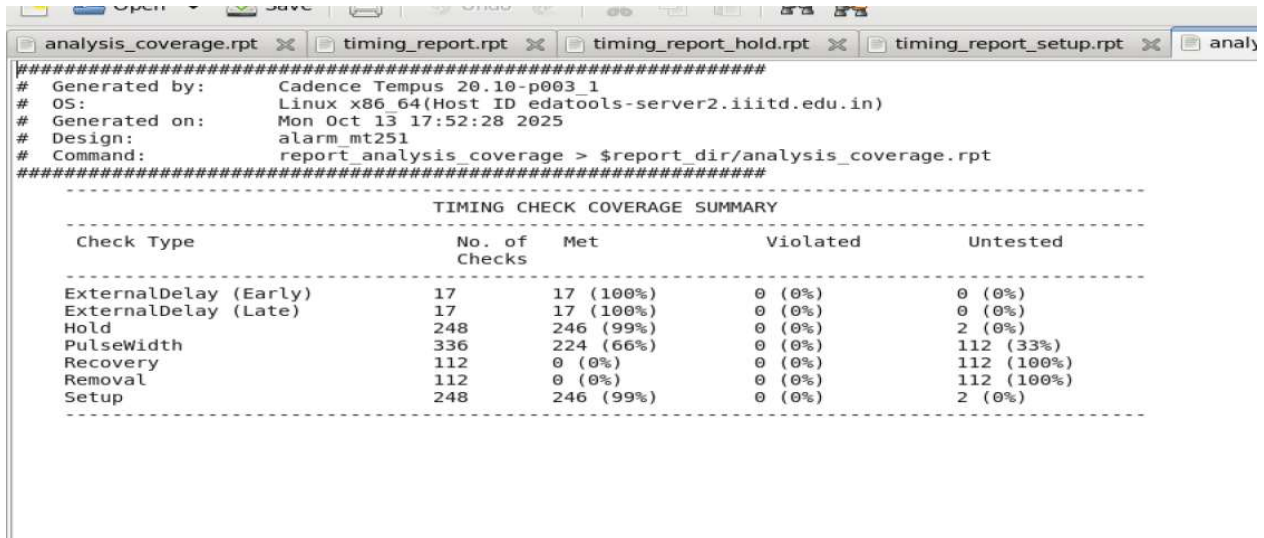
= Beginpoint Arrival Time

0.000

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	15.945
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	16.790
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	17.147
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	17.339
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	17.629
g6888	C v -> Y ^	NOR3X1	0.215	1.900	17.844
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	18.136
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	18.318
g6857	B ^ -> Y v	NAND2XL	0.265	2.638	18.583
g6844	C v -> Y ^	NOR3BX1	0.176	2.815	18.759
g6841	B ^ -> Y v	NAND2XL	0.253	3.068	19.012
g6825	D v -> Y ^	NOR4X1	0.354	3.422	19.367
g6820	C ^ -> Y v	NAND3X1	0.168	3.590	19.535
g6817	B v -> Y ^	NOR2XL	0.202	3.792	19.737
g6810	A ^ -> Y v	INVXL	0.074	3.866	19.811
h1_rp38_reg[3]	SI v	SDDFRHQX1	0.000	3.866	19.811

Path 3: MET Setup Check with Pin h1 rp38 reg[2]/CK

Coverage



```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Mon Oct 13 17:52:28 2025
# Design:            alarm_mt251
# Command:           report analysis_coverage > $report_dir/analysis_coverage.rpt
#####

-----
                        TIMING CHECK COVERAGE SUMMARY
-----
Check Type              No. of  Met          Violated      Untested
                        Checks
-----
ExternalDelay (Early)    17      17 (100%)     0 (0%)        0 (0%)
ExternalDelay (Late)     17      17 (100%)     0 (0%)        0 (0%)
Hold                    248     246 (99%)     0 (0%)        2 (0%)
PulseWidth              336     224 (66%)     0 (0%)        112 (33%)
Recovery                 112      0 (0%)        0 (0%)        112 (100%)
Removal                  112      0 (0%)        0 (0%)        112 (100%)
Setup                   248     246 (99%)     0 (0%)        2 (0%)
-----
```

5.2 STA for BEST TIMING

1. HOLD Time Violation


```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Mon Oct 13 17:56:47 2025
# Design: alarm_mt251
# Command: report_timing -nworst 10000 -early > $report_dir/timing_report_hold.rpt
#####
Path 1: MET Hold Check with Pin is_q_pb34_reg[1]/CK
Endpoint: is_q_pb34_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: is_q_pb34_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
is_q_pb34_reg[0] CK ^ - - 0.000 -0.178
is_q_pb34_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
is_q_pb34_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----
Path 2: MET Hold Check with Pin ic_q_rp38_reg[1]/CK
Endpoint: ic_q_rp38_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: ic_q_rp38_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}

```

```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Mon Oct 13 17:56:47 2025
# Design: alarm_mt251
# Command: report_timing -nworst 10000 -early > $report_dir/timing_report_ho
#####
Path 1: MET Hold Check with Pin is_q_pb34_reg[1]/CK
Endpoint: is_q_pb34_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: is_q_pb34_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
is_q_pb34_reg[0] CK ^ - - 0.000 -0.178
is_q_pb34_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
is_q_pb34_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----
Path 2: MET Hold Check with Pin ic_q_rp38_reg[1]/CK
Endpoint: ic_q_rp38_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: ic_q_rp38_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}

```

Set-up Time Violation

timing_report_setup.rpt (~Downloads/STA/besttiming/sta_besttiming/r

File Edit View Search Tools Documents Help

Open Save Undo

analysis_coverage.rpt timing_report.rpt timing_report_hold.rpt

Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
 Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
 Path Groups: {clk_pb34}
 Other End Arrival Time 0.000
 - Setup 0.281
 + Phase Shift 2.000
 - Uncertainty 0.050
 = Required Time 1.669
 - Arrival Time 3.792
 = Slack Time -2.124
 Clock Rise Edge 0.000
 + Clock Network Latency (Ideal) 0.000
 = Beginpoint Arrival Time 0.000

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	-2.124
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	-1.279
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	-0.921
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	-0.729
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	-0.439
g6888	C v -> Y ^	NOR3X1	0.215	1.900	-0.224
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	0.067
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	0.250
g6857	B ^ -> Y v	NAND2XL	0.265	2.638	0.515
g6844	C v -> Y ^	NOR3BX1	0.176	2.815	0.691
g6841	B ^ -> Y v	NAND2XL	0.253	3.068	0.944
g6825	D v -> Y ^	NOR4X1	0.354	3.422	1.298
g6820	C ^ -> Y v	NAND3X1	0.168	3.590	1.467
g6817	B v -> Y ^	NOR2XL	0.202	3.792	1.669
h1_rp38_reg[3]	D ^	SDDFRHGX1	0.000	3.792	1.669

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Open Save Undo

analysis_coverage.rpt timing_report.rpt timing_report_hold.rpt timing_re

Path 2: VIOLATED Setup Check with Pin h1_rp38_reg[3]/CK
 Endpoint: h1_rp38_reg[3]/SI (v) checked with leading edge of 'clk_pb34'
 Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
 Path Groups: {clk_pb34}
 Other End Arrival Time 0.000
 - Setup 0.139
 + Phase Shift 2.000
 - Uncertainty 0.050
 = Required Time 1.811
 - Arrival Time 3.866
 = Slack Time -2.055
 Clock Rise Edge 0.000
 + Clock Network Latency (Ideal) 0.000
 = Beginpoint Arrival Time 0.000

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	-2.055
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	-1.210
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	-0.853
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	-0.661
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	-0.371
g6888	C v -> Y ^	NOR3X1	0.215	1.900	-0.156
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	0.136
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	0.318
g6857	B ^ -> Y v	NAND2XL	0.265	2.638	0.583
g6844	C v -> Y ^	NOR3BX1	0.176	2.815	0.759
g6841	B ^ -> Y v	NAND2XL	0.253	3.068	1.012
g6825	D v -> Y ^	NOR4X1	0.354	3.422	1.367
g6820	C ^ -> Y v	NAND3X1	0.168	3.590	1.535
g6817	B v -> Y ^	NOR2XL	0.202	3.792	1.737
g6810	A ^ -> Y v	INVX1	0.074	3.866	1.811

Coverage

analysis_coverage.rpt

timing_report.rpt

timing_report_hold.rpt

timing_report_setup.rpt

ana

#####

#

Generated by:

Cadence Tempus 20.10-p003.1

#

OS:

Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)

#

Generated on:

Mon Oct 13 17:56:47 2025

#

Design:

alarm_mt251

#

Command:

report_analysis_coverage > \$report_dir/analysis_coverage.rpt

#####

TIMING CHECK COVERAGE SUMMARY

Check Type	No. of Checks	Met	Violated	Untested
ExternalDelay (Early)	17	17 (100%)	0 (0%)	0 (0%)
ExternalDelay (Late)	17	17 (100%)	0 (0%)	0 (0%)
Hold	248	246 (99%)	0 (0%)	2 (0%)
PulseWidth	336	224 (66%)	0 (0%)	112 (33%)
Recovery	112	0 (0%)	0 (0%)	112 (100%)
Removal	112	0 (0%)	0 (0%)	112 (100%)
Setup	248	185 (74%)	61 (24%)	2 (0%)

STA for Intermediate Stage

1. Timing Report

synthesis_area_report.rep x analysis_coverage.rpt x check_timing.rpt x timing_report.r					
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'					
Path Groups: {clk_pb34}					
Other End Arrival Time 0.000					
- Setup 0.281					
+ Phase Shift 20.000					
- Uncertainty 0.050					
= Required Time 19.669					
- Arrival Time 3.792					
= Slack Time 15.876					
Clock Rise Edge 0.000					
+ Clock Network Latency (Ideal) 0.000					
= Beginpoint Arrival Time 0.000					
Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	15.876
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	16.721
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	17.079
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	17.271
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	17.561
g6888	C v -> Y ^	NOR3X1	0.215	1.900	17.776
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	18.067
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	18.250
g6857	B ^ -> Y v	NAND2XL	0.265	2.638	18.515
g6844	C v -> Y ^	NOR3BX1	0.176	2.815	18.691
g6841	B ^ -> Y v	NAND2XL	0.253	3.068	18.944
g6825	D v -> Y ^	NOR4X1	0.354	3.422	19.298
g6820	C ^ -> Y v	NAND3X1	0.168	3.590	19.467
g6817	B v -> Y ^	NOR2XL	0.202	3.792	19.669
h1_rp38_reg[3]	D ^	SDFRHXQ1	0.000	3.792	19.669

2. Hold time Report


```

synthesis_area_report.rep x analysis_coverage.rpt x check_timing.rpt x timing_report.rpt
#####
Generated by: Cadence Tempus 20.10-p003_1
OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
Generated on: Mon Oct 13 14:52:56 2025
Design: alarm mt251
Command: report_timing -nworst 10000 -early > $report_dir/timing_report_hold.rpt
#####
Path 1: MET Hold Check with Pin is_q_pb34_reg[1]/CK
Endpoint: is_q_pb34_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: is_q_pb34_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
- Hold 0.111
- Phase Shift 0.000
- Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
is_q_pb34_reg[0] CK ^ - - 0.000 -0.178
is_q_pb34_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
is_q_pb34_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----
Path 2: MET Hold Check with Pin ic_q_rp38_reg[1]/CK
Endpoint: ic_q_rp38_reg[1]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: ic_q_rp38_reg[0]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
- Hold 0.111
- Phase Shift 0.000
- Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
ic_q_rp38_reg[0] CK ^ - - 0.000 -0.178
ic_q_rp38_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
ic_q_rp38_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----

```

```

Open Save Undo
synthesis_area_report.rep x analysis_coverage.rpt x check_timing.rpt x timing_report.rpt
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
ic_q_rp38_reg[0] CK ^ - - 0.000 -0.178
ic_q_rp38_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
ic_q_rp38_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----
Path 3: MET Hold Check with Pin bs_q_cs11_reg[0]/CK
Endpoint: bs_q_cs11_reg[0]/D (v) checked with leading edge of 'clk_pb34'
Beginpoint: is_q_pb34_reg[1]/Q (v) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
+ Hold 0.111
+ Phase Shift 0.000
+ Uncertainty 0.050
= Required Time 0.161
Arrival Time 0.339
Slack Time 0.178
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
bs_q_cs11_reg[0] CK ^ - - 0.000 -0.178
bs_q_cs11_reg[0] CK ^ -> Q v DFFRHQX1 0.339 0.339 0.161
bs_q_cs11_reg[1] D v DFFRHQX1 0.000 0.339 0.161
-----

```

3.Set-up Time Report


```

analysis_coverage.rpt  check_timing.rpt  timing_report.rpt  timing_report_hold
#####
# Generated by: Cadence Tempus 20.10-p003.1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Mon Oct 13 14:52:56 2025
# Design: alarm_mt251
# Command: report_timing -nworst 10000 > $report_dir/timing_report_setup.rpt
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time 0.000
- Setup 0.281
+ Phase Shift 20.000
- Uncertainty 0.050
= Required Time 19.669
- Arrival Time 3.792
= Slack Time 15.876
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000

```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	15.876
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	16.721
g6946	A ^ -> Y v	NAND2XL	0.357	1.202	17.079
g6903	C v -> Y ^	NOR3BX1	0.192	1.394	17.271
g6897	B ^ -> Y v	NAND2XL	0.290	1.685	17.561
g6888	C v -> Y ^	NOR3X1	0.215	1.900	17.776
g6876	B ^ -> Y v	NAND2XL	0.291	2.191	18.067
g6859	C v -> Y ^	NOR3BX1	0.182	2.373	18.250

```

analysis_coverage.rpt  check_timing.rpt  timing_report.rpt  timing_report_hold.rpt
g6903 C v -> Y ^ NOR3BX1 0.192 1.394 17.271
g6897 B ^ -> Y v NAND2XL 0.290 1.685 17.561
g6888 C v -> Y ^ NOR3X1 0.215 1.900 17.776
g6876 B ^ -> Y v NAND2XL 0.291 2.191 18.067
g6859 C v -> Y ^ NOR3BX1 0.182 2.373 18.250
g6857 B ^ -> Y v NAND2XL 0.265 2.638 18.515
g6844 C v -> Y ^ NOR3BX1 0.176 2.815 18.691
g6841 B ^ -> Y v NAND2XL 0.253 3.068 18.944
g6825 D v -> Y ^ NOR4X1 0.354 3.422 19.298
g6820 C ^ -> Y v NAND3X1 0.168 3.590 19.467
g6817 B v -> Y ^ NOR2XL 0.202 3.792 19.669
h1_rp38_reg[3] D ^ SDDFRHQX1 0.000 3.792 19.669

```

Path 2: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/SI (v) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}

Other End Arrival Time 0.000
- Setup 0.139
+ Phase Shift 20.000
- Uncertainty 0.050
= Required Time 19.811
- Arrival Time 3.866
= Slack Time 15.945
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	15.945
t1_cs11_reg	CK ^ -> Q ^	DFFRX1	0.845	0.845	16.790

Coverage Report

```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)
# Generated on: Mon Oct 13 14:52:56 2025
# Design: alarm_mt251
# Command: report_analysis_coverage > $report_dir/analysis_coverage.rpt
#####

-----
TIMING CHECK COVERAGE SUMMARY
-----
Check Type           No. of Checks  Met           Violated       Untested
-----
ExternalDelay (Early) 17             17 (100%)     0 (0%)         0 (0%)
ExternalDelay (Late) 17             17 (100%)     0 (0%)         0 (0%)
Hold                  248            246 (99%)     0 (0%)         2 (0%)
PulseWidth            336            224 (66%)     0 (0%)         112 (33%)
Recovery              112            0 (0%)        0 (0%)         112 (100%)
Removal               112            0 (0%)        0 (0%)         112 (100%)
Setup                 248            246 (99%)     0 (0%)         2 (0%)
-----
```

After STA Analysis

Optimization Stage	Required Time (ns)	Arrival Time (ns)	Slack (ns)
Minimum Area	19.669	3.792	+15.876
Intermediate State (Hold Check)	1.669	3.792	-2.124
Best Timing	19.669	3.792	+15.876

- The minimum area and best timing stages both meet setup timing (positive slack).
- The intermediate state shows a hold violation (negative slack = -2.124 ns).
- Hence, the final “best timing” optimization successfully fixed the hold issue while maintaining good setup timing.

TEST INSERTION

Scan cells are special parts inside digital circuits that help in **testing and debugging** the design. They are mainly used in **Design for Testability (DFT)** to make sure the circuit works properly after manufacturing. These cells allow engineers to **control and observe** what's happening inside flip-flops and registers without needing to run the whole circuit functionally.

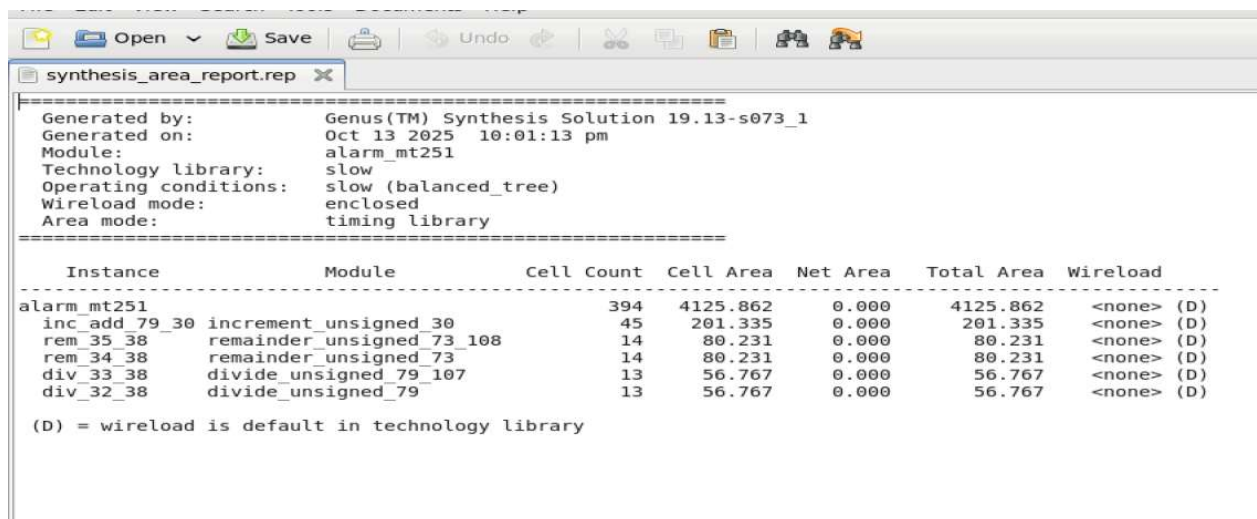
During testing, scan cells are connected in a **chain** so that test data can be shifted into the circuit and results can be shifted out easily. This helps in finding faults like **stuck-at** or **transition** errors much faster.

In debugging, scan cells give engineers direct access to internal signals, making it easier to **find and fix logic errors** in complex circuits. They also help locate exactly **which part of the chip is faulty**, saving time during troubleshooting.

Using tools like **Genus**, the netlist is built with scan cells to ensure good **reliability**, **easy testing**, and **efficient debugging**. Overall, scan cells make digital circuits more **testable**, **dependable**, and **easier to repair or modify**.

MINIMUM AREA

1. AREA REPORT



The screenshot shows a window titled 'synthesis_area_report.rep' with a table of area statistics. The table has columns for Instance, Module, Cell Count, Cell Area, Net Area, Total Area, and Wireload. The data is as follows:

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
alarm_mt251		394	4125.862	0.000	4125.862	<none> (D)
inc_add_79_30	increment_unsigned_30	45	201.335	0.000	201.335	<none> (D)
rem_35_38	remainder_unsigned_73_108	14	80.231	0.000	80.231	<none> (D)
rem_34_38	remainder_unsigned_73	14	80.231	0.000	80.231	<none> (D)
div_33_38	divide_unsigned_79_107	13	56.767	0.000	56.767	<none> (D)
div_32_38	divide_unsigned_79	13	56.767	0.000	56.767	<none> (D)

(D) = wireload is default in technology library

2.CELL Report

File Edit View Search Tools Documents Help				
Open Save Undo				
synthesis_area_report.rep synthesis_cell_report.rep				
NOR3X1	1	4.541	slow	
NOR4BX1	3	20.436	slow	
NOR4X1	2	12.110	slow	
NOR4XL	1	6.055	slow	
OA21X1	4	27.248	slow	
OA21XL	11	74.933	slow	
OAI211X1	5	26.492	slow	
OAI21X1	18	81.745	slow	
OAI21XL	1	4.541	slow	
OAI221X1	2	15.138	slow	
OAI2BB1X1	3	15.895	slow	
OAI31X1	1	6.055	slow	
OAI32X1	1	6.812	slow	
OR2X1	1	4.541	slow	
OR2XL	2	9.083	slow	
SDFFRHQX1	112	2797.502	slow	
XNOR2X1	1	8.326	slow	
XNOR2XL	5	41.630	slow	
XOR2XL	2	16.652	slow	

total	394	4125.862		

Type	Instances	Area	Area %	

sequential	112	2797.502	67.8	
inverter	34	77.204	1.9	
logic	248	1251.156	30.3	
physical_cells	0	0.000	0.0	

total	394	4125.862	100.0	

3. Power Report

Open Save Undo					
synthesis_area_report.rep synthesis_cell_report.rep synthesis_power_report.rep					
Instance: /alarm_mt251					
Power Unit: W					
PDB Frames: /stim#0/frame#0					
Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.70937e-05	4.56874e-05	1.58253e-06	6.43636e-05	82.34%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	5.99558e-06	3.01804e-06	1.61140e-06	1.06250e-05	13.59%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	3.17520e-06	3.17520e-06	4.06%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%

Subtotal	2.30893e-05	4.87054e-05	6.36913e-06	7.81638e-05	99.99%
Percentage	29.54%	62.31%	8.15%	100.00%	100.00%

4. Timing Report

File Edit View Search Tools Documents Help									
Open Save Undo									
synthesis_area_report.rep synthesis_cell_report.rep synthesis_power_report.rep synthesis_timing_report.rep									
g3330/B								+0	1182
g3330/Y	NOR3BX1	2	3.3	185				+191	1373 R
g3325/B								+0	1373
g3325/Y	NAND2XL	5	13.8	350				+287	1660 F
g3323/C								+0	1660
g3323/Y	NOR3X1	3	5.0	214				+211	1872 R
g3320/B								+0	1872
g3320/Y	NAND2XL	6	12.5	325				+274	2146 F
g3315/C								+0	2146
g3315/Y	NOR3BX1	2	3.3	178				+177	2323 R
g3313/B								+0	2323
g3313/Y	NAND2XL	6	12.6	323				+266	2589 F
g3304/C								+0	2589
g3304/Y	NOR3BX1	2	3.3	177				+176	2766 R
g3303/B								+0	2766
g3303/Y	NAND2XL	6	13.4	340				+279	3044 F
g3298/B								+0	3044
g3298/Y	NOR2XL	3	10.6	349				+335	3379 R
g3296/C								+0	3379
g3296/Y	NAND3X1	2	4.5	191				+176	3556 F
g3292/B								+0	3556
g3292/Y	NOR2XL	3	7.2	243				+224	3780 R
h1_rp38_reg[3]/D <<<	SDFFRHQX1							+0	3780
h1_rp38_reg[3]/CK	setup							0	4075 R

(clock clk_pb34)	capture								50000 R
	uncertainty							-20	49980 R

Cost Group : 'clk_pb34' (path_group 'clk_pb34')									
Timing slack : 45905ps									
Start-point : t1_cs11_reg/CK									
End-point : h1_rp38_reg[3]/D									

ANALYSIS -

Analysis (Minimum Area Constraint)

1. Area Efficiency

Minimum area constraint clearly reduced combinational logic duplication, but increased dependency on fewer gates, which can stretch timing paths.

2. Cell Composition & Design Balance

The design uses a good balance of AOI/OAI complex gates to optimize logic depth. High number of DFFs indicates this design is register-heavy (storage and state elements are significant).

Buffers are minimal suggesting synthesis didn't add too much extra buffering—this is expected in min-area runs, but may hurt timing in high fan-out paths.

3. Power Perspective

Dynamic power is proportional to switching activity; since sequential cells dominate, clock power is likely the main contributor. Leakage is minimized due to smaller area (fewer buffers and smaller drive strength cells).

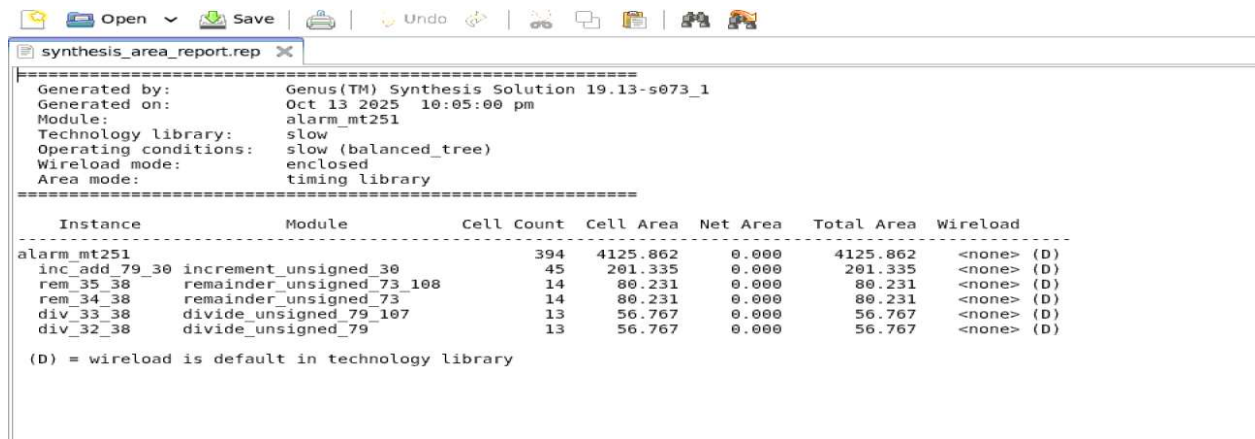
4. Timing Closure

Timing report shows delays close to constraints (critical path ~2.3–2.4 ns).

With minimum area focus, slack margins are tight, meaning if tighter timing constraints are applied, violations may occur. For a robust design, area-optimized netlist may need buffer insertion in physical design to recover slack.

INTERMEDIATE

1. AREA REPORT



The screenshot shows a software window titled "synthesis_area_report.rep". It contains a header section with metadata and a table of instance areas. The metadata includes: Generated by: Genus(TM) Synthesis Solution 19.13-s073_1, Generated on: Oct 13 2025 10:05:00 pm, Module: alarm_mt251, Technology library: slow, Operating conditions: slow (balanced_tree), Wireload mode: enclosed, and Area mode: timing library. The table has columns for Instance, Module, Cell Count, Cell Area, Net Area, Total Area, and Wireload. The data rows show the area breakdown for the alarm_mt251 module and its sub-components.

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
alarm_mt251		394	4125.862	0.000	4125.862	<none> (D)
inc_add_79_30	increment_unsigned_30	45	201.335	0.000	201.335	<none> (D)
rem_35_38	remainder_unsigned_73_108	14	80.231	0.000	80.231	<none> (D)
rem_34_38	remainder_unsigned_73	14	80.231	0.000	80.231	<none> (D)
div_33_38	divide_unsigned_79_107	13	56.767	0.000	56.767	<none> (D)
div_32_38	divide_unsigned_79	13	56.767	0.000	56.767	<none> (D)

(D) = wireload is default in technology library

2. CELL REPORT

File Edit View Search Tools Documents Help

Open Save Undo

synthesis_area_report.rep synthesis_cell_report.rep

NOR3X1	1	4.541	slow
NOR4BX1	3	20.436	slow
NOR4X1	2	12.110	slow
NOR4XL	1	6.055	slow
OA21X1	4	27.248	slow
OA21XL	11	74.933	slow
OAI211X1	5	26.492	slow
OAI21X1	18	81.745	slow
OAI21XL	1	4.541	slow
OAI221X1	2	15.138	slow
OAI2BB1X1	3	15.895	slow
OAI31X1	1	6.055	slow
OAI32X1	1	6.812	slow
OR2X1	1	4.541	slow
OR2XL	2	9.083	slow
SDFFRHQX1	112	2797.502	slow
XNOR2X1	1	8.326	slow
XNOR2XL	5	41.630	slow
XOR2XL	2	16.652	slow

total	394	4125.862	

Type	Instances	Area	Area %

sequential	112	2797.502	67.8
inverter	34	77.204	1.9
logic	248	1251.156	30.3
physical_cells	0	0.000	0.0

total	394	4125.862	100.0

3. POWER REPORT

File Edit View Search Tools Documents Help

Open Save Undo

synthesis_area_report.rep synthesis_cell_report.rep synthesis_power_report.rep

Instance: /alarm_mt251
Power Unit: W
PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%

memory	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
register	1.70937e-05	1.14300e-04	5.81529e-06	1.37209e-04	84.32%
latch	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
logic	5.99558e-06	7.54509e-06	4.02851e-06	1.75692e-05	10.80%
bbox	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
clock	0.000000e+00	0.000000e+00	7.93800e-06	7.93800e-06	4.88%
pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pm	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%

Subtotal	2.30893e-05	1.21845e-04	1.77818e-05	1.62716e-04	100.00%
Percentage	14.19%	74.88%	10.93%	100.00%	100.00%

4. TIMING REPORT

File Edit View Search Tools Documents Help							
synthesis_area_report.rep X synthesis_cell_report.rep X synthesis_power_report.rep							
g3330/B					+0	1182	
g3330/Y	NOR3BX1	2	3.3	185	+191	1373	R
g3325/B					+0	1373	
g3325/Y	NAND2XL	5	13.8	350	+287	1660	F
g3323/C					+0	1660	
g3323/Y	NOR3X1	3	5.0	214	+211	1872	R
g3320/B					+0	1872	
g3320/Y	NAND2XL	6	12.5	325	+274	2146	F
g3315/C					+0	2146	
g3315/Y	NOR3BX1	2	3.3	178	+177	2323	R
g3313/B					+0	2323	
g3313/Y	NAND2XL	6	12.6	323	+266	2589	F
g3304/C					+0	2589	
g3304/Y	NOR3BX1	2	3.3	177	+176	2766	R
g3303/B					+0	2766	
g3303/Y	NAND2XL	6	13.4	340	+279	3044	F
g3298/B					+0	3044	
g3298/Y	NOR2XL	3	10.6	349	+335	3379	R
g3296/C					+0	3379	
g3296/Y	NAND3X1	2	4.5	191	+176	3556	F
g3292/B					+0	3556	
g3292/Y	NOR2XL	3	7.2	243	+224	3780	R
h1_rp38_reg[3]/D <<<	SDFFRHQX1				+0	3780	
h1_rp38_reg[3]/CK	setup			0	+295	4075	R

(clock clk_pb34)	capture					20000	R
	uncertainty				-50	19950	R

Cost Group : 'clk_pb34' (path_group 'clk_pb34')							
Timing slack : 15875ps							
Start-point : t1_cs11_reg/CK							
End-point : h1_rp38_reg[3]/D							

ANALYSIS -

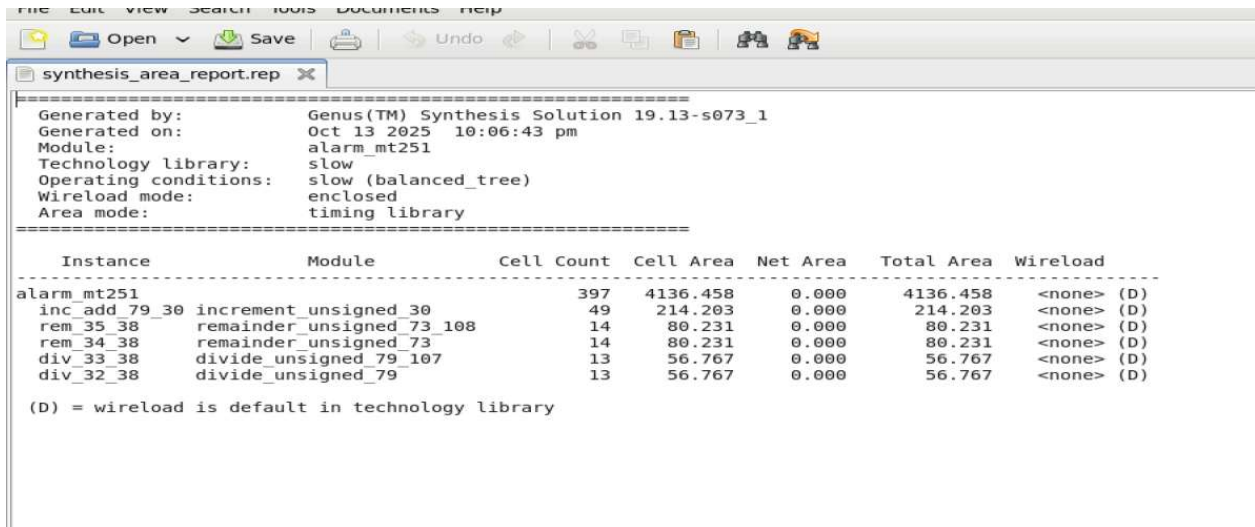
1. Total area and instance counts are the same as your min-area run. The tool did not upsize or replicate logic under the nominal constraint, which means your timing target and design loads are mild enough that the min-area solution already meets them.

2. Power & timing are dominated by the 112 flops and ideal clock
With 112 DFFs, register (clocked) power dominates the breakdown shown. Since this is pre-layout with an ideal clock, the clock tree contribution is understated; expect clock power to rise after CTS.

Timing is met with small margins; the critical path still walks through simple NAND/NOR cells—consistent with an area-oriented mapping.

BEST TIMING

1. AREA REPORT



Generated by: Genus(TM) Synthesis Solution 19.13-s073_1
Generated on: Oct 13 2025 10:06:43 pm
Module: alarm_mt251
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode: enclosed
Area mode: timing library

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
alarm_mt251		397	4136.458	0.000	4136.458	<none> (D)
inc_add_79_30	increment_unsigned_30	49	214.203	0.000	214.203	<none> (D)
rem_35_38	remainder_unsigned_73_108	14	80.231	0.000	80.231	<none> (D)
rem_34_38	remainder_unsigned_73	14	80.231	0.000	80.231	<none> (D)
div_33_38	divide_unsigned_79_107	13	56.767	0.000	56.767	<none> (D)
div_32_38	divide_unsigned_79	13	56.767	0.000	56.767	<none> (D)

(D) = wireload is default in technology library

2. CELL REPORT

File Edit View Search Tools Documents Help				
synthesis_area_report.rep X synthesis_cell_report.rep X				
NOR4BX1	4	27.248	slow	
NOR4BXL	1	6.812	slow	
NOR4X1	1	6.055	slow	
NOR4XL	1	6.055	slow	
OA21X1	4	27.248	slow	
OA21XL	10	68.121	slow	
OA22X1	1	7.569	slow	
OAI211X1	2	10.597	slow	
OAI21X1	19	86.287	slow	
OAI21XL	1	4.541	slow	
OAI221X1	2	15.138	slow	
OAI22X1	1	6.055	slow	
OAI2BB1X1	3	15.895	slow	
OAI33X1	1	8.326	slow	
OR2X1	1	4.541	slow	
OR2XL	2	9.083	slow	
SDFFRHQX1	111	2772.525	slow	
SDFFRX1	1	27.248	slow	
XNOR2XL	4	33.304	slow	

total	397	4136.458		

Type	Instances	Area	Area %	

sequential	112	2799.773	67.7	
inverter	33	74.933	1.8	
logic	252	1261.752	30.5	
physical_cells	0	0.000	0.0	

total	397	4136.458	100.0	

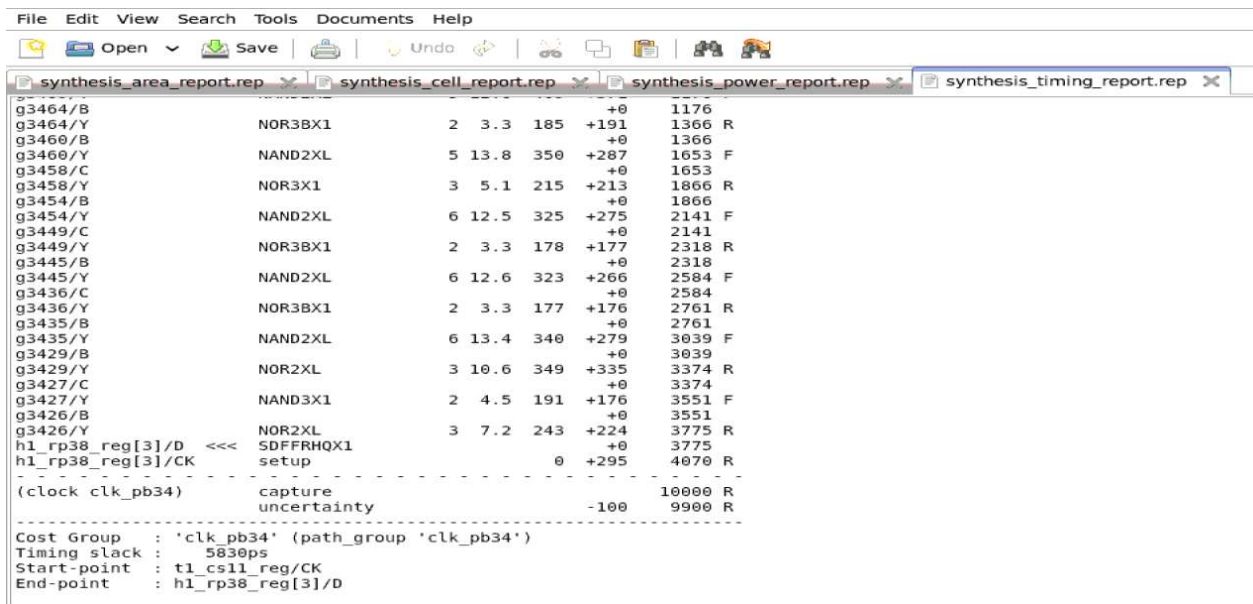
3. POWER REPORT

File Edit View Search Tools Documents Help						
synthesis_area_report.rep X synthesis_cell_report.rep X synthesis_power_report.rep X						
Instance: /alarm_mt251						
Power Unit: W						
PDB Frames: /stim#0/frame#0						
Category	Leakage	Internal	Switching	Total	Row%	

memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%	
register	1.70934e-05	2.30414e-04	1.55597e-05	2.63067e-04	84.95%	
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%	
logic	5.80456e-06	1.63283e-05	8.61280e-06	3.07457e-05	9.93%	
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%	
clock	0.00000e+00	0.00000e+00	1.58760e-05	1.58760e-05	5.13%	
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%	
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%	

Subtotal	2.28980e-05	2.46742e-04	4.00485e-05	3.09689e-04	100.01%	
Percentage	7.39%	79.67%	12.93%	100.00%	100.00%	

4. TIMING REPORT



The screenshot shows a software window titled 'synthesis_timing_report.rep' with a menu bar (File, Edit, View, Search, Tools, Documents, Help) and a toolbar. The main content is a table of synthesis data. The table has columns for gate names, cell types, and various timing metrics. The data is organized into sections for different gate types (NOR3BX1, NAND2XL, NOR3X1, NAND2XL, NOR2XL, NAND3X1, NOR2XL) and a summary section for the clock path (clk_pb34). The summary section includes cost group, timing slack, start-point, end-point, and capture uncertainty.

Gate Name	Cell Type	Area	Delay	Setup	Hold	Transition	Power	Comments
g3464/B	NOR3BX1	2	3.3	185	+191	1366	R	
g3460/B	NAND2XL	5	13.8	350	+287	1653	F	
g3458/C	NOR3X1	3	5.1	215	+213	1866	R	
g3454/B	NAND2XL	6	12.5	325	+275	2141	F	
g3449/C	NOR3BX1	2	3.3	178	+177	2318	R	
g3445/B	NAND2XL	6	12.6	323	+266	2584	F	
g3436/C	NOR3BX1	2	3.3	177	+176	2761	R	
g3435/B	NAND2XL	6	13.4	340	+279	3039	F	
g3429/B	NOR2XL	3	10.6	349	+335	3374	R	
g3427/C	NAND3X1	2	4.5	191	+176	3551	F	
g3426/B	NOR2XL	3	7.2	243	+224	3775	R	
h1_rp38_reg[3]/D	SDFFRHQX1				+0	3775		
h1_rp38_reg[3]/CK	setup			0	+295	4070	R	
(clock clk_pb34)	capture					10000	R	
	uncertainty			-100		9900	R	

Cost Group : 'clk_pb34' (path_group 'clk_pb34')

Timing slack : 5830ps

Start-point : t1_cs11_reg/CK

End-point : h1_rp38_reg[3]/D

ANALYSIS -

Synthesis behavior changed from “area/nominal-timing-driven.”

The tool upsized a small set of gates on the critical cone (e.g., NAND/NOR/INV variants to higher-drive cells) and inserted a couple of extra buffers/reconvergent cells. +0.26%).

Critical path is shortened by drive strength, not by logic restructuring.

The timing report shows the same style of path but with higher-drive stages and one or two added buffers to control transition/fanout.

Combinational depth is roughly the same; delays per stage drop. Slack turns positive / improves, indicating the new netlist meets your tighter period with a thin but real margin.

Power tracks the area change (slight rise).

Internal + switching power inch up due to upsized cells and extra buffers.

Leakage changes negligibly.

VDF GROUP PROJECT

PART-2 REPORT

Digital Alarm Clock

Group 15

Submitted to:

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2024-25

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INTERPRETATION OF RESULTS:

The table below shows the PPA obtained for various steps of physical design with core utilization of 0.5 and 0.8.

Table 1: PPA for different floorplan

	Setup Slack (ps)		Hold Slack (ps)		Area (um ²)		Power (uW)	
	0.5	0.8	0.5	0.8	0.5	0.8	0.5	0.8
Core Utilization	0.5	0.8	0.5	0.8	0.5	0.8	0.5	0.8
After Placement	4.773	4.930	-0.183	-0.181	4136.458	4136.458	466	462.8
After CTS	4.786	4.944	0.041	0.057	4181.873	4175.060	392.6	384.6
After Routing	4.785	4.952	0.036	0.058	4220.474	4213.662	399.3	391.2

EXPLANATION:

Comparing the 0.5 and 0.8 floorplans, the **setup and hold slacks** show slight improvement after CTS and routing for both, indicating effective clock balancing and minimal delay impact. The **0.8 utilization** consistently achieves marginally better timing due to shorter interconnect lengths, though with tighter placement. The **area** slightly increases after each stage due to added buffers and routing overhead, while the 0.8 design remains more area-efficient. The **power consumption** decreases after CTS and routing for both cases, showing optimization in the clock network and reduced switching activity.

Figure below shows the layout with core utilization of 0.5.

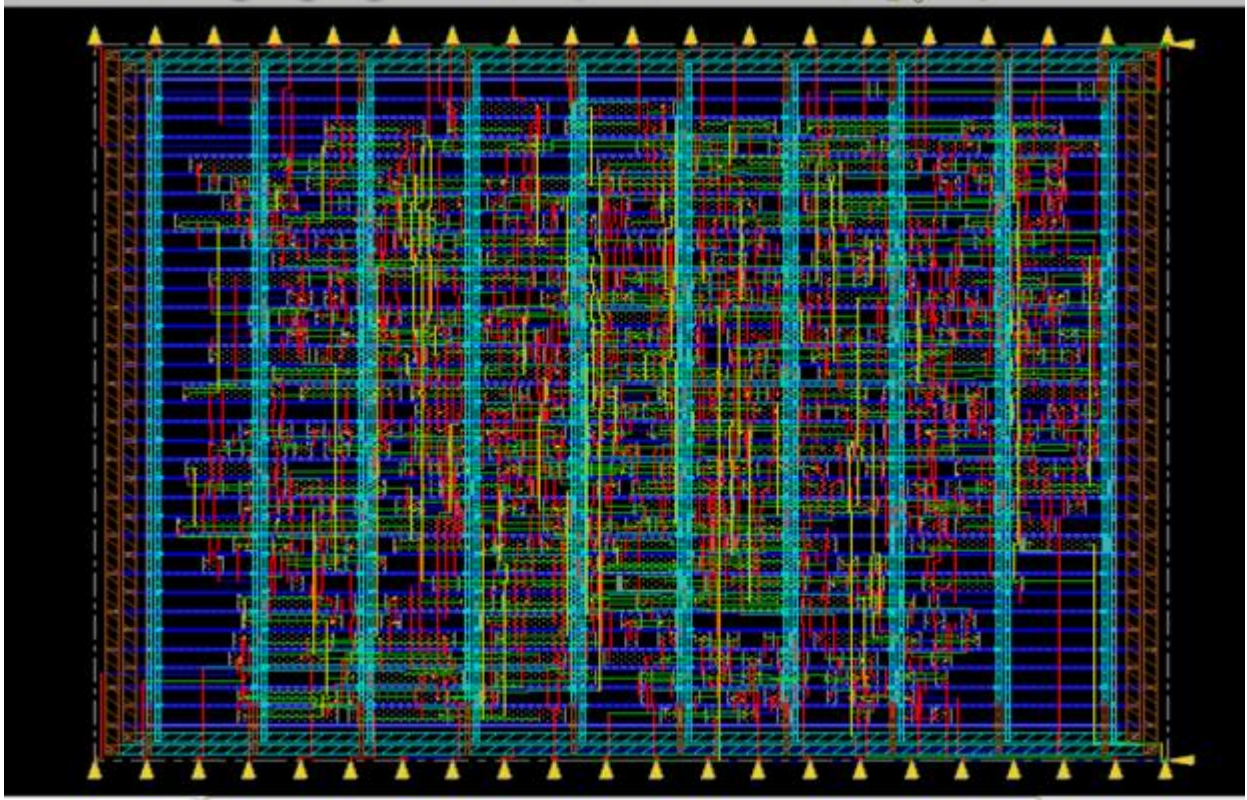


Figure 1: Layout with core utilization 0.5

Figure below shows the layout with core utilization of 0.8.

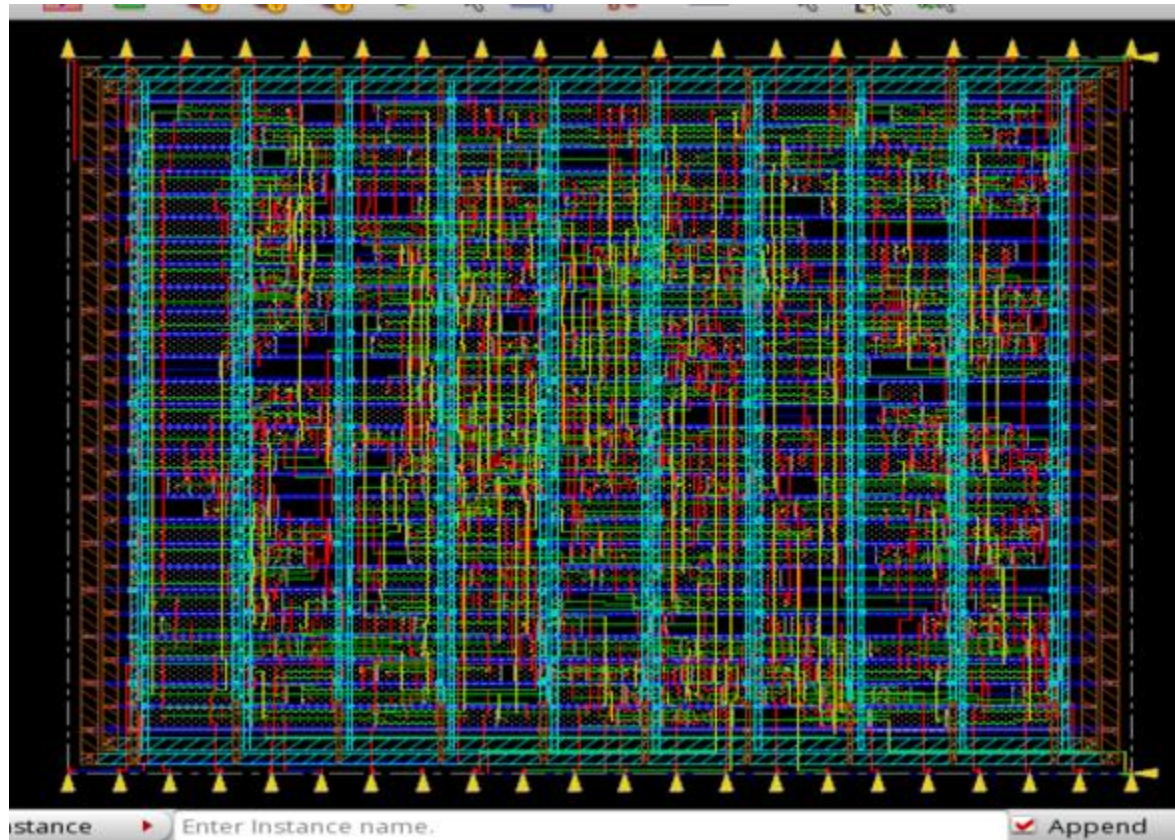


Figure 2: Layout with core utilization 0.8

BEFORE PHYSICAL DESIGN

This section shows the timing report of worst path with analysis of each component and how it is computed. It also explains the effect of slew, load, positive/negative unate-ness and PBA/GBA in the computation of timing of each stage. Further it reports the area and power before starting the physical design.

TIMING REPORT OF WORST PATH

SETUP:

Figure below shows the timing report of path with worst setup slack.

```
# Design:          aldim ml231
# Command:         report_timing -nworst 10000 > $report_dir/timing_report_setup.rpt
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint:  h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q    (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time          0.000
- Setup                        0.293
+ Phase Shift                  10.000
- Uncertainty                  0.100
= Required Time                 9.607
- Arrival Time                 3.795
= Slack Time                    5.812
Clock Rise Edge                0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time      0.000
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^	-	-	0.000	5.812
t1_cs11_reg	CK ^ -> Q ^	SDFFRHQX1	0.819	0.819	6.630
g3469	A ^ -> Y v	NAND2XL	0.372	1.190	7.002
g3464	B v -> Y ^	NOR3BX1	0.192	1.383	7.194
g3460	B ^ -> Y v	NAND2XL	0.287	1.670	7.481
g3458	C v -> Y ^	NOR3X1	0.214	1.884	7.696
g3454	B ^ -> Y v	NAND2XL	0.275	2.159	7.970
g3449	C v -> Y ^	NOR3BX1	0.177	2.336	8.148

Figure 3: Timing report of path with worst setup slack

The following equations show how the setup slack and its component is computed.

Setup Slack Time = Required Time – Arrival Time

Arrival Time = $T_{\text{launch}} + T_{\text{clk-q}} + T_{\text{data}}$

Required Time = $T_{\text{period}} + T_{\text{capture}} - T_{\text{setup-c}} - \text{Uncertainty}$

HOLD:

Figure below shows the timing report of path with worst hold slack.

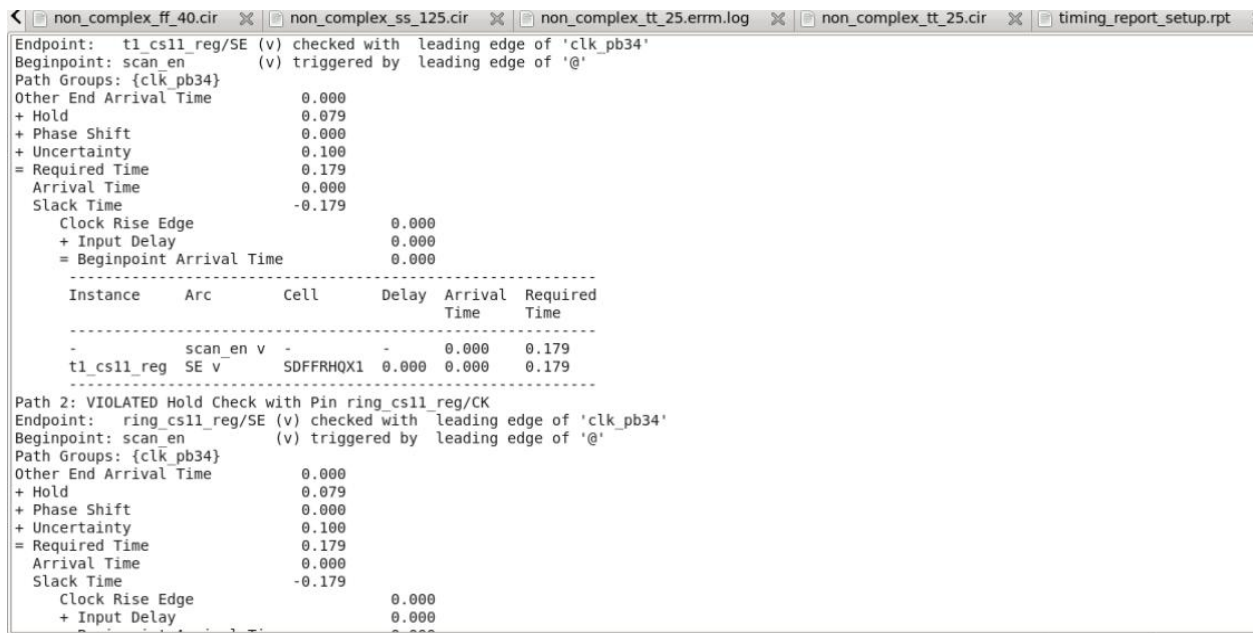


Figure 4: Timing report of path with worst hold slack

The following equations show how the hold slack and its component is computed.

Hold Slack Time = Arrival Time - Required Time

Arrival Time = $T_{\text{launch}} + T_{\text{clk-q}} + T_{\text{data}}$

Required Time = $T_{\text{capture}} + T_{\text{hold-c}} + \text{Uncertainty}$

EFFECT OF LOAD

SETUP WITH LOAD 1 pf

```
Path 1: MET Setup Check with Pin d3_nm30_reg[3]/CK
Endpoint: d3_nm30_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: d3_nm30_reg[3]/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Other End Arrival Time          0.000
- Setup                        1.064
+ Phase Shift                  10.000
- Uncertainty                  0.100
= Required Time                 8.836
- Arrival Time                 6.281
= Slack Time                    2.555
  Clock Rise Edge              0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time      0.000
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
d3_nm30_reg[3]	CK ^	-	-	0.000	2.555
d3_nm30_reg[3]	CK ^ -> Q ^	SDDFRHGX1	6.281	6.281	8.836
d3_nm30_reg[3]	D ^	SDDFRHGX1	0.000	6.281	8.836

Figure 9: Setup report with load 1 pF

HOLD WITH LOAD 1 pF

Path 1: VIOLATED Hold Check with Pin t1_cs11_reg/CK					
Endpoint: t1_cs11_reg/SE (v) checked with leading edge of 'clk_pb34'					
Beginpoint: scan_en (v) triggered by leading edge of '@'					
Path Groups: {clk_pb34}					
Other End Arrival Time			0.000		
+ Hold			0.079		
+ Phase Shift			0.000		
+ Uncertainty			0.100		
= Required Time			0.179		
Arrival Time			0.000		
Slack Time			-0.179		
Clock Rise Edge			0.000		
+ Input Delay			0.000		
= Beginpoint Arrival Time			0.000		
Instance	Arc	Cell	Delay	Arrival Time	Required Time
-	scan_en v	-	-	0.000	0.179
t1_cs11_reg	SE v	SDFFRHQX1	0.000	0.000	0.179

Figure 11: Hold report with load 1 pF

EXPLANATION:

With **1 pF load** added, the data path gets much slower, so **setup slack drops** from about **+5.812 ps** to roughly **+2.555 ps**—still meeting setup but with far less margin. The **hold slack** stays around **-0.179 ps** (still a violation) because the extra load was on the downstream data net and **didn't meaningfully slow the fast, short hold path.**

PBA FOR SETUP SLACK:

```
# OS: Linux x86_64 (Host ID edatools-server2.111td.edu.in)
# Generated on: Mon Nov 10 20:44:13 2025
# Design: alarm_mt251
# Command: report_timing -retime path_slew_propagation -late -max_path 50 -path_type full_cl
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Retime Analysis { Data Path-Slew }
Other End Arrival Time 0.000
- Setup 0.278
+ Phase Shift 10.000
- Uncertainty 0.100
= Required Time 9.622
- Arrival Time 3.810
= Slack Time 5.812
= Slack Time(original) 5.812
Clock Rise Edge 0.000
= Beginpoint Arrival Time 0.000
Timing Path:
```

Instance	Arc	Cell	Retime Delay	Arrival Time	Required Time
-	clk_pb34 ^	-	-	0.000	5.812
t1_cs11_reg	-	SDFFRHQX1	0.000	0.000	5.812
t1_cs11_reg	CK ^ -> Q ^	SDFFRHQX1	0.834	0.834	6.646
g3469	-	NAND2XL	0.000	0.834	6.646
g3469	A ^ -> Y v	NAND2XL	0.372	1.206	7.018
g3464	-	NOR3BX1	0.000	1.206	7.018

Figure 16: Setup Report using PBA

PBA FOR HOLD SLACK:

```
# Design:          alarm_mt251
# Command:         report_timing -retime path_slew_propagation -early -max_path 50 -path_type
#####
Path 1: VIOLATED Hold Check with Pin t1_cs11_reg/CK
Endpoint:  t1_cs11_reg/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
Retime Analysis { Data Path-Slew }
Other End Arrival Time      0.000
+ Hold                      0.060
+ Phase Shift               0.000
+ Uncertainty               0.100
= Required Time              0.161
  Arrival Time              0.000
  Slack Time                 -0.161
= Slack Time(original)      -0.160
  Clock Rise Edge           0.000
  + Input Delay              0.000
  = Beginpoint Arrival Time  0.000
Timing Path:
-----
Instance   Arc      Cell      Retime  Arrival  Required
          Delay    Time    Time
-----
-          scan_en v -      -      0.000   0.161
t1_cs11_reg -      SDFFRHQX1 0.000   0.000   0.161
-----
Clock Rise Edge           0.000
= Beginpoint Arrival Time  0.000
```

Figure 17: Hold Report using PBA

FLOORPLANNING

Floorplanning is the initial stage of the physical design process, where the chip is divided into various subsystems or blocks. During this step, area is allocated for standard cells, macros, and memory components, and the locations of input/output pads are also defined. In our design, floorplanning was carried out for two different core utilizations: 0.5 and 0.8.

The figure below illustrates the floorplan corresponding to a core utilization of 0.5



Figure 22: Floorplanning with core utilization 0.5

Figure below shows floorplanning for 0.8 core utilization.



Figure 23: Floorplanning with core utilization 0.8

In VLSI design, a floorplan of 0.5 means 50% core utilization, while 0.8 means 80%. A 0.5 utilization gives more whitespace, making routing, timing, and power easier to manage but increases chip area. In contrast, 0.8 utilization reduces area and cost but causes higher congestion, timing challenges, and IR drop issues. Lower utilization favors easier design closure, while higher utilization focuses on area efficiency.

POWER NETWORK DESIGN

At this stage of the physical design flow, the power distribution network is implemented to ensure a stable supply of power across the chip. The figure below presents the power network layout corresponding to a core utilization of 0.5.

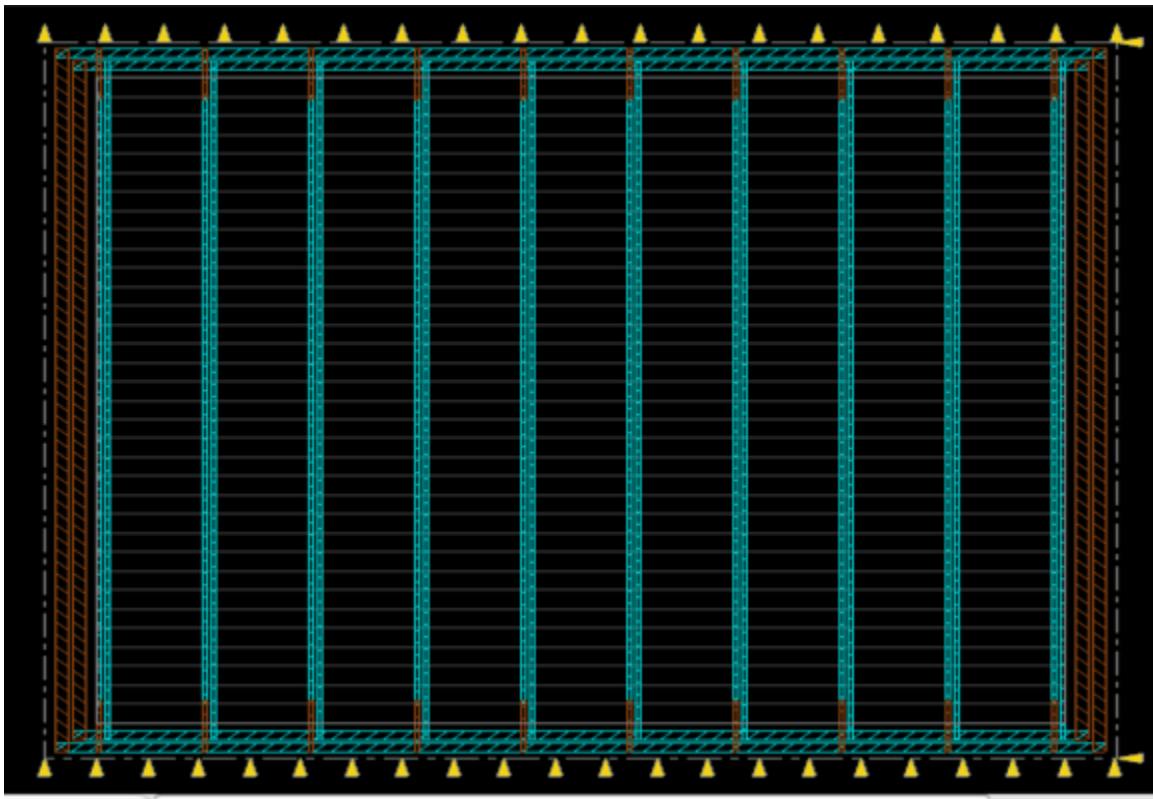


Figure 24: Power design network with core utilization 0.5

Figure below shows power design network for 0.8 core utilization.

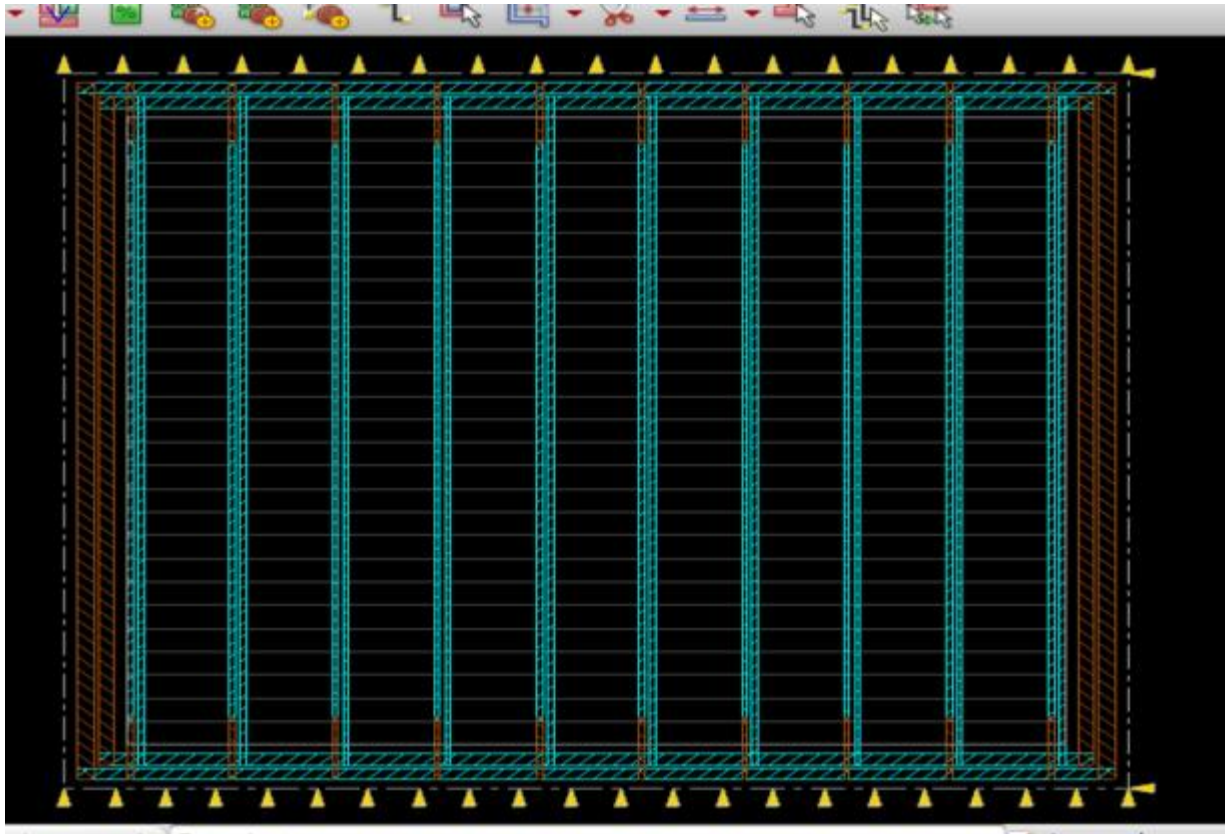


Figure 25: Power design network with core utilization 0.8

COMPARISON:

The **first image** shows a **denser floorplan** (likely 0.8 utilization) with tighter power grid and smaller area, suited for area efficiency but harder timing/routing.

The **second image** shows a **looser floorplan** (around 0.5 utilization), with more whitespace and easier routing, ideal for early design or timing-critical circuits.

PLACEMENT

FOR CORE UTILIZATION 0.5

Placement determines the precise locations of standard cells within the design. The main objectives of this step are to minimize total wire length, meet timing constraints, and avoid routing congestion.

SETUP TIMING REPORT:

Figure below shows the worst setup slack after placement.

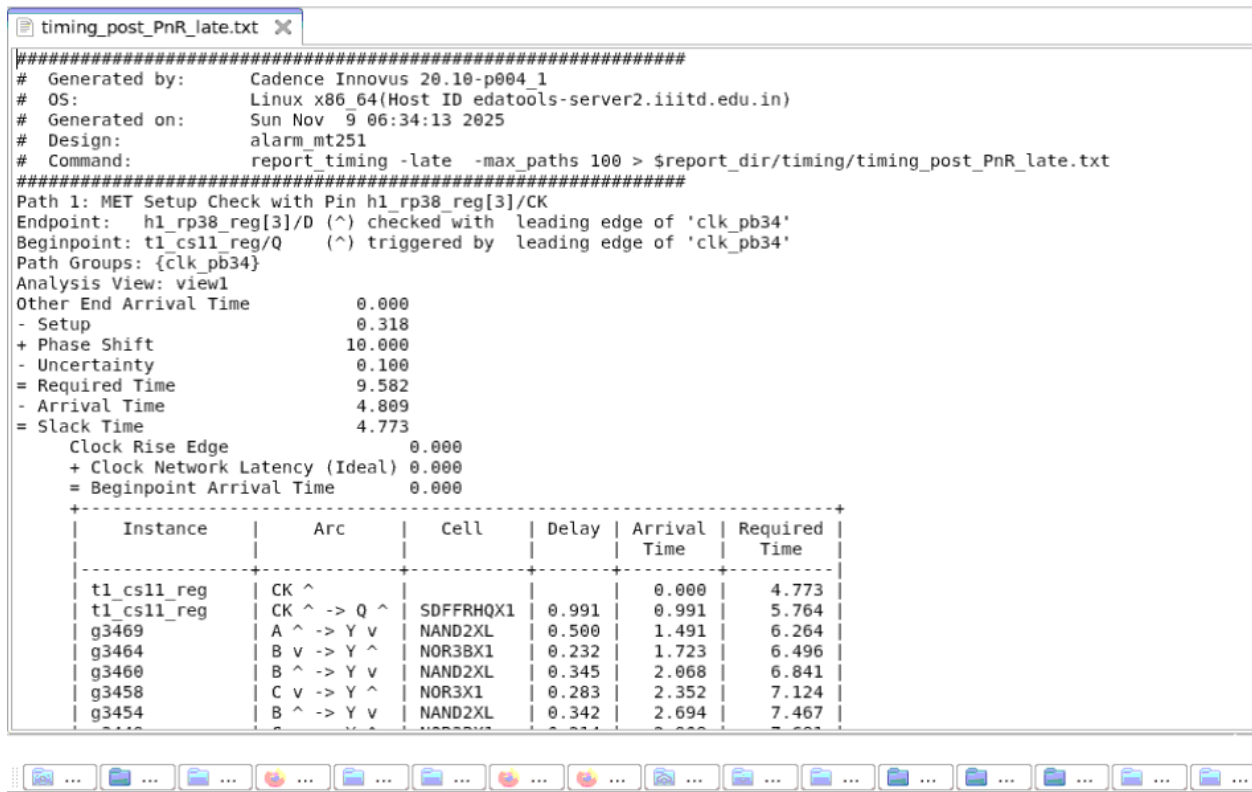


Figure 26: Setup Report after placement for 0.5

HOLD TIMING REPORT:

Figure below shows the worst hold slack after placement.

```
# OS: Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 06:34:13 2025
# Design: alarm_mt251
# Command: report_timing -early -view {view1} -max_paths 100 > $report_dir/timing/timing_post_PnR_early.txt
#####
Path 1: VIOLATED Hold Check with Pin s0_pb34_reg[3]/CK
Endpoint: s0_pb34_reg[3]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time 0.000
+ Hold 0.087
+ Phase Shift 0.000
+ Uncertainty 0.100
= Required Time 0.187
Arrival Time 0.004
Slack Time -0.183
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
|-----|-----|-----|-----|-----|-----|
| s0_pb34_reg[3] | scan_en v | SDFFRHQX1 | 0.004 | 0.004 | 0.187 |
+-----+-----+-----+-----+-----+-----+
Path 2: VIOLATED Hold Check with Pin m0_cs11_reg[3]/CK
Endpoint: m0_cs11_reg[3]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
```

Plain T

Figure 27: Hold Report after placement for 0.5

Table below compares worst setup and hold slack before and after placement.

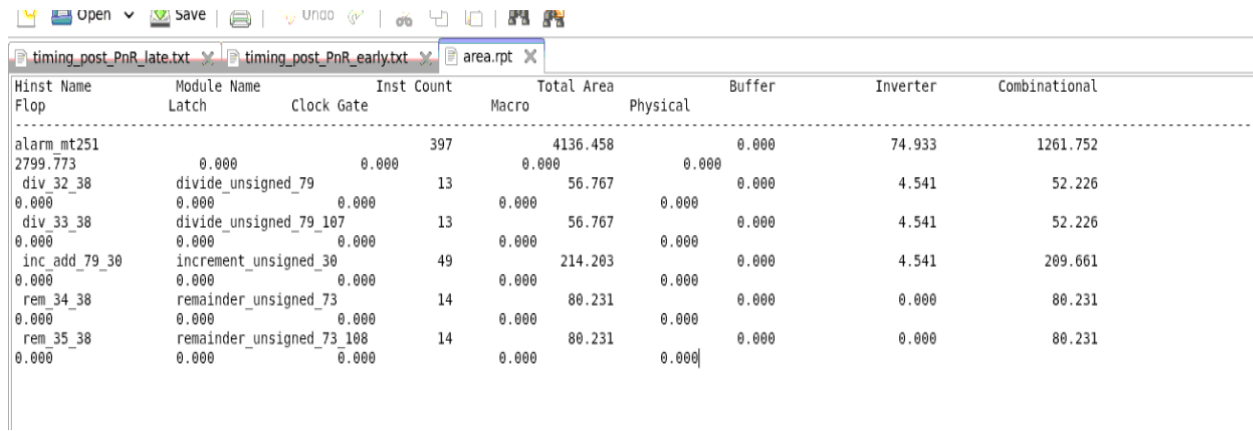
Table 3: Worst Setup and Hold slack before and after placement for 0.5

	Setup Slack (ps)	Hold Slack (ps)
Before Placement	5.812	-0.179
After Placement	4.773	-0.183

After placement, the setup slack decreased slightly from 5.812 ps to 4.773 ps, indicating a minor increase in data path delay due to added wire parasitics, but it still meets timing requirements. However, the hold slack degraded significantly from -0.179 ps to -1.83 ps, showing that data is arriving too early at the capture flip-flop, causing hold time violations. This means that while setup timing remains safe, hold violations increased after placement and need to be corrected through hold fixing techniques such as buffer insertion.

AREA REPORT:

Figure below shows the area report and breakup of different kinds of cells.



The screenshot shows a software window titled 'area.rpt' with a menu bar (Open, Save, Undo) and a toolbar. The main area contains a table with the following data:

Hinst Name	Module Name	Inst Count	Total Area	Physical	Buffer	Inverter	Combinational
Flop	Latch	Clock Gate	Macro				
alarm_mt251		397	4136.458		0.000	74.933	1261.752
2799.773	0.000	0.000	0.000	0.000			
div_32_38	divide_unsigned_79	13	56.767		0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000			
div_33_38	divide_unsigned_79_107	13	56.767		0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000			
inc_add_79_30	increment_unsigned_30	49	214.203		0.000	4.541	209.661
0.000	0.000	0.000	0.000	0.000			
rem_34_38	remainder_unsigned_73	14	80.231		0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000			
rem_35_38	remainder_unsigned_73_108	14	80.231		0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000			

Figure 28: Area report after placement for 0.5

Total Area = 4136.458 μm^2

GATES REPORT:

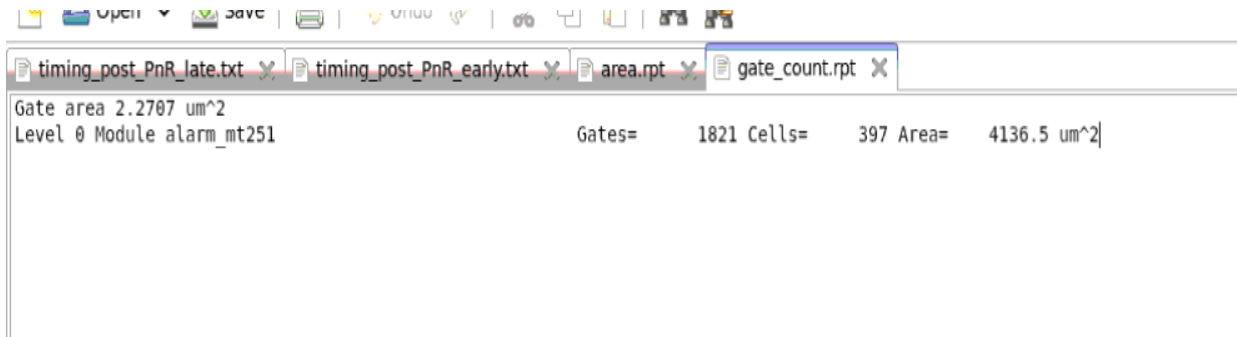


Figure 29: Gate count after placement for 0.5

Total Gates = 1821

POWER REPORT:

g3561	1.036e-08	9.911e-10	1.656e-05	1.655e-05	NOR48XL
g2023	2.059e-09	8.25e-10	1.511e-05	1.511e-05	NAND3X1
g3427	3.186e-10	2.043e-10	1.511e-05	1.511e-05	NAND3X1
g3462	4.359e-07	1.916e-07	1.473e-05	1.41e-05	OAI211X1
inc_add_79_30/g788	1.298e-06	6.215e-07	1.428e-05	1.236e-05	NOR2XL
g3450	1.288e-06	5.318e-07	1.418e-05	1.236e-05	NOR2XL
g3452	8.647e-07	8.479e-07	1.407e-05	1.236e-05	NOR2XL
g3454	8.158e-07	3.056e-06	1.316e-05	9.289e-06	NAND2XL
g3478	3.952e-09	3.572e-08	1.305e-05	1.301e-05	CLKINX1
inc_add_79_30/g768	1.196e-06	2.17e-06	1.266e-05	9.289e-06	NAND2XL
g3438	1.134e-07	4.301e-08	1.251e-05	1.236e-05	NOR2XL
g3442	7.191e-08	6.684e-08	1.25e-05	1.236e-05	NOR2XL
g3431	8.056e-09	9.634e-09	1.238e-05	1.236e-05	NOR2XL
g3429	1.291e-09	2.037e-09	1.236e-05	1.236e-05	NOR2XL
g3426	1.46e-10	1.65e-10	1.236e-05	1.236e-05	NOR2XL
inc_add_79_30/g782	3.55e-07	1.024e-06	1.067e-05	9.289e-06	NAND2XL
g1991	5.051e-10	4.132e-10	1.046e-05	1.046e-05	NAND4XL
g3443	2.34e-10	2.192e-10	1.046e-05	1.046e-05	NAND4XL
g3445	6.881e-08	3.019e-07	9.66e-06	9.289e-06	NAND2XL
inc_add_79_30/g766	1.061e-07	1.954e-07	9.591e-06	9.289e-06	NAND2XL
g2056	4.311e-07	2.831e-07	9.471e-06	8.757e-06	INVXL
inc_add_79_30/g779	2.861e-08	8.233e-08	9.4e-06	9.289e-06	NAND2XL
g2030	1.868e-08	8.312e-09	9.316e-06	9.289e-06	NAND2XL
g3435	4.364e-09	2.041e-08	9.314e-06	9.289e-06	NAND2XL
inc_add_79_30/g764	7.142e-09	1.234e-08	9.309e-06	9.289e-06	NAND2XL
g1450	7.449e-11	6.585e-11	8.757e-06	8.757e-06	INVXL
g1989	5.498e-10	3.29e-10	8.729e-06	8.728e-06	OAI21XL

Total (397 of 397)	0.3666	0.07653	0.466	0.0229	
Total Capacitance	5.14e-12 F				
Power Density	*** No Die Area ***				

Disi

Figure 30: Power report after placement for 0.5

Total Power = 466uW

LAYOUT:

Figure below shows the layout of design and connectivity showing fly-lines.

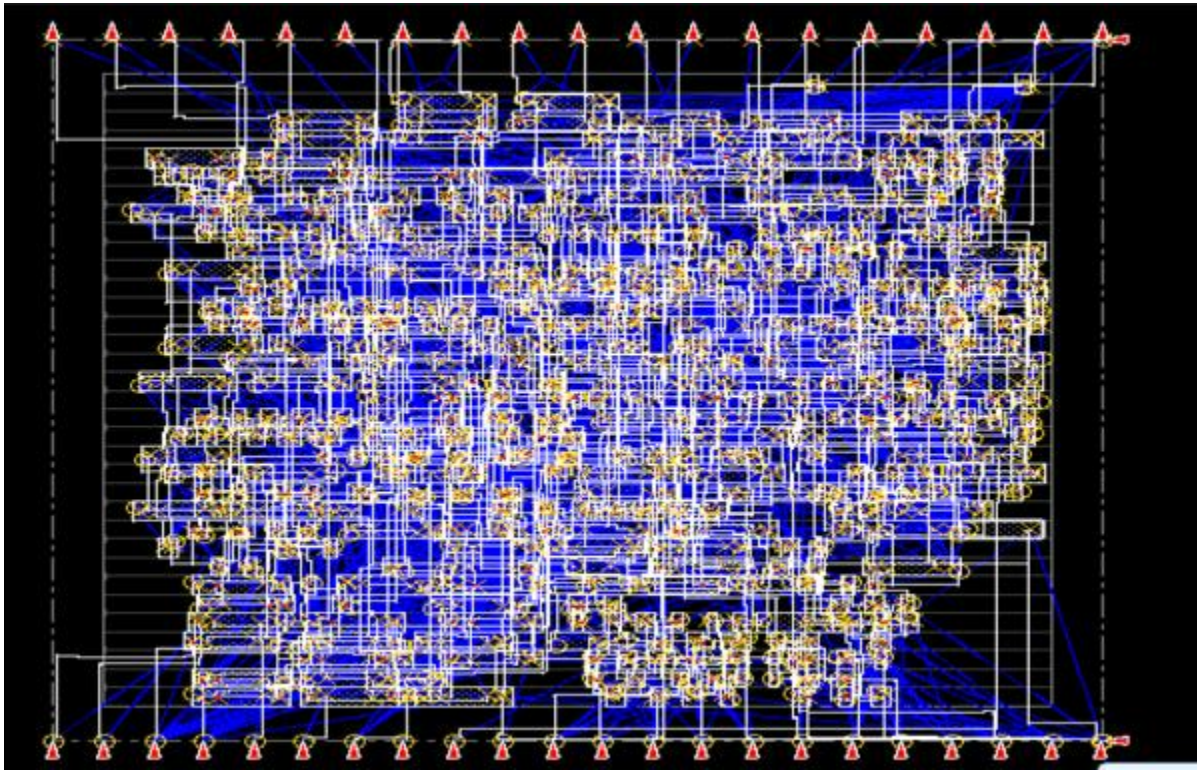


Figure 31: Layout after placement showing fly-lines for 0.5

The layout with fly-lines shows the physical placement of blocks and their logical connections before routing. The fly-lines are thin lines that indicate how different modules or cells are connected based on the netlist. They help visualize connectivity, optimize block placement, and reduce wirelength and congestion. Short, organized fly-lines indicate a good floorplan, while long or crossed ones suggest the need for placement improvement.

FOR CORE UTILIZATION 0.8

SETUP TIMING REPORT:

```
# Generated by: Cadence Innovus 20.10-p004_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 06:46:31 2025
# Design: alarm_mt251
# Command: report_timing -late -max_paths 100 > $report_dir/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time 0.000
- Setup 0.310
+ Phase Shift 10.000
- Uncertainty 0.100
= Required Time 9.590
- Arrival Time 4.660
= Slack Time 4.930
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
|-----|-----|-----|-----|-----|-----|
| t1_cs11_reg | CK ^ |  |  | 0.000 | 4.930 |
```

Figure 32: Setup Report after placement for 0.8

HOLD TIMING REPORT:

```
# Generated on: Sun Nov 9 06:46:31 2025
# Design: alarm_mt251
# Command: report_timing -early -view {view1} -max_paths 100 > $report_dir/timing/timing_post_PnR_earl
#####
Path 1: VIOLATED Hold Check with Pin m1_nm30_reg[0]/CK
Endpoint: m1_nm30_reg[0]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time 0.000
+ Hold 0.085
+ Phase Shift 0.000
+ Uncertainty 0.100
= Required Time 0.185
Arrival Time 0.004
Slack Time -0.181
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
|-----|-----|-----|-----|-----|-----|
| m1_nm30_reg[0] | scan_en v | SDFFRHGX1 | 0.004 | 0.004 | 0.185 |
+-----+-----+-----+-----+-----+-----+
Path 2: VIOLATED Hold Check with Pin m0_cs11_reg[3]/CK
Endpoint: m0_cs11_reg[3]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
```

Figure 33: Hold Report after placement for 0.8

Table below compares worst setup and hold slack before and after placement.

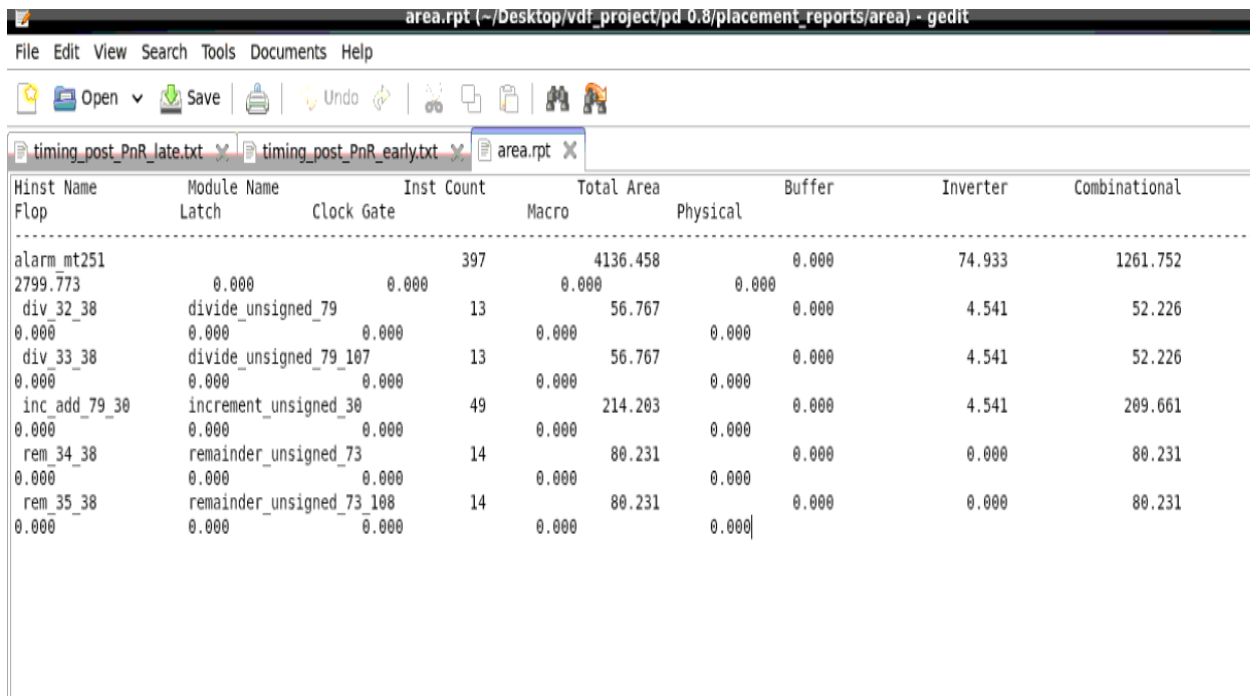
Table 4: Worst Setup and Hold slack before and after placement for 0.8

	Setup Slack (ps)	Hold Slack (ps)
Before Placement	5.812	-0.179
After Placement	4.930	-0.181

.EXPLANATION :

For the 0.8 floorplan, the **setup slack** decreased slightly from **5.812 ps** to **4.930 ps** after placement because, at higher utilization, cells are placed more compactly, leading to **increased routing complexity and parasitic effects** such as wire capacitance and resistance. These parasitics increase the overall data path delay, causing a small reduction in setup slack. The **hold slack** changed minimally from **-0.179 ps** to **-0.181 ps**, indicating that **short data paths were not significantly affected** by placement, and clock skew or local interconnect variations remained almost the same

AREA REPORT:



Hinst Name	Module Name	Inst Count	Total Area	Buffer	Inverter	Combinational
Flop	Latch	Clock Gate	Macro	Physical		
alarm_mt251		397	4136.458	0.000	74.933	1261.752
2799.773	0.000	0.000	0.000	0.000		
div_32_38	divide_unsigned_79	13	56.767	0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000		
div_33_38	divide_unsigned_79_107	13	56.767	0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000		
inc_add_79_30	increment_unsigned_30	49	214.203	0.000	4.541	209.661
0.000	0.000	0.000	0.000	0.000		
rem_34_38	remainder_unsigned_73	14	80.231	0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000		
rem_35_38	remainder_unsigned_73_108	14	80.231	0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000		

Figure 34: Area report after placement for 0.8

Total Area = 4136.458 μm^2

GATES REPORT:

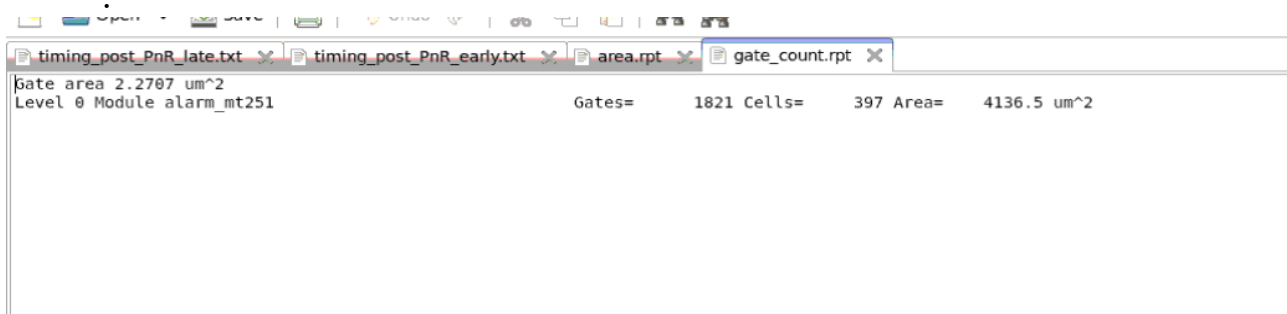


Figure 35: Gate count after placement for 0.8

Total Gates = 1821

POWER REPORT:

Open

Save

Undo

Redo

Print

Copy

Paste

Find

Help

timing_post_PnR_late.txt timing_post_PnR_early.txt area.rpt gate_count.rpt power.rpt

g3561	1.387e-08	2.089e-09	1.656e-05	1.655e-05	NOR48XL
g2023	1.767e-09	7.015e-10	1.511e-05	1.511e-05	NAND3X1
g3427	2.646e-10	1.872e-10	1.511e-05	1.511e-05	NAND3X1
g3462	4.159e-07	1.837e-07	1.47e-05	1.41e-05	OAI211X1
inc_add_79_30/g788	1.457e-06	7.101e-07	1.453e-05	1.236e-05	NOR2XL
g3450	1.157e-06	4.942e-07	1.401e-05	1.236e-05	NOR2XL
g3452	7.798e-07	7.765e-07	1.391e-05	1.236e-05	NOR2XL
g3478	3.475e-09	2.797e-08	1.305e-05	1.301e-05	CLKINVX1
g3454	7.257e-07	3.007e-06	1.302e-05	9.289e-06	NAND2XL
inc_add_79_30/g768	1.331e-06	2.336e-06	1.296e-05	9.289e-06	NAND2XL
g3438	1.082e-07	3.944e-08	1.251e-05	1.236e-05	NOR2XL
g3442	6.957e-08	6.936e-08	1.25e-05	1.236e-05	NOR2XL
g3431	7.851e-09	8.77e-09	1.237e-05	1.236e-05	NOR2XL
g3429	1.114e-09	1.614e-09	1.236e-05	1.236e-05	NOR2XL
g3426	1.343e-10	1.388e-10	1.236e-05	1.236e-05	NOR2XL
inc_add_79_30/g782	3.745e-07	9.231e-07	1.059e-05	9.289e-06	NAND2XL
g1991	5.568e-10	5.318e-10	1.046e-05	1.046e-05	NAND4XL
g3443	3.012e-10	2.644e-10	1.046e-05	1.046e-05	NAND4XL
g3445	6.801e-08	2.553e-07	9.613e-06	9.289e-06	NAND2XL
inc_add_79_30/g766	9.92e-08	1.888e-07	9.577e-06	9.289e-06	NAND2XL
inc_add_79_30/g779	2.851e-08	7.079e-08	9.389e-06	9.289e-06	NAND2XL
g2056	3.943e-07	2.032e-07	9.354e-06	8.757e-06	INVXL
g2030	1.818e-08	1.032e-08	9.318e-06	9.289e-06	NAND2XL
g3435	4.225e-09	1.823e-08	9.312e-06	9.289e-06	NAND2XL
inc_add_79_30/g764	6.63e-09	1.14e-08	9.307e-06	9.289e-06	NAND2XL
g1450	6.792e-11	5.302e-11	8.757e-06	8.757e-06	INVXL
g1989	1.367e-09	9.031e-10	8.731e-06	8.728e-06	OAI21XL

Total (397 of 397)	0.3656	0.07438	0.4628	0.0229
Total Capacitance	5.043e-12 F			
Power Density	*** No Die Area ***			

Plain Text Tab Width: 8

Figure 36: Power report after placement for 0.8

Total Power consumption = 462.8 uW

LAYOUT:

Figure below shows the layout of design and connectivity showing fly-lines.

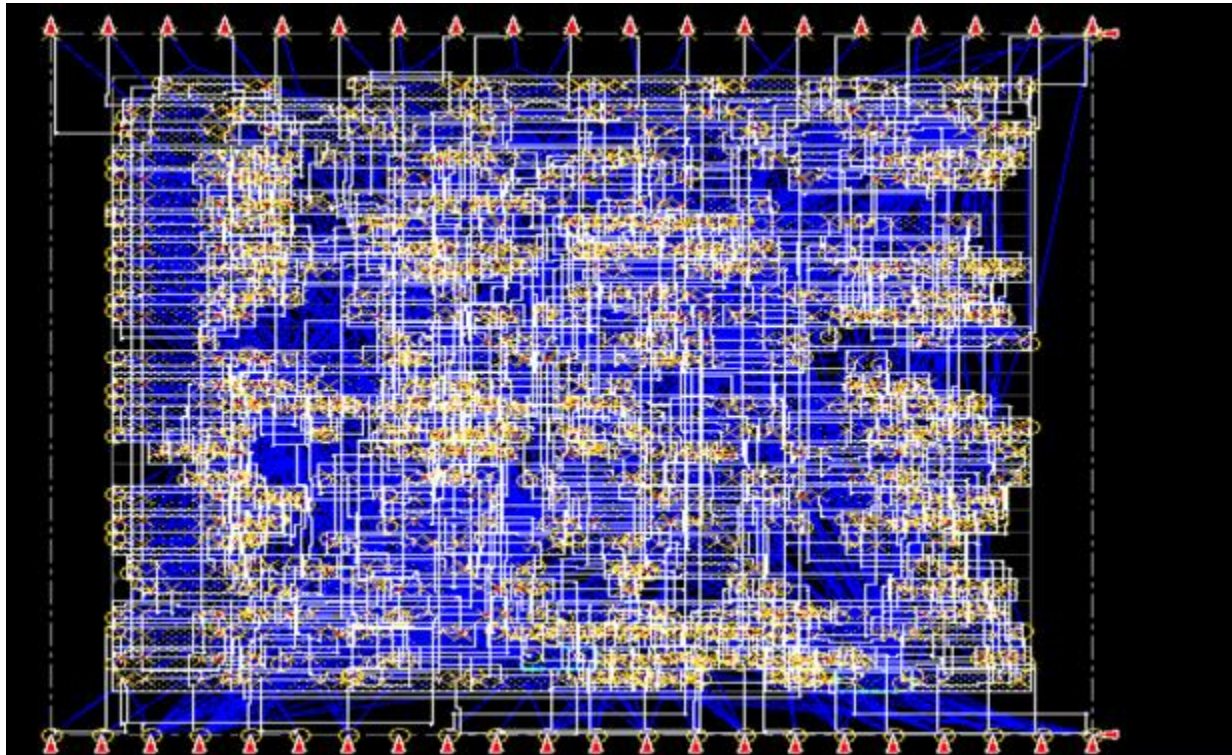


Figure 37: Layout after placement showing fly-lines for 0.8

CLOCK TREE SYNTHESIS (CTS)

Clock tree synthesis (CTS) decides the topology of clock network and path of clock to reach every clocked element. It performs wiring of clock network. It ensures minimum skew and symmetrical structure. It also tries to minimize the power dissipation as clock consumed large amount of power.

FOR CORE UTILIZATION 0.5

SETUP TIMING REPORT:

```
# Generated by: Cadence innovus 20.10-p004_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 06:35:19 2025
# Design: alarm_mt251
# Command: report_timing -late -max_paths 100 > optimised_cts_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time -0.003
- Setup 0.303
+ Phase Shift 10.000
- Uncertainty 0.100
= Required Time 9.594
- Arrival Time 4.808
= Slack Time 4.786
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.003
= Beginpoint Arrival Time -0.003
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
t1_cs11_reg	CK ^			-0.003	4.783
t1_cs11_reg	CK ^ -> Q ^	SDFFRHQX1	1.000	0.997	5.783
g3469	A ^ -> Y v	NAND2XL	0.499	1.496	6.283
g3464	B v -> Y ^	NOR3BX1	0.232	1.729	6.515
g3460	B ^ -> Y v	NAND2XL	0.345	2.074	6.860
g3458	C v -> Y ^	NOR3X1	0.284	2.358	7.144
g3454	B ^ -> Y v	NAND2XL	0.342	2.699	7.485

Figure 38: Setup Report after CTS for 0.5

HOLD TIMING REPORT:

```

# Generated on: Sun Nov 9 06:35:19 2025
# Design: alarm_mt251
# Command: report_timing -early -view {view1} -max_paths 100 > optimised_cts_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Hold Check with Pin ic_q_rp38_reg[1]/CK
Endpoint: ic_q_rp38_reg[1]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time -0.004
+ Hold 0.075
+ Phase Shift 0.000
+ Uncertainty 0.100
= Required Time 0.172
Arrival Time 0.213
Slack Time 0.041
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
+-----+
| Instance | Arc | Cell | Delay | Arrival | Required | |
|---|---|---|---|---|---|---|
| FE_PHC5_scan_en | scan_en v | A v -> Y v | DLY1X1 | 0.213 | 0.213 | 0.172 |
| ic_q_rp38_reg[1] | SE v | SDDFRHQX1 | 0.000 | 0.213 | 0.213 | 0.172 |
+-----+
Path 2: MET Hold Check with Pin bs_q_cs11_reg[1]/CK
Endpoint: bs_q_cs11_reg[1]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'

```

Figure 39: Hold Report after CTS for 0.5

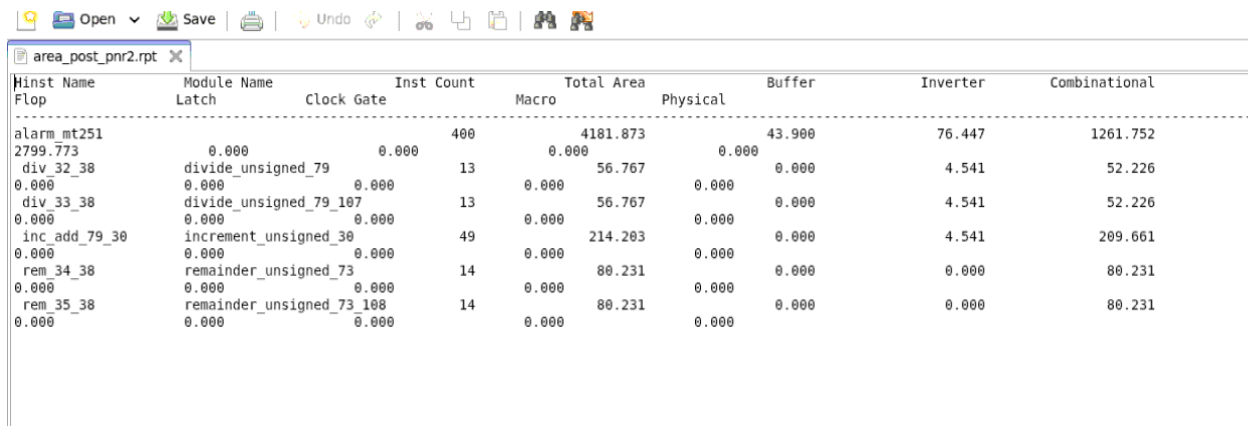
Table below compares worst setup and hold slack before and after CTS.

Table 5: Worst Setup and Hold slack before and after CTS for 0.5

	Setup Slack (ps)	Hold Slack (ps)
Before CTS	5.812	-0.179
After CTS	4.786	0.041

For the 0.5 floorplan, the setup slack decreased slightly from 5.812 ps to 4.786 ps after CTS because the insertion of the clock tree added clock latency and skew, slightly increasing the setup path delay. However, the hold slack improved from -0.179 ps to 0.041 ps, turning positive, as clock tree buffering helped balance the clock arrival times at launch and capture flip-flops, reducing early data arrival issues.

AREA REPORT:



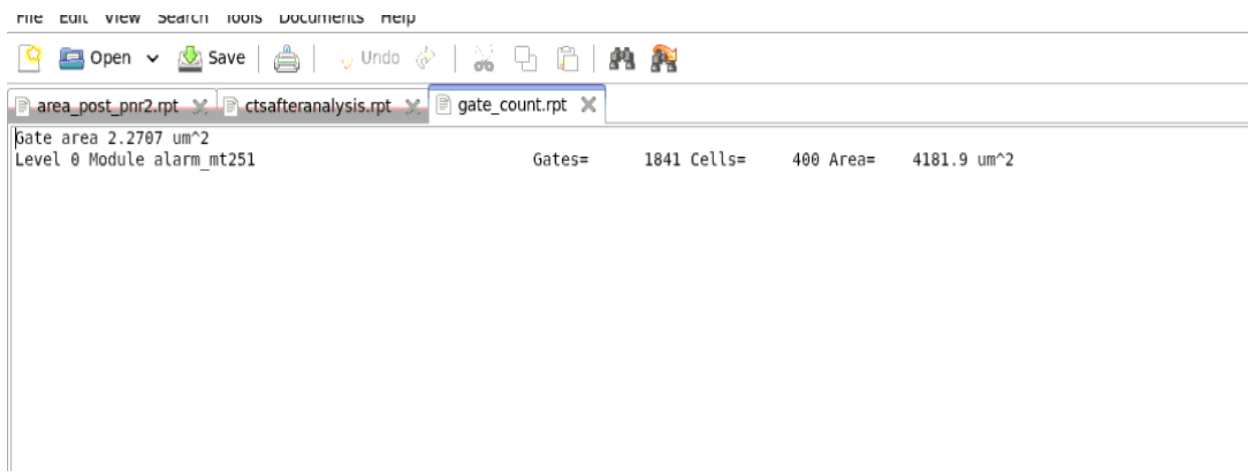
Hinst Name	Flop	Module Name	Latch	Clock Gate	Inst Count	Macro	Total Area	Physical	Buffer	Inverter	Combinational
alarm_mt251					400		4181.873		43.900	76.447	1261.752
2799.773				0.000		0.000		0.000			
div_32_38		divide_unsigned_79			13		56.767		0.000	4.541	52.226
0.000		0.000		0.000		0.000		0.000			
div_33_38		divide_unsigned_79_107			13		56.767		0.000	4.541	52.226
0.000		0.000		0.000		0.000		0.000			
inc_add_79_30		increment_unsigned_30			49		214.203		0.000	4.541	209.661
0.000		0.000		0.000		0.000		0.000			
rem_34_38		remainder_unsigned_73			14		80.231		0.000	0.000	80.231
0.000		0.000		0.000		0.000		0.000			
rem_35_38		remainder_unsigned_73_108			14		80.231		0.000	0.000	80.231
0.000		0.000		0.000		0.000		0.000			

Figure 40: Area report after CTS for 0.5

Total Area = 4181.873 um^2

A noticeable increase in buffer area, and consequently the overall chip area, is observed after Clock Tree Synthesis (CTS) compared to before CTS. This increase occurs due to the insertion of the clock tree, which is designed to minimize clock skew across the circuit.

GATES REPORT:



Gate area	Level	Module	Gates	Cells	Area
2.2707 um^2	0	alarm_mt251	1841	400	4181.9 um^2

Figure 41: Gate count after CTS for 0.5

Total Gates = 1841

POWER REPORT:

g3450	1.116e-06	4.936e-07	1.397e-05	1.236e-05	NOR2XL
g3452	7.494e-07	7.428e-07	1.385e-05	1.236e-05	NOR2XL
g3478	1.917e-09	1.722e-08	1.303e-05	1.301e-05	CLKINVX1
inc_add_79_30/g788	4.068e-07	1.95e-07	1.296e-05	1.236e-05	NOR2XL
g3438	9.298e-08	3.516e-08	1.249e-05	1.236e-05	NOR2XL
g3442	6.007e-08	5.512e-08	1.247e-05	1.236e-05	NOR2XL
g3454	6.642e-07	2.468e-06	1.242e-05	9.289e-06	NAND2XL
g3431	6.77e-09	8.085e-09	1.237e-05	1.236e-05	NOR2XL
g3429	9.286e-10	1.459e-09	1.236e-05	1.236e-05	NOR2XL
g3426	1.102e-10	1.254e-10	1.236e-05	1.236e-05	NOR2XL
g1991	4.929e-10	3.927e-10	1.046e-05	1.046e-05	NAND4XL
g3443	2.027e-10	2.51e-10	1.046e-05	1.046e-05	NAND4XL
inc_add_79_30/g768	3.664e-07	6.662e-07	1.032e-05	9.289e-06	NAND2XL
inc_add_79_30/g782	1.048e-07	3.022e-07	9.696e-06	9.289e-06	NAND2XL
g3445	5.704e-08	2.513e-07	9.598e-06	9.289e-06	NAND2XL
inc_add_79_30/g766	2.777e-08	5.027e-08	9.367e-06	9.289e-06	NAND2XL
g2056	3.749e-07	2.353e-07	9.367e-06	8.757e-06	INVXL
inc_add_79_30/g779	8.258e-09	2.378e-08	9.321e-06	9.289e-06	NAND2XL
g2030	1.558e-08	6.911e-09	9.312e-06	9.289e-06	NAND2XL
g3435	3.599e-09	1.682e-08	9.31e-06	9.289e-06	NAND2XL
inc_add_79_30/g764	2.031e-09	3.511e-09	9.295e-06	9.289e-06	NAND2XL
g1450	5.631e-11	4.978e-11	8.757e-06	8.757e-06	INVXL
g1989	1.15e-09	6.832e-10	8.73e-06	8.728e-06	OAI21XL

Total (400 of 400)	0.3062	0.06297	0.3926	0.0234	
Total Capacitance	5.476e-12 F				
Power Density	*** No Die Area ***				

Plain Text ▾ Tab Width: 8 ▾ Ln 1, Col 1

Figure 42: Power report after CTS for 0.5

Total Power = 392.6 uW

LAYOUT:

Figure below shows the layout of design showing clock tree.

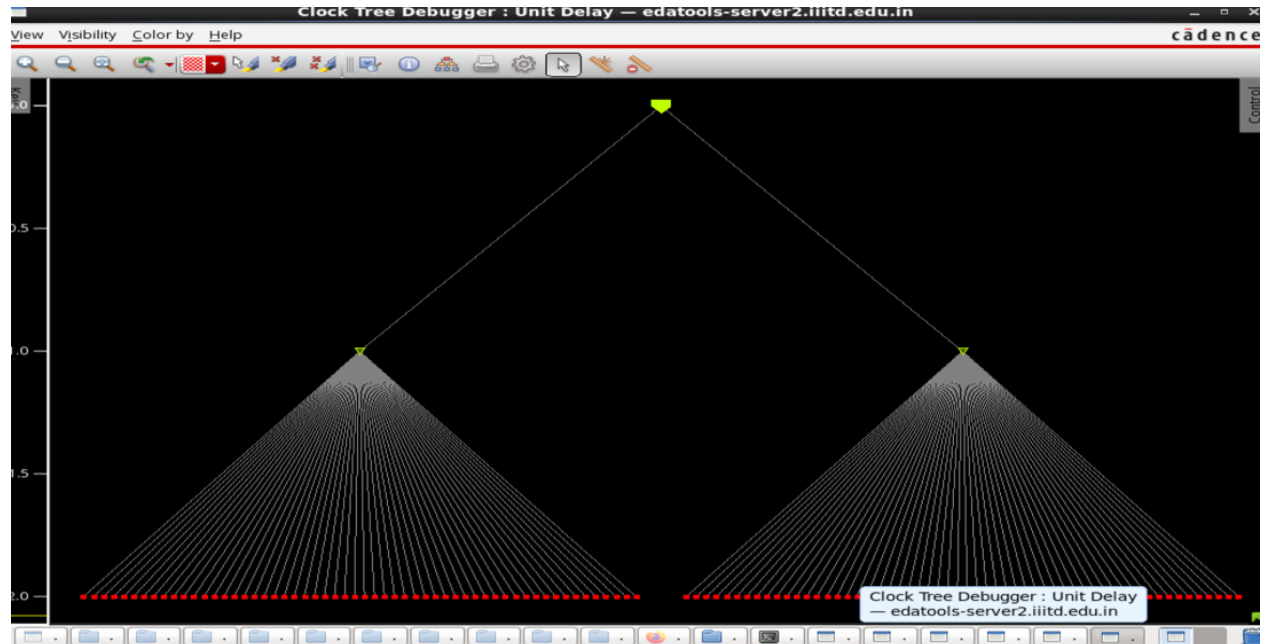


Figure 43: Layout after CTS showing clock tree for 0.5

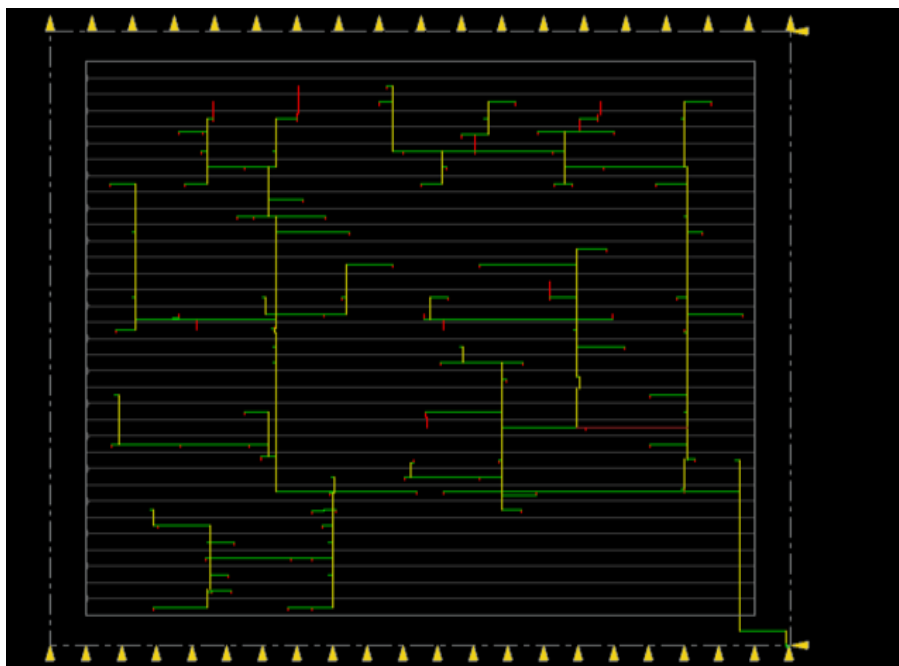


Figure 44: Layout after CTS showing clock tree for 0.5

SETUP TIMING REPORT:



HOLD TIMING REPORT:

```
#####
# Generated by: Cadence Innovus 20.10-p004_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 06:47:20 2025
# Design: alarm_mt251
# Command: report_timing -early -view {view1} -max_paths 100 > optimised_cts_reports/timing/timing_r
#####
Path 1: MET Hold Check with Pin bs_q_cs11_reg[1]/CK
Endpoint: bs_q_cs11_reg[1]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time -0.005
+ Hold 0.069
+ Phase Shift 0.000
+ Uncertainty 0.100
= Required Time 0.165
Arrival Time 0.222
Slack Time 0.057
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| scan_en v | scan_en v | CLKBUX2 | 0.096 | 0.096 | -0.057 |
| FE_PHC6_scan_en | A v -> Y v | CLKBUX2 | 0.125 | 0.222 | 0.165 |
| FE_PHC4_scan_en | A v -> Y v | CLKBUX2 | 0.125 | 0.222 | 0.165 |
| bs_q_cs11_reg[1] | SE v | SDDFRHGX1 | 0.000 | 0.222 | 0.165 |
+-----+-----+-----+-----+-----+-----+
Path 2: MET Hold Check with Pin is_q_pb34_reg[1]/CK
Endpoint: is_q_pb34_reg[1]/SE (v) checked with leading edge of 'clk_pb34'
```

Figure 46: Hold Report after CTS for 0.8

Table below compares worst setup and hold slack before and after CTS.

Table 6: Worst Setup and Hold slack before and after CTS for 0.8

	Setup Slack (ps)	Hold Slack (ps)
Before CTS	4.930	-0.181
After CTS	4.944	0.057

For this design, the **setup slack** slightly improved from **4.930 ps** to **4.944 ps** after CTS, showing that clock tree insertion had a minimal but positive effect on setup timing due to better clock balancing and reduced skew. The **hold slack** also improved from **-0.181 ps** to **0.057 ps**, becoming positive, which indicates that the CTS process effectively delayed the clock at the capture flip-flop or adjusted skew to fix early data arrival issues.

AREA REPORT:

File Edit View Search Tools Documents Help

timing_post_PnR_late.txt ctsafteranalysis.rpt area_post_pnr2.rpt

Hinst Name	Module Name	Inst Count	Total Area	Buffer	Inverter	Combinational
Flop	Latch	Clock Gate	Macro	Physical		
alarm_mt251		400	4175.060	36.331	77.204	1261.752
2799.773	0.000	0.000	0.000	0.000		
div_32_38	divide_unsigned_79	13	56.767	0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000		
div_33_38	divide_unsigned_79_107	13	56.767	0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000		
inc_add_79_30	increment_unsigned_30	49	214.203	0.000	4.541	209.661
0.000	0.000	0.000	0.000	0.000		
rem_34_38	remainder_unsigned_73	14	80.231	0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000		
rem_35_38	remainder_unsigned_73_108	14	80.231	0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000		

Figure 47: Area report after CTS for 0.8

Total Area = 4175.060 μm^2

There is significant increase in buffer area and thus overall area after CTS as compared to before CTS. This is because of addition of clock tree, which tries to minimize clock skew.

GATES REPORT:

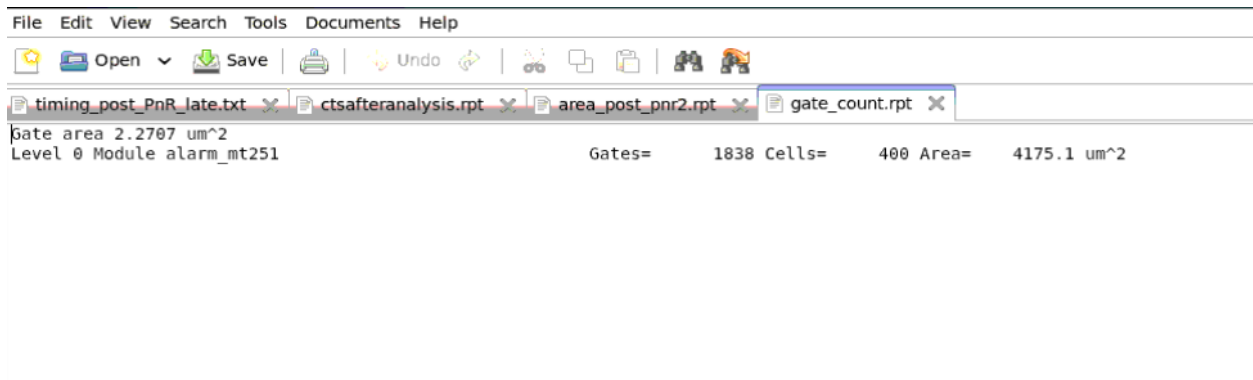


Figure 48: Gate count after CTS for 0.8

Total Gates = 1838

POWER REPORT:

timing_post_PnR_late.txt					
ctsafteranalysis.rpt					
g2023	1.545e-09	6.136e-10	1.511e-05	1.511e-05	NAND3X1
g3427	2.181e-10	1.606e-10	1.511e-05	1.511e-05	NAND3X1
inc_add_79_30/g776	1.296e-06	1.216e-06	1.487e-05	1.236e-05	NOR2XL
g3462	3.147e-07	1.346e-07	1.455e-05	1.41e-05	OAI211X1
g3450	1.129e-06	4.81e-07	1.397e-05	1.236e-05	NOR2XL
g3452	7.556e-07	7.509e-07	1.386e-05	1.236e-05	NOR2XL
g3478	1.895e-09	1.521e-08	1.303e-05	1.301e-05	CLKINVX1
inc_add_79_30/g788	4.09e-07	1.963e-07	1.296e-05	1.236e-05	NOR2XL
g3454	6.634e-07	2.709e-06	1.266e-05	9.289e-06	NAND2XL
g3438	9.106e-08	3.294e-08	1.248e-05	1.236e-05	NOR2XL
g3442	5.894e-08	5.836e-08	1.248e-05	1.236e-05	NOR2XL
g3431	6.732e-09	7.44e-09	1.237e-05	1.236e-05	NOR2XL
g3429	9.228e-10	1.332e-09	1.236e-05	1.236e-05	NOR2XL
g3426	1.103e-10	1.121e-10	1.236e-05	1.236e-05	NOR2XL
g1991	4.854e-10	4.535e-10	1.046e-05	1.046e-05	NAND4XL
g3443	2.888e-10	2.543e-10	1.046e-05	1.046e-05	NAND4XL
inc_add_79_30/g768	3.665e-07	6.424e-07	1.03e-05	9.289e-06	NAND2XL
inc_add_79_30/g782	1.049e-07	2.555e-07	9.65e-06	9.289e-06	NAND2XL
g3445	5.718e-08	2.148e-07	9.561e-06	9.289e-06	NAND2XL
inc_add_79_30/g766	2.773e-08	5.171e-08	9.369e-06	9.289e-06	NAND2XL
g2056	3.822e-07	1.948e-07	9.334e-06	8.757e-06	INVXL
inc_add_79_30/g779	8.279e-09	2.048e-08	9.318e-06	9.289e-06	NAND2XL
g2030	1.54e-08	8.582e-09	9.313e-06	9.289e-06	NAND2XL
g3435	3.614e-09	1.541e-08	9.308e-06	9.289e-06	NAND2XL
inc_add_79_30/g764	2.027e-09	3.479e-09	9.295e-06	9.289e-06	NAND2XL
g1450	5.563e-11	4.323e-11	8.757e-06	8.757e-06	INVXL
g1989	1.141e-09	7.362e-10	8.73e-06	8.728e-06	OAI21XL

Total (400 of 400)	0.3002	0.06098	0.3846	0.02341	
Total Capacitance	5.337e-12 F				
Power Density	*** No Die Area ***				

Figure 49: Power report after CTS for 0.8

$$\text{Power} = 384.6 \text{ uW}$$

LAYOUT:

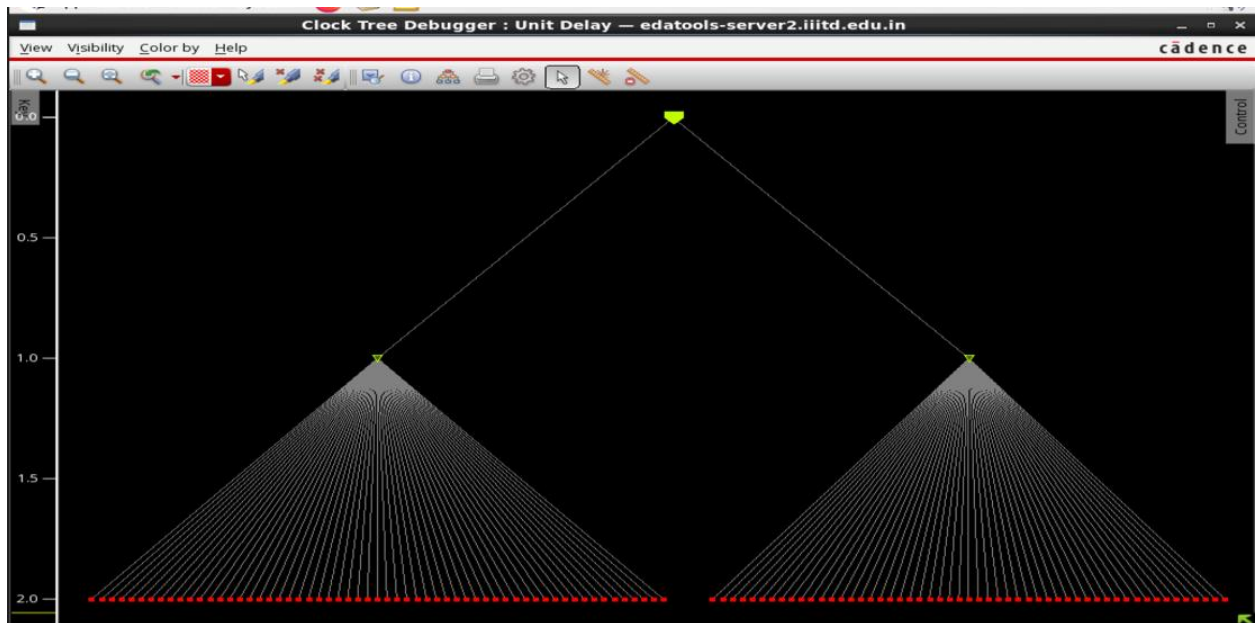


Figure 50: Layout after CTS showing clock tree for 0.8

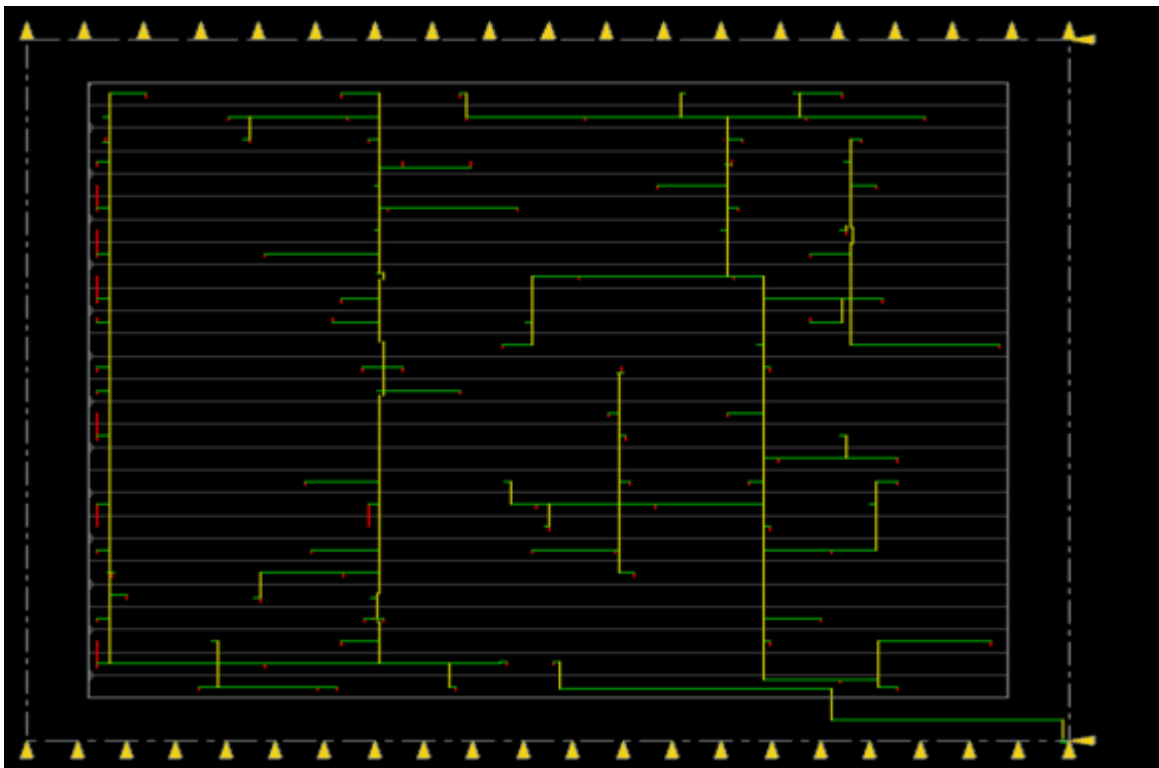


Figure 51: Layout after CTS showing clock tree for 0.8

DETAILED ROUTING

In this stage, the routing of all signal nets—excluding clock and power lines—is performed while meeting design constraints. The objective is to minimize total wire length and efficiently utilize resources such as vias.

FOR CORE UTILIZATION 0.5

SETUP TIMING REPORT:

```
# OS: Linux x86_64 (HOST ID: edatools-server2.111td.edu.in)
# Generated on: Sun Nov 9 06:35:28 2025
# Design: alarm_mt251
# Command: report_timing -late -max_paths 100 > post_route_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time -0.004
- Setup 0.305
+ Phase Shift 10.000
- Uncertainty 0.100
= Required Time 9.591
- Arrival Time 4.806
= Slack Time 4.785
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.004
= Beginpoint Arrival Time -0.004
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
| | | | | Time | Time |
+-----+-----+-----+-----+-----+-----+
| t1_cs11_reg | CK ^ | | | -0.004 | 4.781 |
| t1_cs11_reg | CK ^ -> Q ^ | S0FFRHQX1 | 1.001 | 0.997 | 5.782 |
| g3469 | A ^ -> Y v | NAND2XL | 0.504 | 1.501 | 6.286 |
| g3464 | B v -> Y ^ | NOR3BX1 | 0.229 | 1.730 | 6.515 |
| g3460 | B ^ -> Y v | NAND2XL | 0.346 | 2.076 | 6.861 |
| g3458 | C v -> Y ^ | NOR3X1 | 0.285 | 2.361 | 7.146 |
| g3454 | B ^ -> Y v | NAND2XL | 0.340 | 2.701 | 7.486 |
```

Figure 52: Setup Report after detailed routing for 0.5

HOLD TIMING REPORT:

# Design: alarm_mt251					
# Command: report_timing -early -view {view1} -max_paths 100 > post_route_reports/timing/timing_post_PnR_early.t					
#####					
Path 1: MET Hold Check with Pin ic_q_rp38_reg[1]/CK					
Endpoint: ic_q_rp38_reg[1]/SE (v) checked with leading edge of 'clk_pb34'					
Beginpoint: scan_en (v) triggered by leading edge of '@'					
Path Groups: {clk_pb34}					
Analysis View: view1					
Other End Arrival Time -0.002					
+ Hold 0.078					
+ Phase Shift 0.000					
+ Uncertainty 0.100					
= Required Time 0.176					
Arrival Time 0.212					
Slack Time 0.036					
Clock Rise Edge 0.000					
+ Input Delay 0.000					
= Beginpoint Arrival Time 0.000					
+-----+-----+-----+-----+-----+-----+					
Instance Arc Cell Delay Arrival Required					
Time Time					
+-----+-----+-----+-----+-----+-----+					
FE_PHC5_scan_en scan_en v DLY1X1 0.212 0.000 -0.036					
A v -> Y v					
ic_q_rp38_reg[1] SE v SDFFRHGX1 0.000 0.212 0.176					
+-----+-----+-----+-----+-----+-----+					
Path 2: MET Hold Check with Pin bs_q_cs11_reg[1]/CK					
Endpoint: bs_q_cs11_reg[1]/SE (v) checked with leading edge of 'clk_pb34'					
Beginpoint: scan_en (v) triggered by leading edge of '@'					

Plain Te					

Figure 53: Hold Report after detailed routing for 0.5

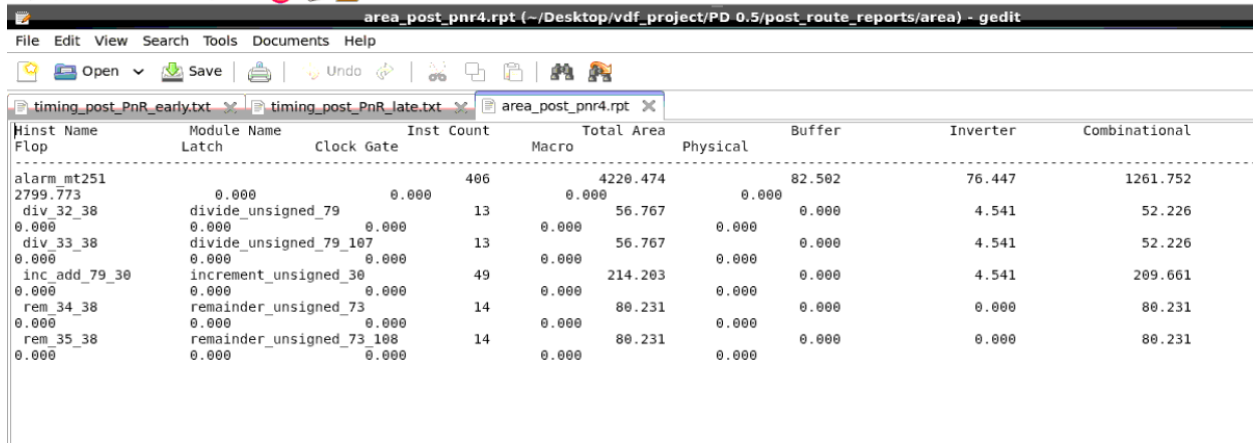
Table below compares worst setup and hold slack before and after detailed routing.

Table 7: Worst Setup and Hold slack before and after detailed routing for 0.5

	Setup Slack (ps)	Hold Slack (ps)
Before routing	4.786	0.041
After routing	4.785	0.036

After detailed routing, the **setup slack** changed slightly from **4.786 ps** to **4.785 ps**, and the **hold slack** decreased marginally from **0.041 ps** to **0.036 ps**. These minimal variations indicate that routing parasitics, such as wire resistance and capacitance, had a very small impact on the timing. Overall, both setup and hold timings remained almost the same, showing that the routing stage was well-optimized with stable connectivity and no significant degradation in timing performance.

AREA REPORT:



Hinst Name	Flop	Module Name	Latch	Clock Gate	Inst Count	Macro	Total Area	Physical	Buffer	Inverter	Combinational
alarm_mt251					406		4220.474		82.502	76.447	1261.752
2799.773		0.000		0.000		0.000	0.000	0.000			
div_32_38		divide_unsigned_79			13		56.767		0.000	4.541	52.226
0.000		0.000		0.000		0.000	56.767	0.000	0.000	4.541	52.226
div_33_38		divide_unsigned_79_107			13		56.767	0.000	0.000	4.541	52.226
0.000		0.000		0.000		0.000	214.203	0.000	0.000	4.541	209.661
inc_add_79_30		increment_unsigned_30			49		0.000	0.000	0.000		
0.000		0.000		0.000		0.000	80.231	0.000	0.000	0.000	80.231
rem_34_38		remainder_unsigned_73			14		0.000	0.000	0.000	0.000	80.231
0.000		0.000		0.000		0.000	80.231	0.000	0.000	0.000	80.231
rem_35_38		remainder_unsigned_73_108			14		0.000	0.000	0.000	0.000	80.231
0.000		0.000		0.000		0.000	0.000	0.000	0.000	0.000	80.231

Figure 54: Area report after detailed routing for 0.5

Total Area = 4220.474 μm^2

GATES REPORT:

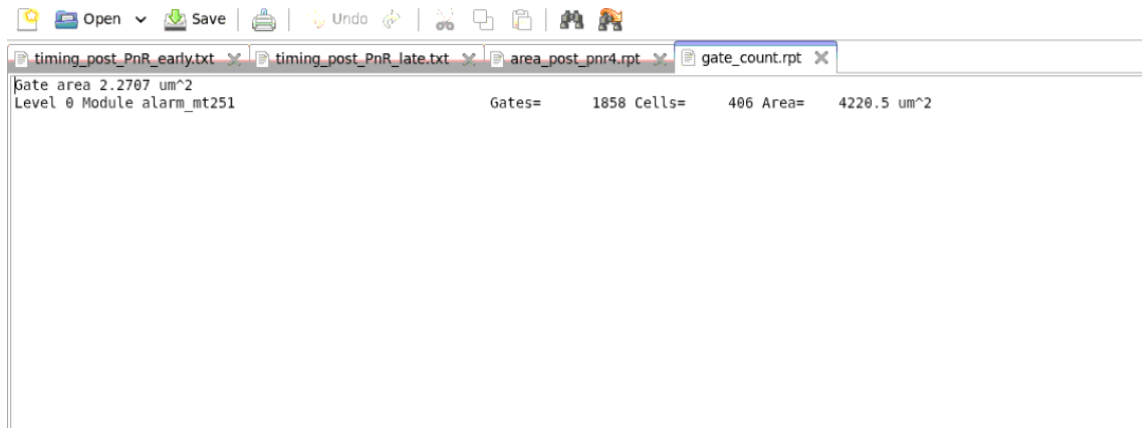


Figure 55: Gate count after detailed routing for 0.5

Total Gates = 1858

POWER REPORT:

g3492	2.515e-08	1.781e-08	1.455e-05	1.840e-05	NAND4XL
g3462	3.159e-07	1.33e-07	1.455e-05	1.41e-05	OAI211X1
g3450	1.118e-06	4.555e-07	1.393e-05	1.236e-05	NOR2XL
g3452	7.495e-07	7.263e-07	1.383e-05	1.236e-05	NOR2XL
g3478	1.915e-09	1.729e-08	1.303e-05	1.301e-05	CLKINVX1
inc_add_79_30/g788	4.072e-07	1.945e-07	1.296e-05	1.236e-05	NOR2XL
g3438	9.253e-08	3.479e-08	1.249e-05	1.236e-05	NOR2XL
g3442	5.989e-08	5.372e-08	1.247e-05	1.236e-05	NOR2XL
g3454	6.644e-07	2.45e-06	1.24e-05	9.289e-06	NAND2XL
g3431	6.783e-09	7.906e-09	1.237e-05	1.236e-05	NOR2XL
g3429	9.29e-10	1.476e-09	1.236e-05	1.236e-05	NOR2XL
g3426	1.102e-10	1.282e-10	1.236e-05	1.236e-05	NOR2XL
g1991	4.922e-10	3.878e-10	1.046e-05	1.046e-05	NAND4XL
g3443	2.829e-10	2.479e-10	1.046e-05	1.046e-05	NAND4XL
inc_add_79_30/g768	3.666e-07	6.627e-07	1.032e-05	9.289e-06	NAND2XL
inc_add_79_30/g782	1.048e-07	3.019e-07	9.696e-06	9.289e-06	NAND2XL
g3445	5.705e-08	2.431e-07	9.59e-06	9.289e-06	NAND2XL
inc_add_79_30/g766	2.776e-08	4.968e-08	9.367e-06	9.289e-06	NAND2XL
g2056	3.739e-07	2.287e-07	9.36e-06	8.757e-06	INVXL
inc_add_79_30/g779	8.261e-09	2.523e-08	9.323e-06	9.289e-06	NAND2XL
g2030	1.549e-08	6.844e-09	9.312e-06	9.289e-06	NAND2XL
g3435	3.595e-09	1.691e-08	9.31e-06	9.289e-06	NAND2XL
inc_add_79_30/g764	2.032e-09	3.489e-09	9.295e-06	9.289e-06	NAND2XL
g1450	5.662e-11	4.807e-11	8.757e-06	8.757e-06	INVXL
g1989	1.15e-09	7.067e-10	8.73e-06	8.728e-06	OAI21XL

Total (406 of 406)	0.3103	0.0653	0.3993	0.02368	
Total Capacitance	5.762e-12 F				
Power Density	*** No Die Area ***				

Plain Text v

Figure 56: Power report after detailed routing for 0.5

Total power consumption = 399.3uW

LAYOUT:

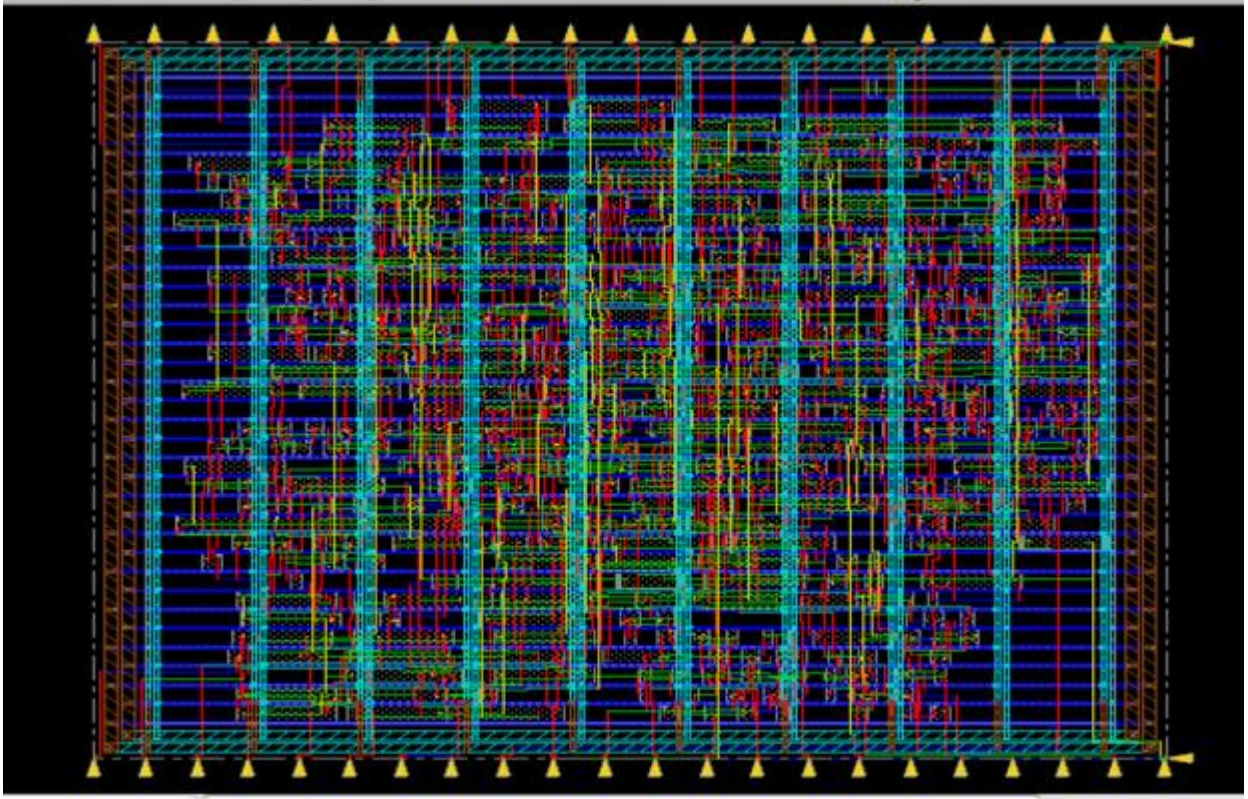


Figure 59: Layout after detailed routing for 0.5

METAL 1:

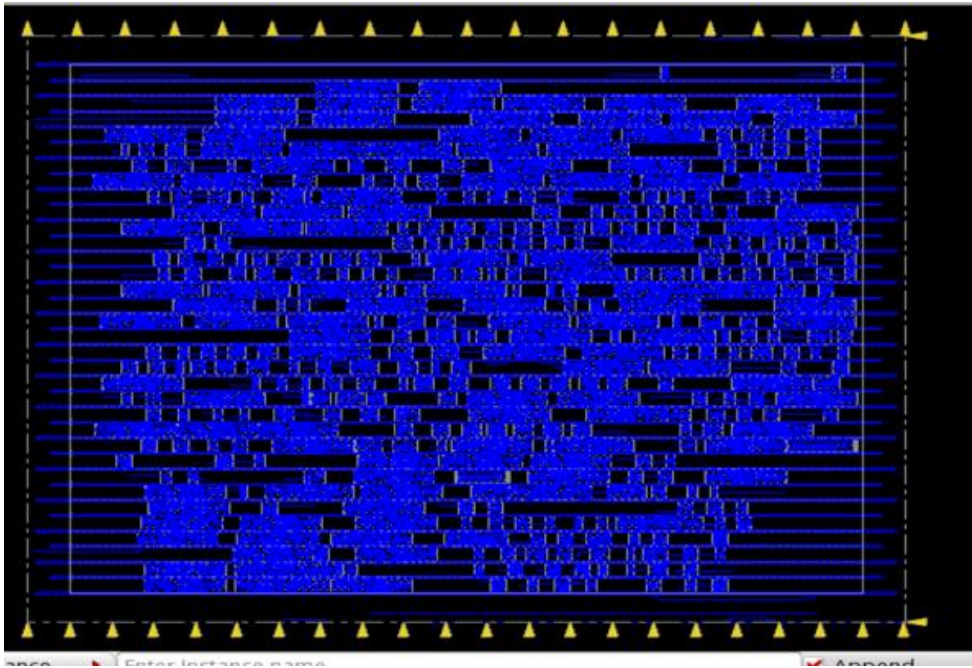


Figure 60: Layout showing metal layer 1 for 0.5

METAL 2:

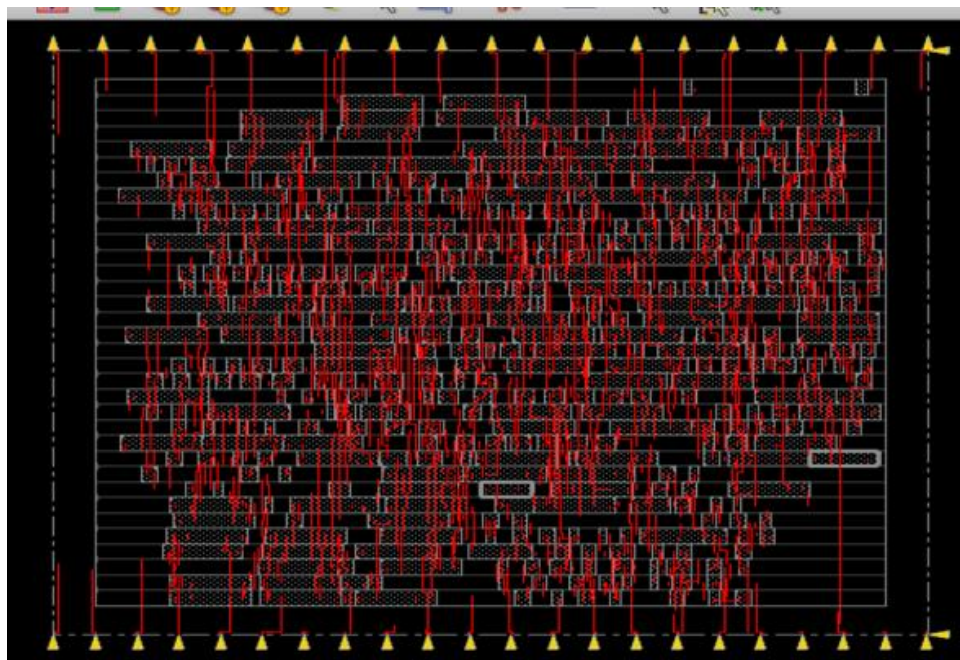


Figure 61: Layout showing metal layer 2 for 0.5

METAL 3:

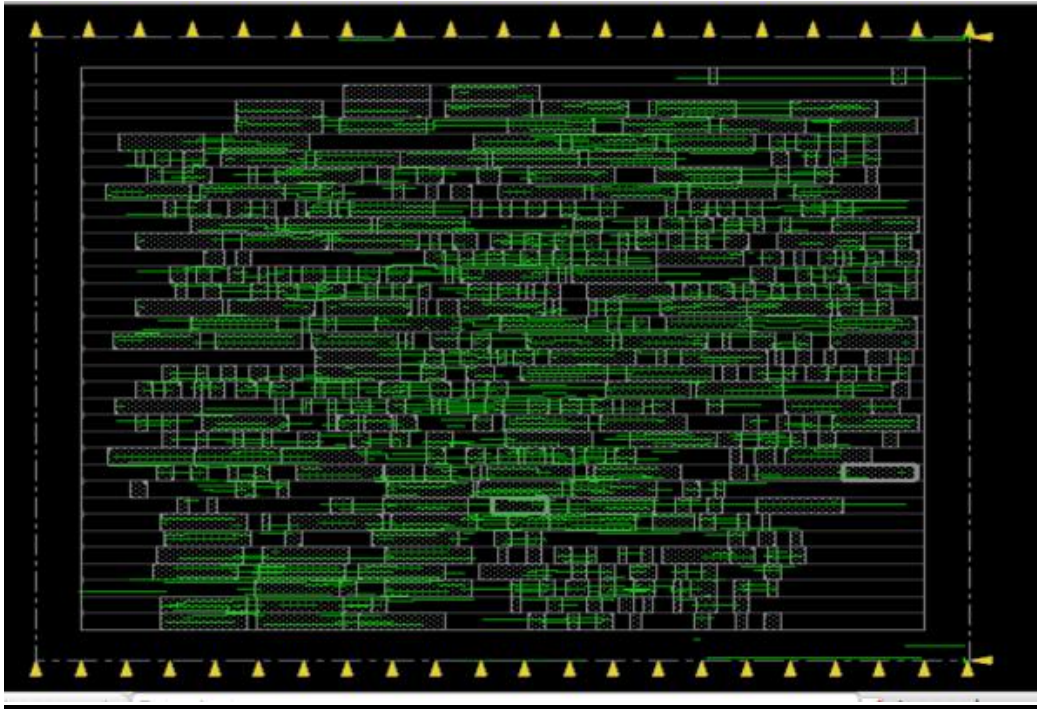


Figure 62: Layout showing metal layer 3 for 0.5

METAL 4:

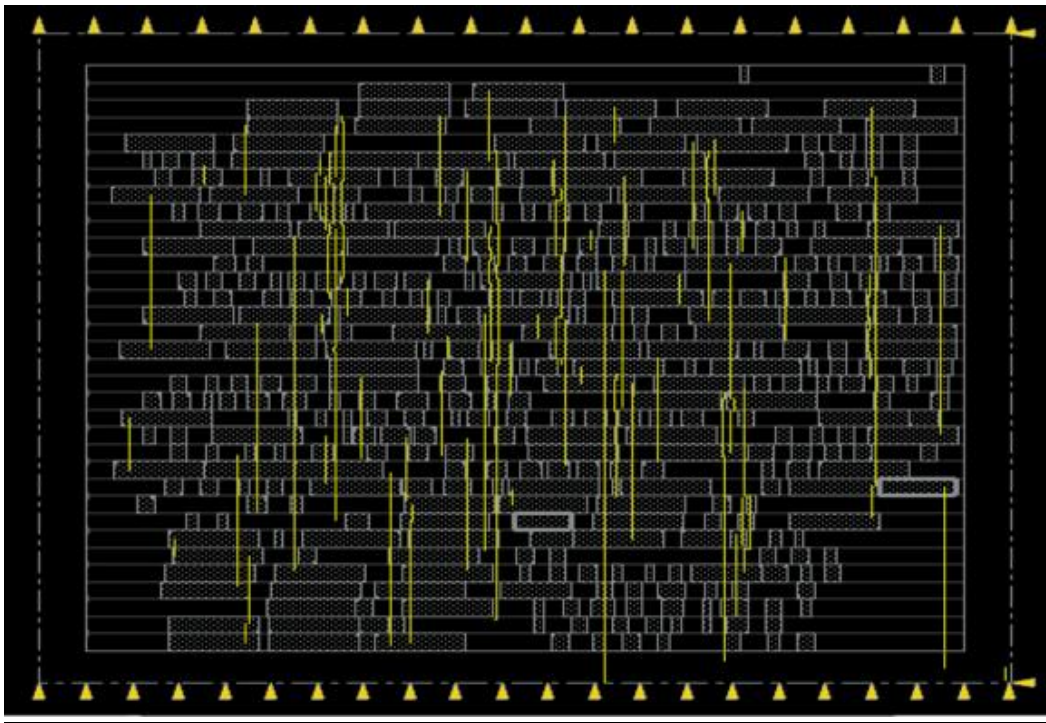


Figure 63: Layout showing metal layer 4 for 0.5

METAL 5:

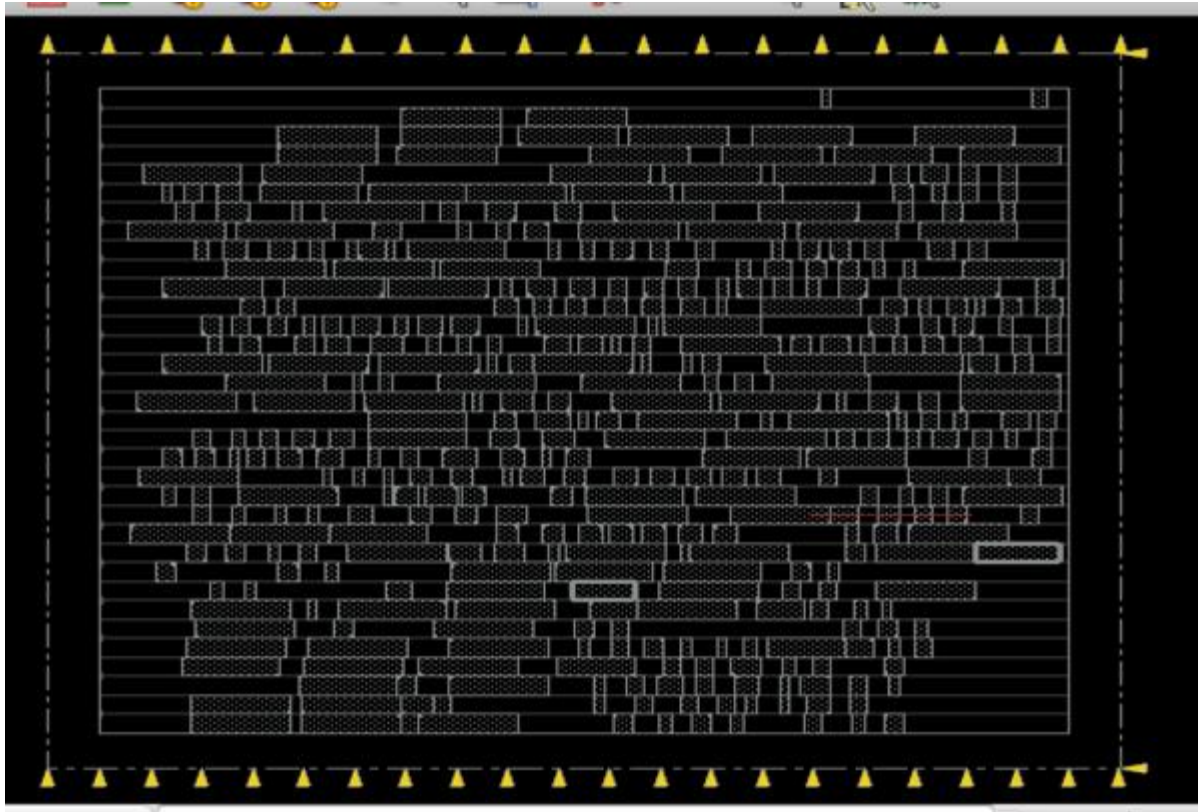


Figure 64: Layout showing metal layer 5 for 0.5

METAL 6:

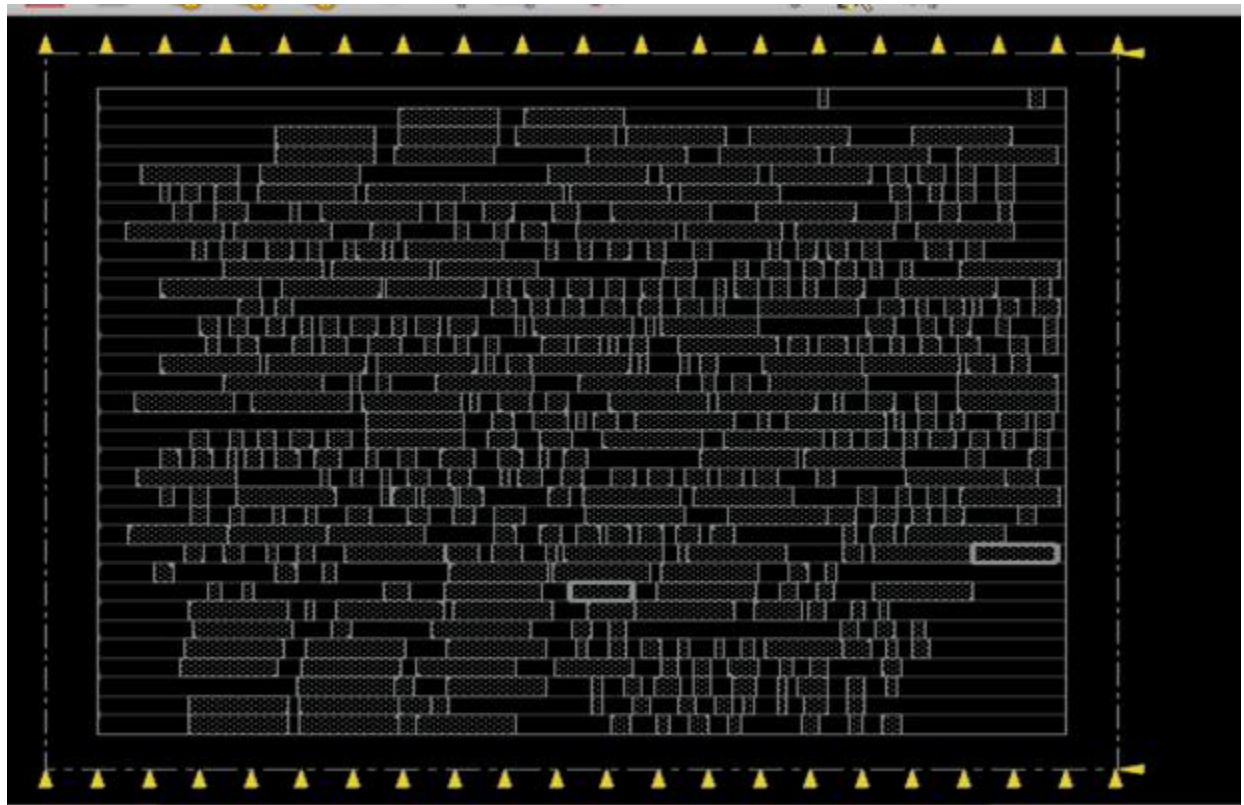


Figure 65: Layout showing metal layer 6 for 0.5

METAL 7:

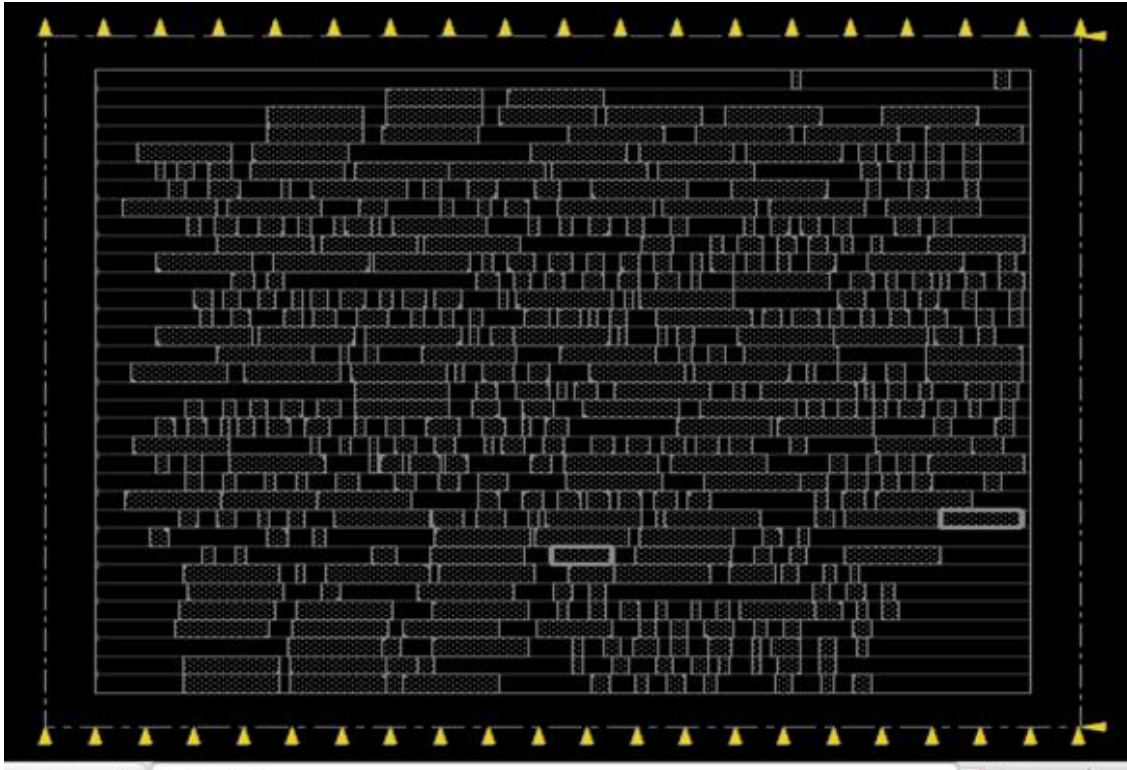


Figure 66: Layout showing metal layer 7 for 0.5

METAL 8:

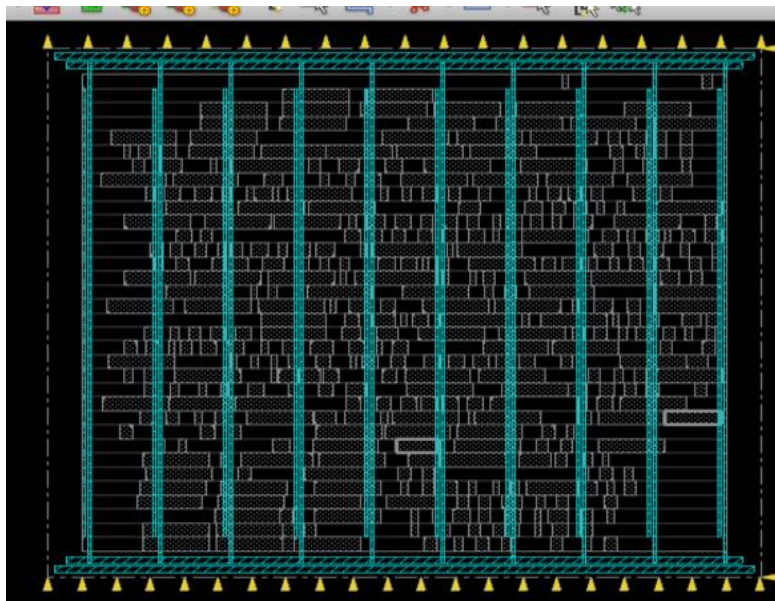


Figure 67: Layout showing metal layer 8 for 0.5

METAL 9:

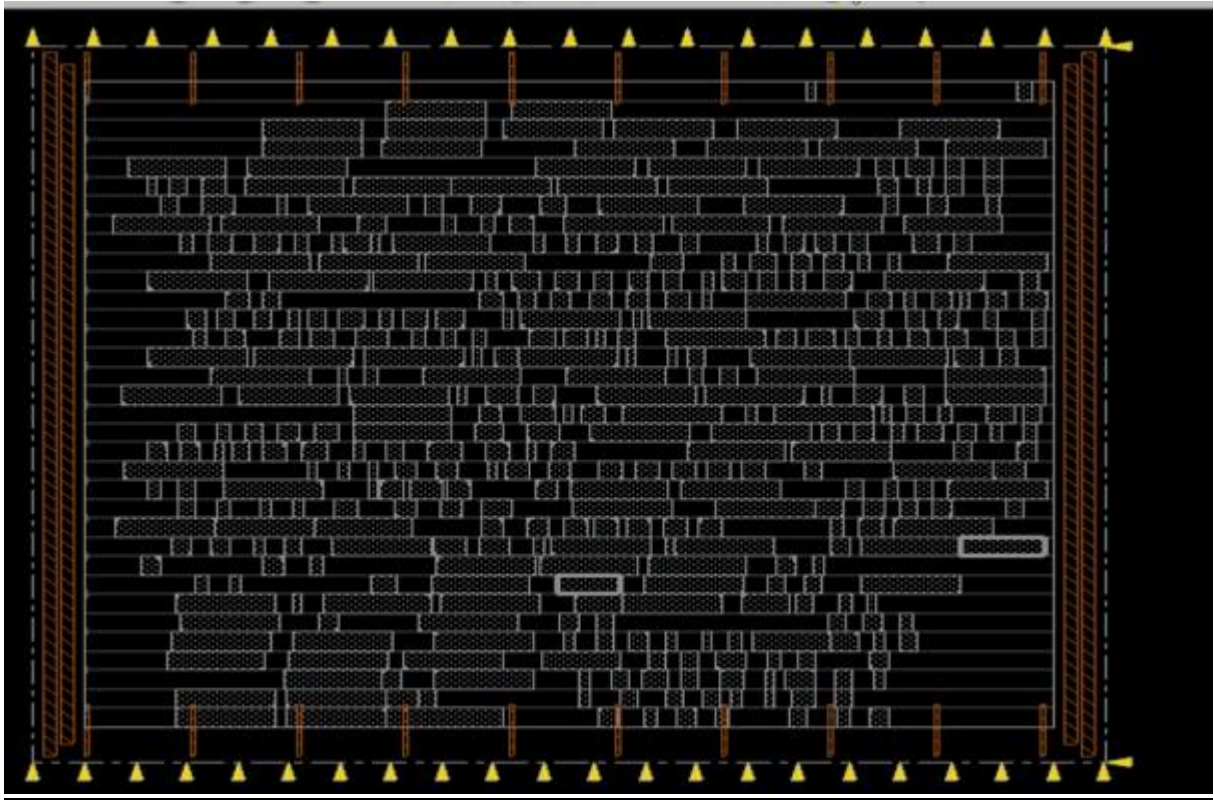


Figure 68: Layout showing metal layer 9 for 0.5

Different metal layers affect **path timing** due to their varying **resistance (R)** and **capacitance (C)** characteristics. **Lower metal layers** (like M1–M3) are thinner and closer to the substrate, having **higher resistance and capacitance**, which increases delay and is suitable for short local connections. **Middle layers** (M4–M6) are used for intermediate routing as they offer a balance between area and performance. **Upper metal layers** (M7–M9) are thicker and wider with **low resistance and capacitance**, making them ideal for **long or critical paths**, power, and clock routing since they minimize RC delay and signal degradation.

FOR CORE UTILIZATION 0.8

SETUP TIMING REPORT:

```
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 06:47:27 2025
# Design: alarm_mt251
# Command: report_timing -late -max_paths 100 > post_route_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin h1_rp38_reg[3]/CK
Endpoint: h1_rp38_reg[3]/D (^) checked with leading edge of 'clk_pb34'
Beginpoint: t1_cs11_reg/Q (^) triggered by leading edge of 'clk_pb34'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time -0.003
- Setup 0.291
+ Phase Shift 10.000
- Uncertainty 0.100
= Required Time 9.606
- Arrival Time 4.654
= Slack Time 4.952
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.003
= Beginpoint Arrival Time -0.003
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
| Time | Time |
+-----+-----+-----+-----+-----+-----+
| t1_cs11_reg | CK ^ | | | -0.003 | 4.949 |
| t1_cs11_reg | CK ^ -> Q ^ | S0FFRHQX1 | 0.980 | 0.977 | 5.929 |
| g3469 | A ^ -> Y v | NAND2XL | 0.452 | 1.429 | 6.381 |
| g3464 | B v -> Y ^ | NOR3BX1 | 0.219 | 1.647 | 6.599 |
| g3460 | B ^ -> Y v | NAND2XL | 0.318 | 1.965 | 6.917 |
| g3458 | C v -> Y ^ | NOR3X1 | 0.288 | 2.253 | 7.205 |
```

Figure 69: Setup Report after detailed routing for 0.8

HOLD TIMING REPORT:

```
# OS: Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 06:47:27 2025
# Design: alarm_mt251
# Command: report_timing -early -view {view1} -max_paths 100 > post_route_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Hold Check with Pin bs_q_cs11_reg[1]/CK
Endpoint: bs_q_cs11_reg[1]/SE (v) checked with leading edge of 'clk_pb34'
Beginpoint: scan_en (v) triggered by leading edge of '@'
Path Groups: {clk_pb34}
Analysis View: view1
Other End Arrival Time -0.004
+ Hold 0.069
+ Phase Shift 0.000
+ Uncertainty 0.100
= Required Time 0.165
Arrival Time 0.222
Slack Time 0.058
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
|-----|-----|-----|-----|-----|-----|
| scan_en v | scan_en v | CLKBUX2 | 0.097 | 0.097 | 0.039 |
| FE_PHC6_scan_en | A v -> Y v | CLKBUX2 | 0.126 | 0.222 | 0.165 |
```

Figure 70: Hold Report after detailed routing for 0.8

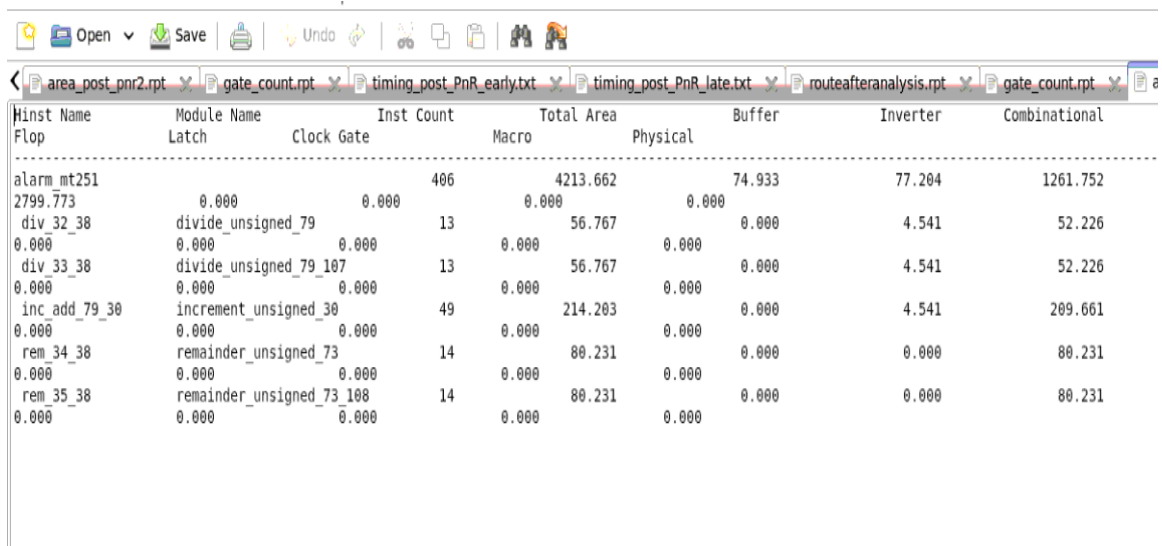
Table below compares worst setup and hold slack before and after detailed routing.

Table 8: Worst Setup and Hold slack before and after detailed routing for 0.8

	Setup Slack (ps)	Hold Slack (ps)
Before routing	4.944	0.057
After routing	4.952	0.058

For the 0.8 floorplan, the **setup slack** slightly improved from **4.944 ps to 4.952 ps** after routing, showing that the detailed routing stage introduced minimal parasitic effects and maintained good timing. The **hold slack** also improved marginally from **0.057 ps to 0.058 ps**, indicating stable short-path timing and effective clock balancing.

AREA REPORT:



Inst Name	Module Name	Inst Count	Total Area	Buffer	Inverter	Combinational
Flop	Latch	Clock Gate	Macro	Physical		
alarm_mt251		406	4213.662	74.933	77.204	1261.752
2799.773	0.000	0.000	0.000	0.000		
div_32_38	divide_unsigned_79	13	56.767	0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000		
div_33_38	divide_unsigned_79_107	13	56.767	0.000	4.541	52.226
0.000	0.000	0.000	0.000	0.000		
inc_add_79_30	increment_unsigned_30	49	214.203	0.000	4.541	209.661
0.000	0.000	0.000	0.000	0.000		
rem_34_38	remainder_unsigned_73	14	80.231	0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000		
rem_35_38	remainder_unsigned_73_108	14	80.231	0.000	0.000	80.231
0.000	0.000	0.000	0.000	0.000		

Figure 71: Area report after detailed routing for 0.8

Total Area = 4213.662 μm^2

GATES REPORT:



Gate area	Level	Module	Gates	Cells	Area
2.2707 μm^2	0	alarm_mt251	1855	406	4213.7 μm^2

Figure 72: Gate count after detailed routing for 0.8

Total Gates = 1855

POWER REPORT:

Figure 73: Power report after detailed routing for 0.8

g3452	7.563e-07	7.55e-07	1.387e-05	1.236e-05	NOR2XL
g3478	1.893e-09	1.563e-08	1.303e-05	1.301e-05	CLKINVX1
inc_add_79_30/g788	4.086e-07	2.034e-07	1.297e-05	1.236e-05	NOR2XL
g3454	6.638e-07	2.741e-06	1.269e-05	9.289e-06	NAND2XL
g3438	9.121e-08	3.284e-08	1.248e-05	1.236e-05	NOR2XL
g3442	5.904e-08	5.748e-08	1.247e-05	1.236e-05	NOR2XL
g3431	6.77e-09	7.541e-09	1.237e-05	1.236e-05	NOR2XL
g3429	9.275e-10	1.361e-09	1.236e-05	1.236e-05	NOR2XL
g3426	1.102e-10	1.138e-10	1.236e-05	1.236e-05	NOR2XL
g1991	4.874e-10	4.509e-10	1.046e-05	1.046e-05	NAND4XL
g3443	2.881e-10	2.673e-10	1.046e-05	1.046e-05	NAND4XL
inc_add_79_30/g768	3.669e-07	6.396e-07	1.03e-05	9.289e-06	NAND2XL
inc_add_79_30/g782	1.05e-07	2.53e-07	9.647e-06	9.289e-06	NAND2XL
g3445	5.723e-08	2.17e-07	9.564e-06	9.289e-06	NAND2XL
inc_add_79_30/g766	2.773e-08	5.174e-08	9.369e-06	9.289e-06	NAND2XL
g2056	3.835e-07	1.924e-07	9.333e-06	8.757e-06	INVXL
inc_add_79_30/g779	8.275e-09	2.078e-08	9.318e-06	9.289e-06	NAND2XL
g2030	1.539e-08	8.402e-09	9.313e-06	9.289e-06	NAND2XL
g3435	3.599e-09	1.629e-08	9.309e-06	9.289e-06	NAND2XL
inc_add_79_30/g764	2.029e-09	3.716e-09	9.295e-06	9.289e-06	NAND2XL
g1450	5.576e-11	4.318e-11	8.757e-06	8.757e-06	INVXL
g1989	1.146e-09	7.304e-10	8.73e-06	8.728e-06	OAI21XL

Total (406 of 406)	0.3041	0.06347	0.3912	0.0237	
Total Capacitance	5.646e-12 F				
Power Density	*** No Die Area ***				

Total power consumption = 391.2uW

LAYOUT:

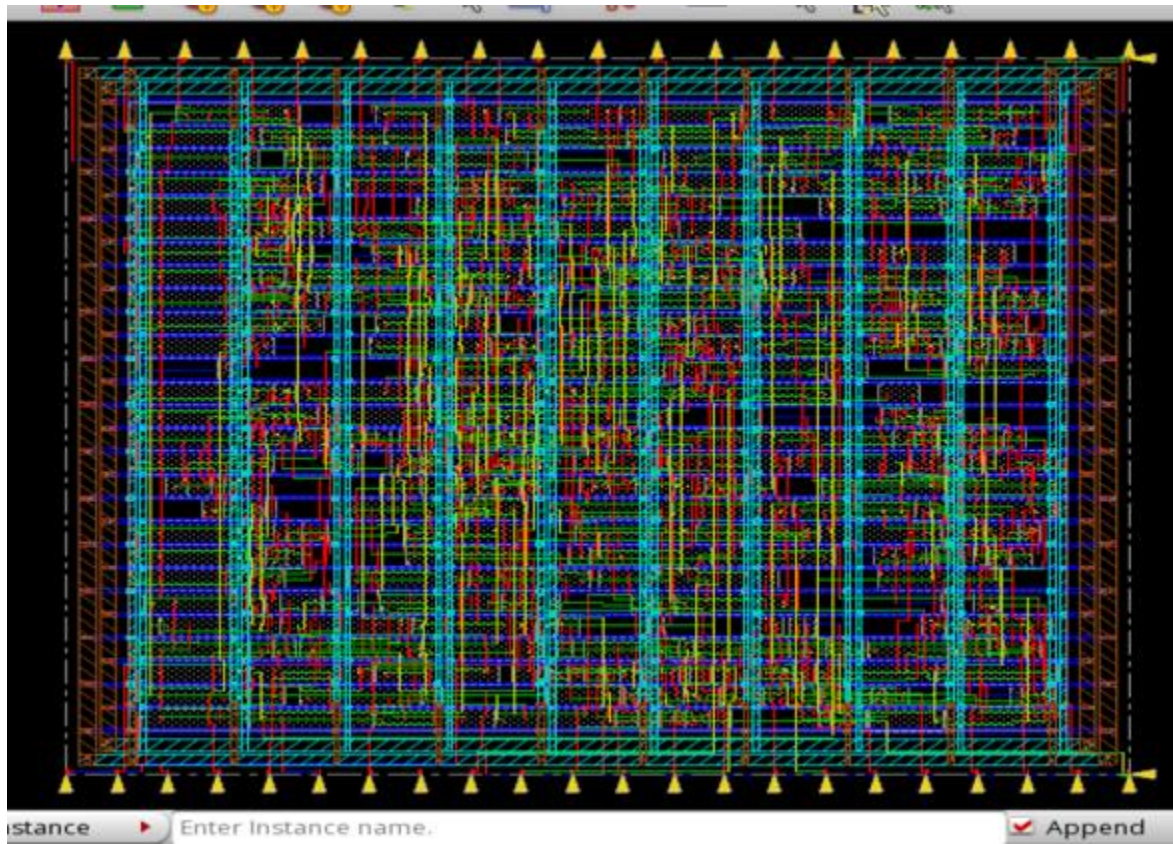


Figure 75: Layout after detailed routing for 0.8

METAL 1:

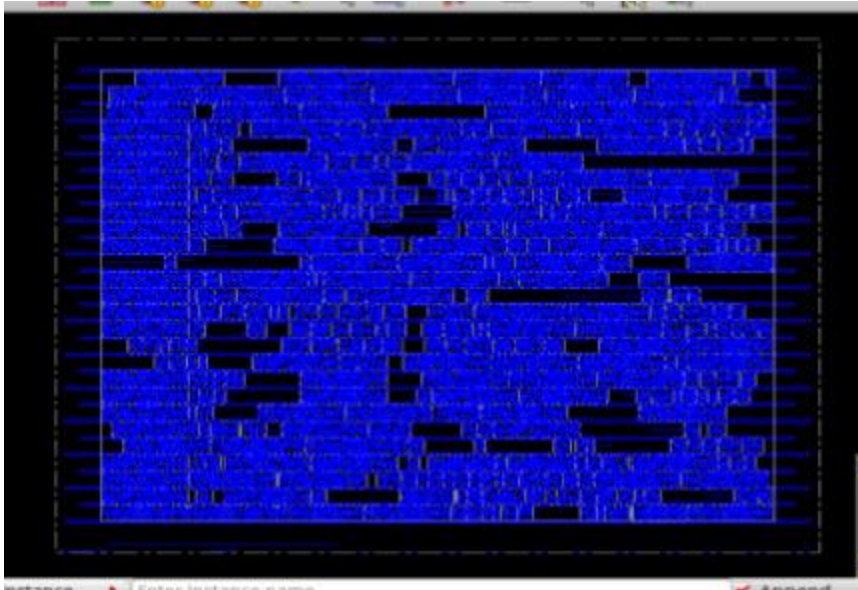


Figure 76: Layout showing metal layer 1 for 0.8

METAL 2:

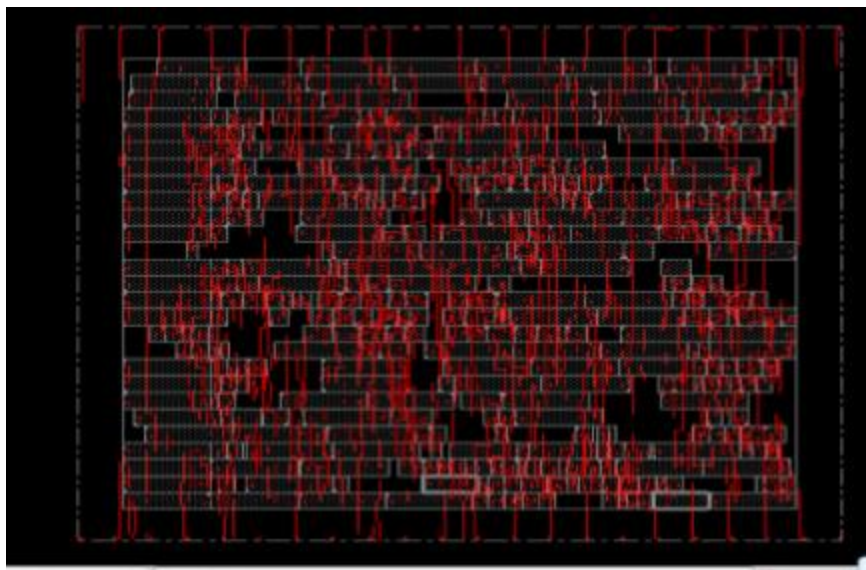


Figure 77: Layout showing metal layer 2 for 0.8

METAL 3:

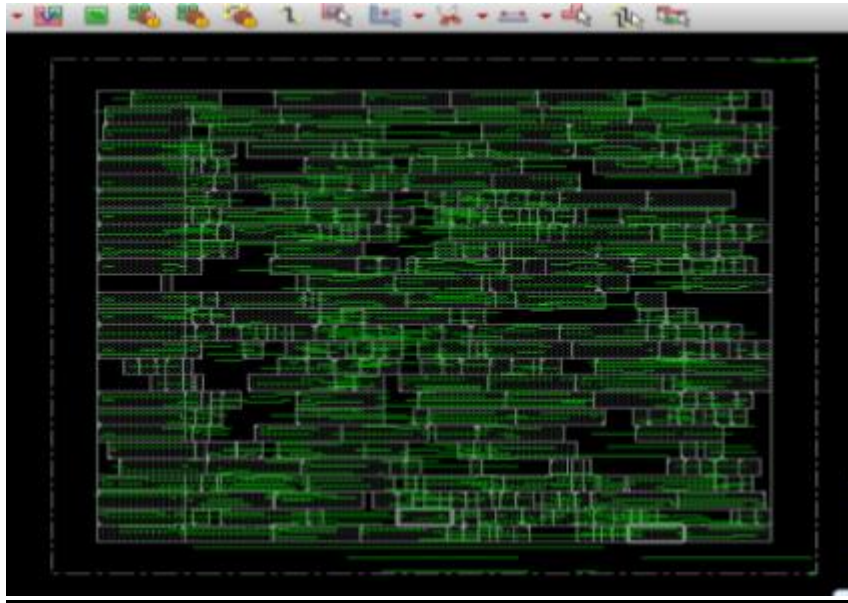


Figure 78: Layout showing metal layer 3 for 0.8

METAL 4:

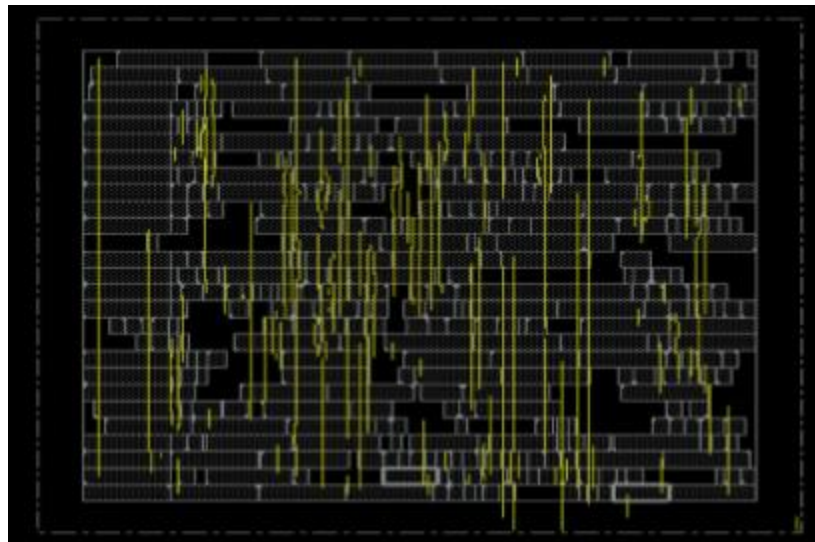


Figure 79: Layout showing metal layer 4 for 0.8

METAL 5:

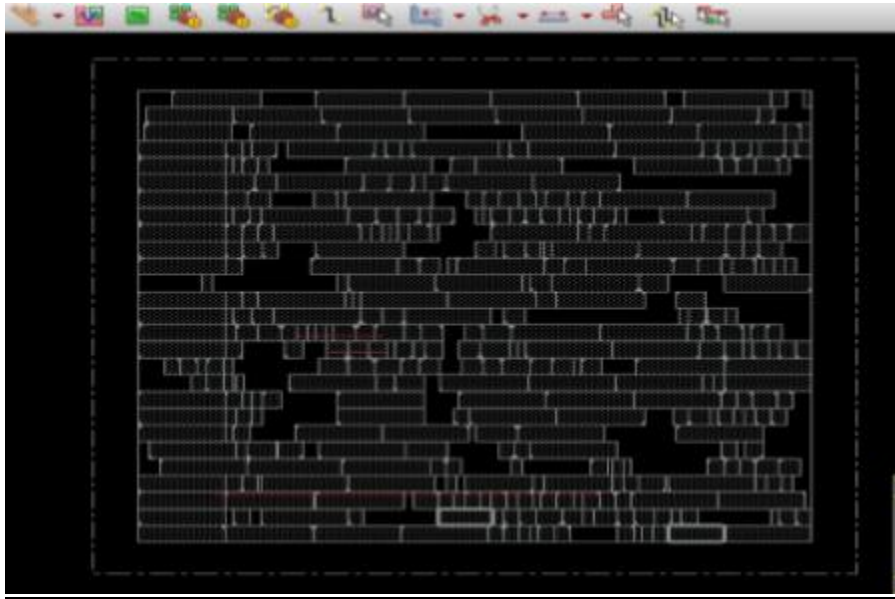


Figure 80: Layout showing metal layer 5 for 0.8

METAL 6:

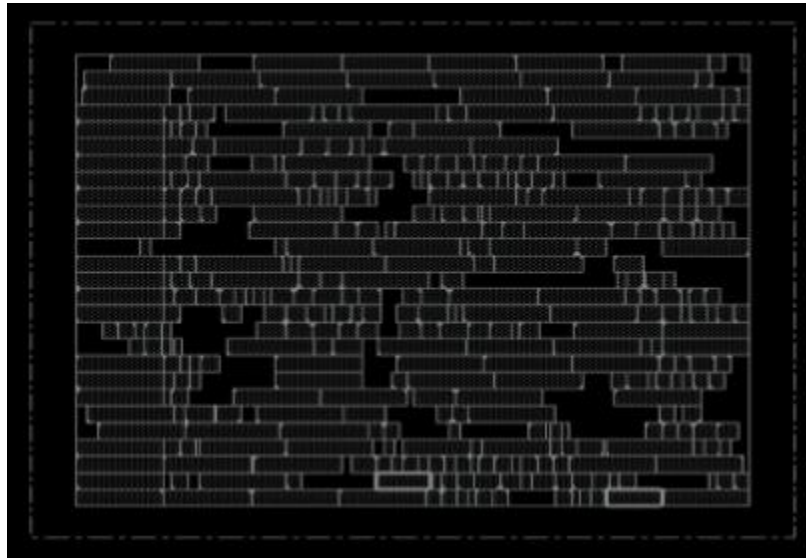


Figure 81: Layout showing metal layer 6 for 0.8

METAL 7:

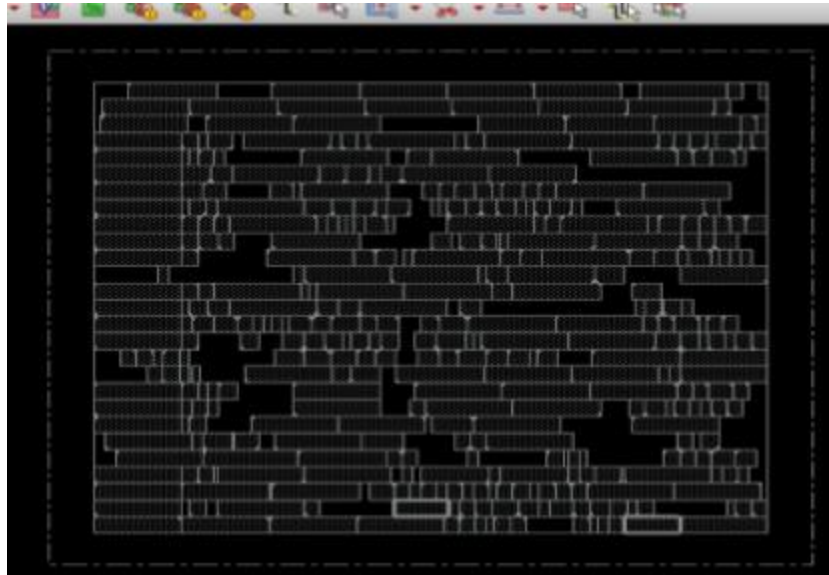


Figure 82: Layout showing metal layer 7 for 0.8

METAL 8:

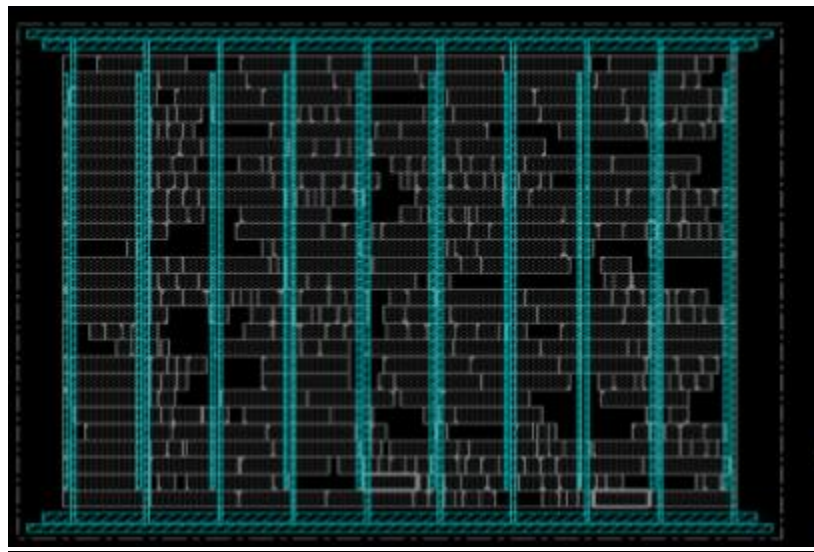


Figure 83: Layout showing metal layer 8 for 0.8

METAL 9:

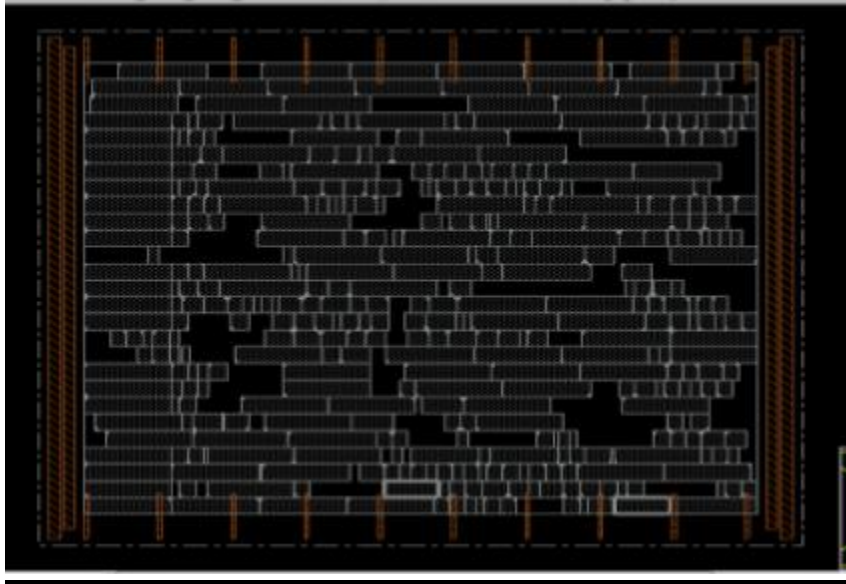


Figure 84: Layout showing metal layer 9 for 0.8